



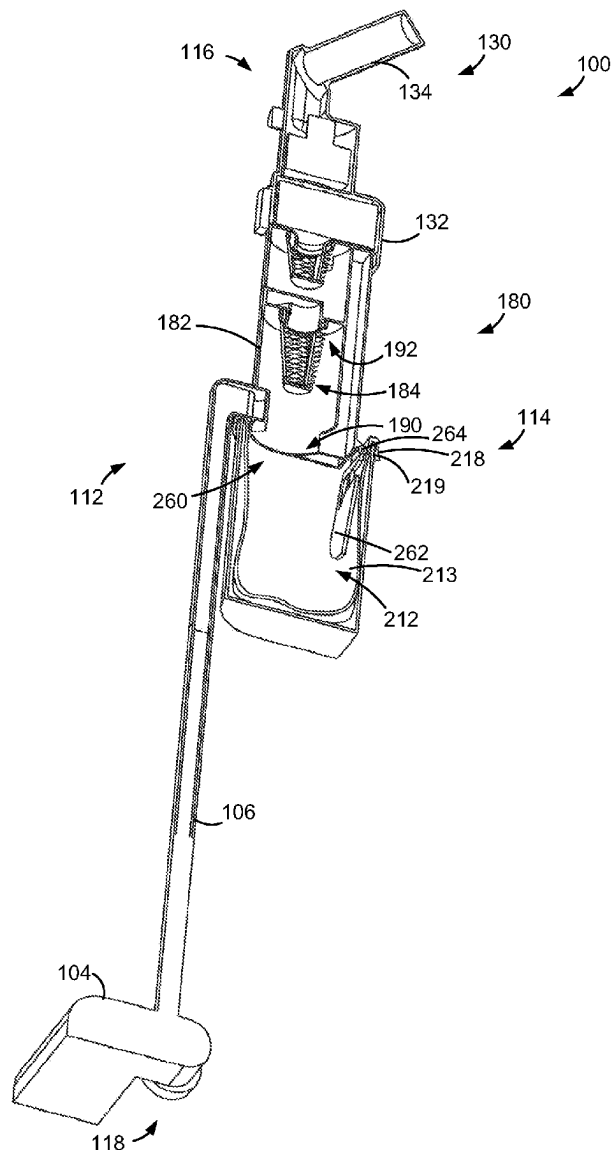
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(19) **United States**(12) **Patent Application Publication**  
**Conrad**(10) **Pub. No.: US 2025/0281016 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **SURFACE CLEANING APPARATUS**(71) Applicant: **Omachron Intellectual Property Inc.,**  
Hampton (CA)(72) Inventor: **Wayne Ernest Conrad, Hampton (CA)**(21) Appl. No.: **18/596,575**(22) Filed: **Mar. 5, 2024****Publication Classification**(51) **Int. Cl.**  
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(2013.01); *A47L 9/1683* (2013.01)

(57)

**ABSTRACT**

A surface cleaning apparatus includes an air treatment chamber having a first end having an air inlet and a dirt outlet and an axially opposed second end having an air outlet. A dirt collection chamber is moveable between an in use position at the first end of the air treatment chamber and an emptying position. A plate is moveably mounted to the air treatment chamber and moveable between a closed position at an interface of the dirt collection chamber and the air treatment chamber and an open position moved away from the interface. When the plate is closed the dirt outlet comprises a space between the plate and a wall of the air treatment chamber. A bag is positionable in the dirt collection chamber and is removable from the dirt collection chamber when the dirt collection chamber is in the emptying position.



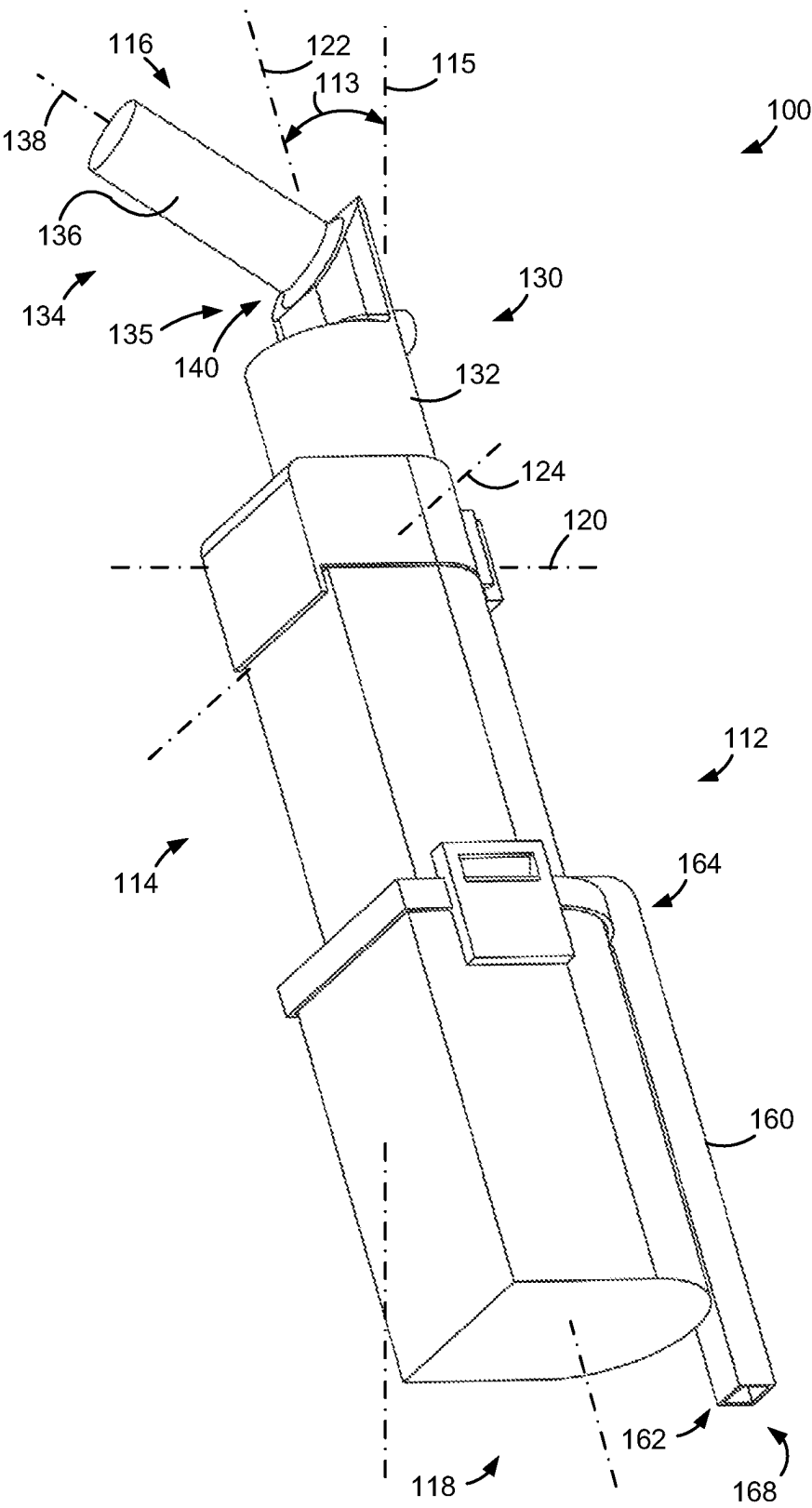
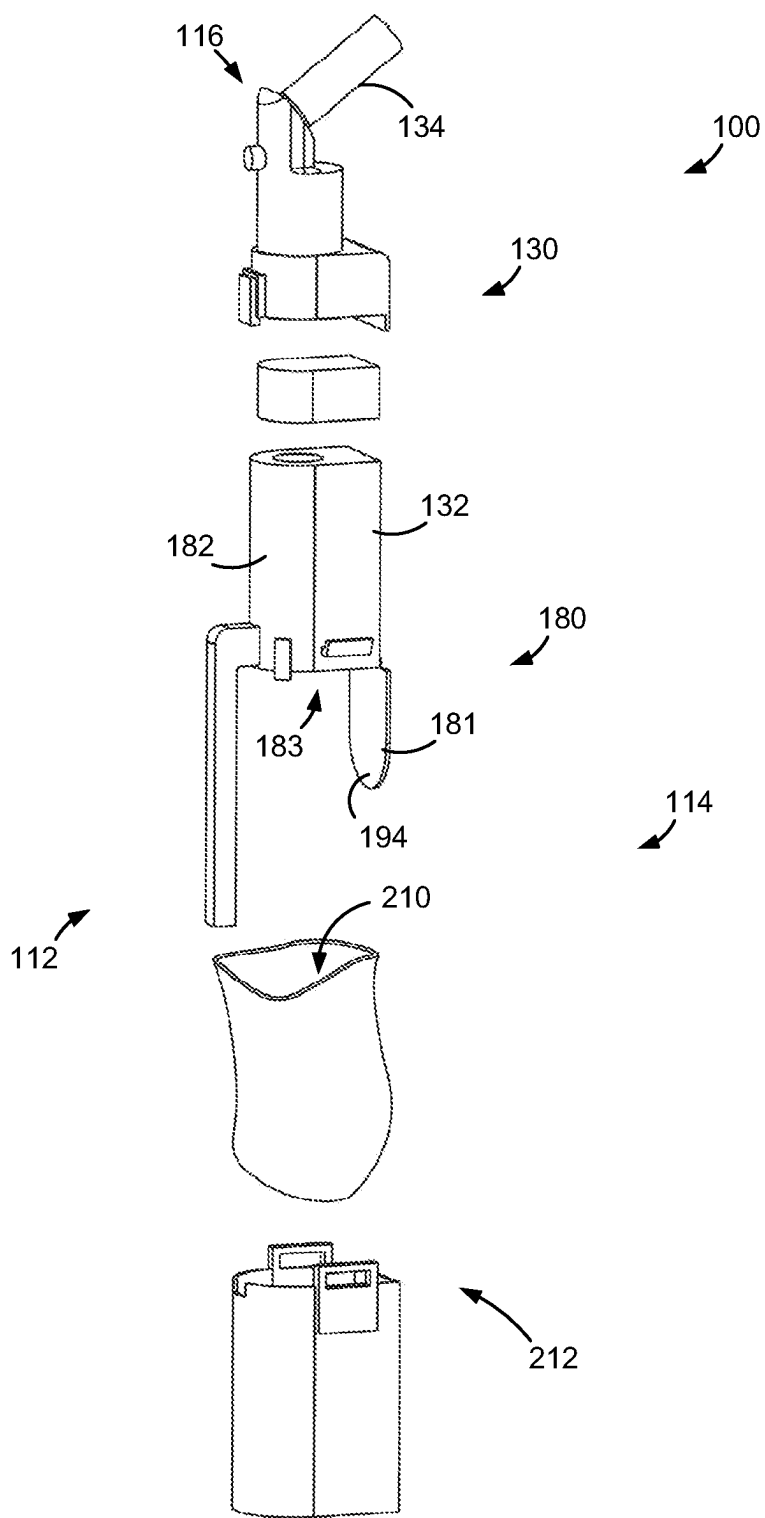
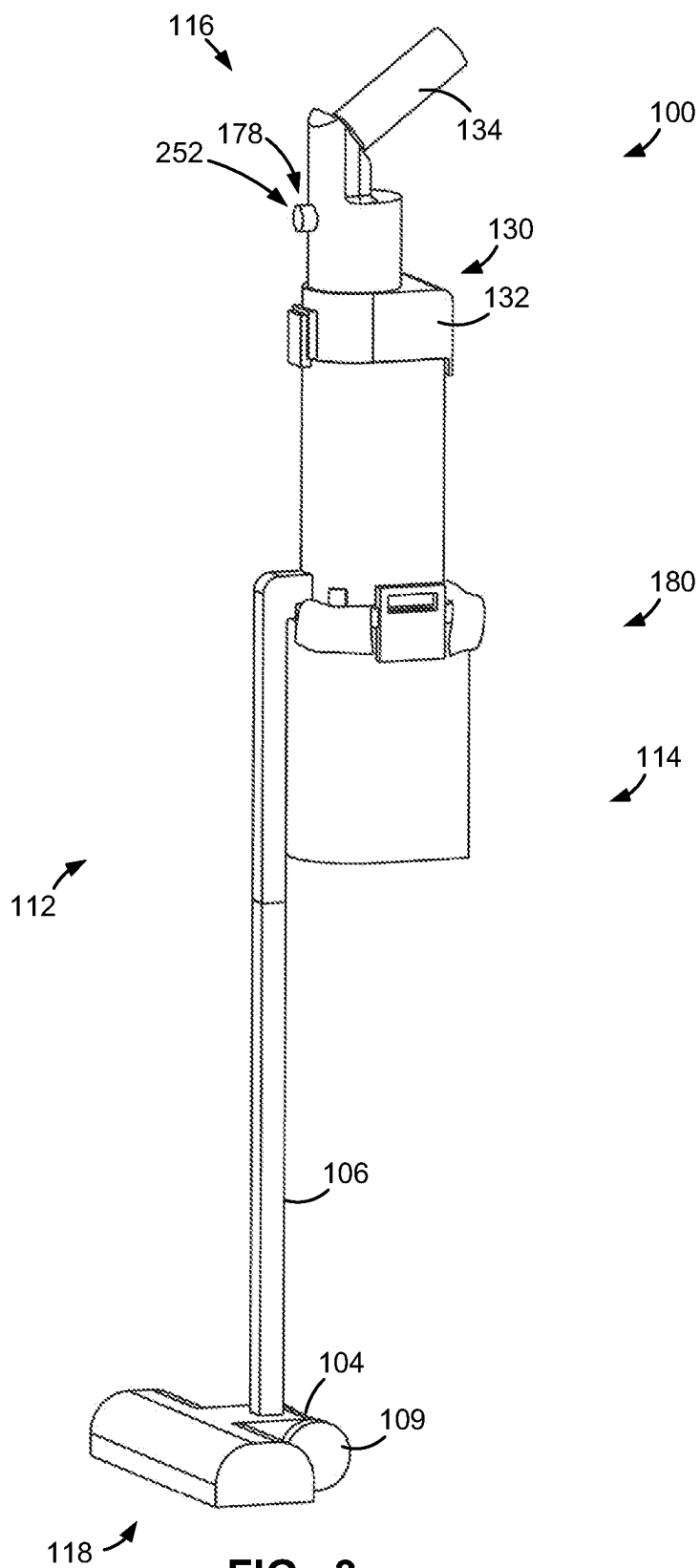


FIG. 1



118  
**FIG. 2**



**FIG. 3**

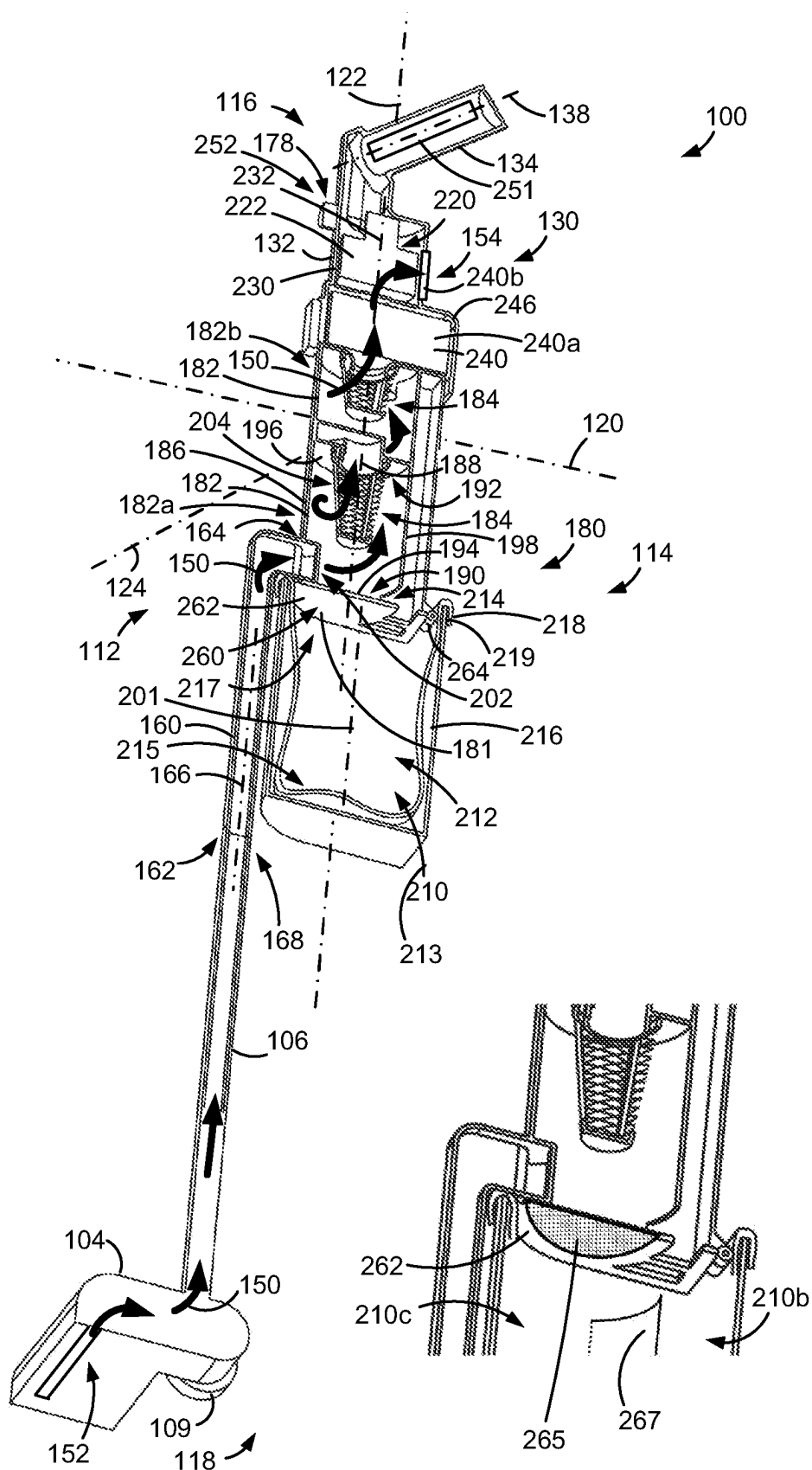
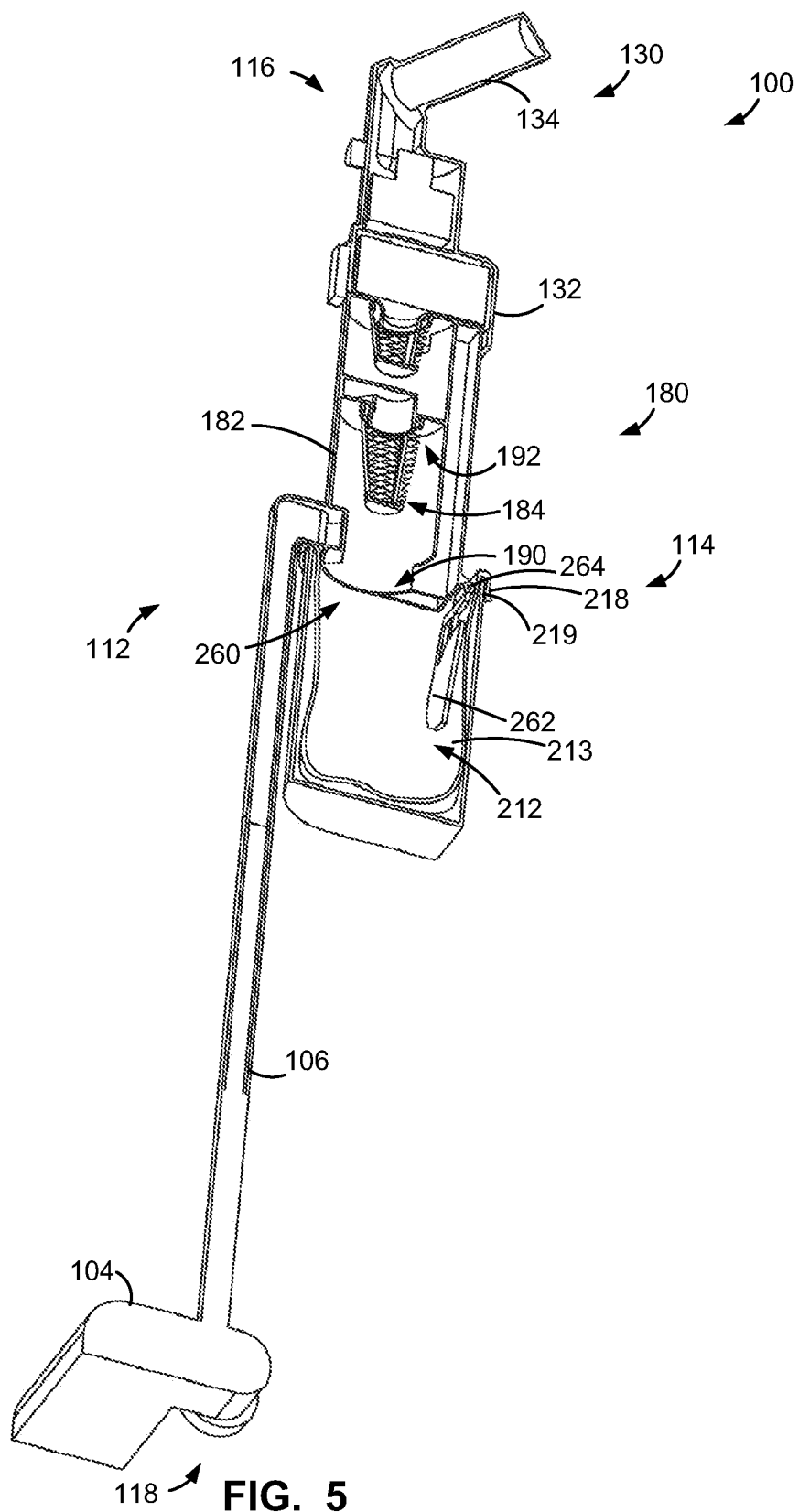


FIG. 4

FIG. 4A



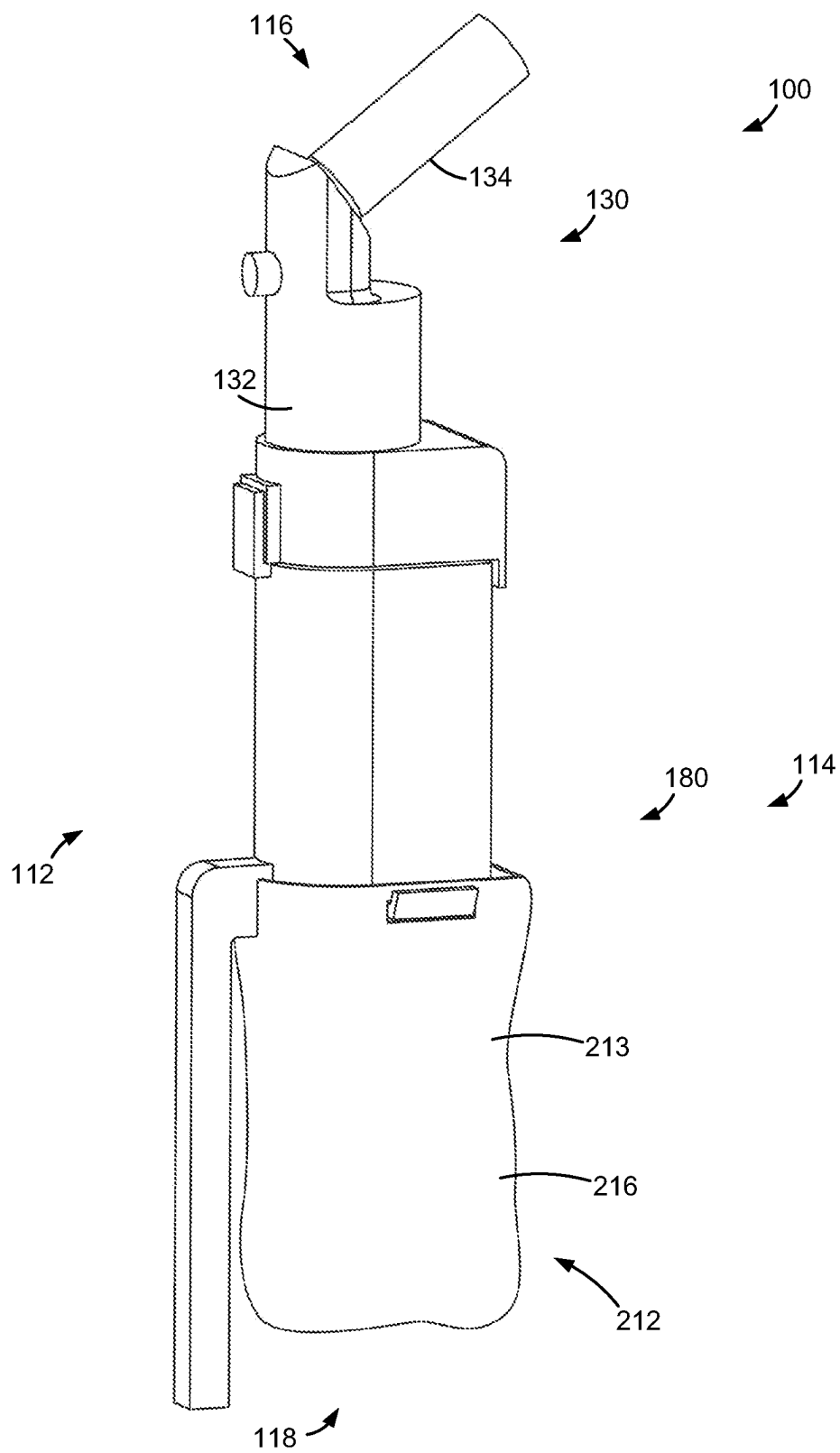
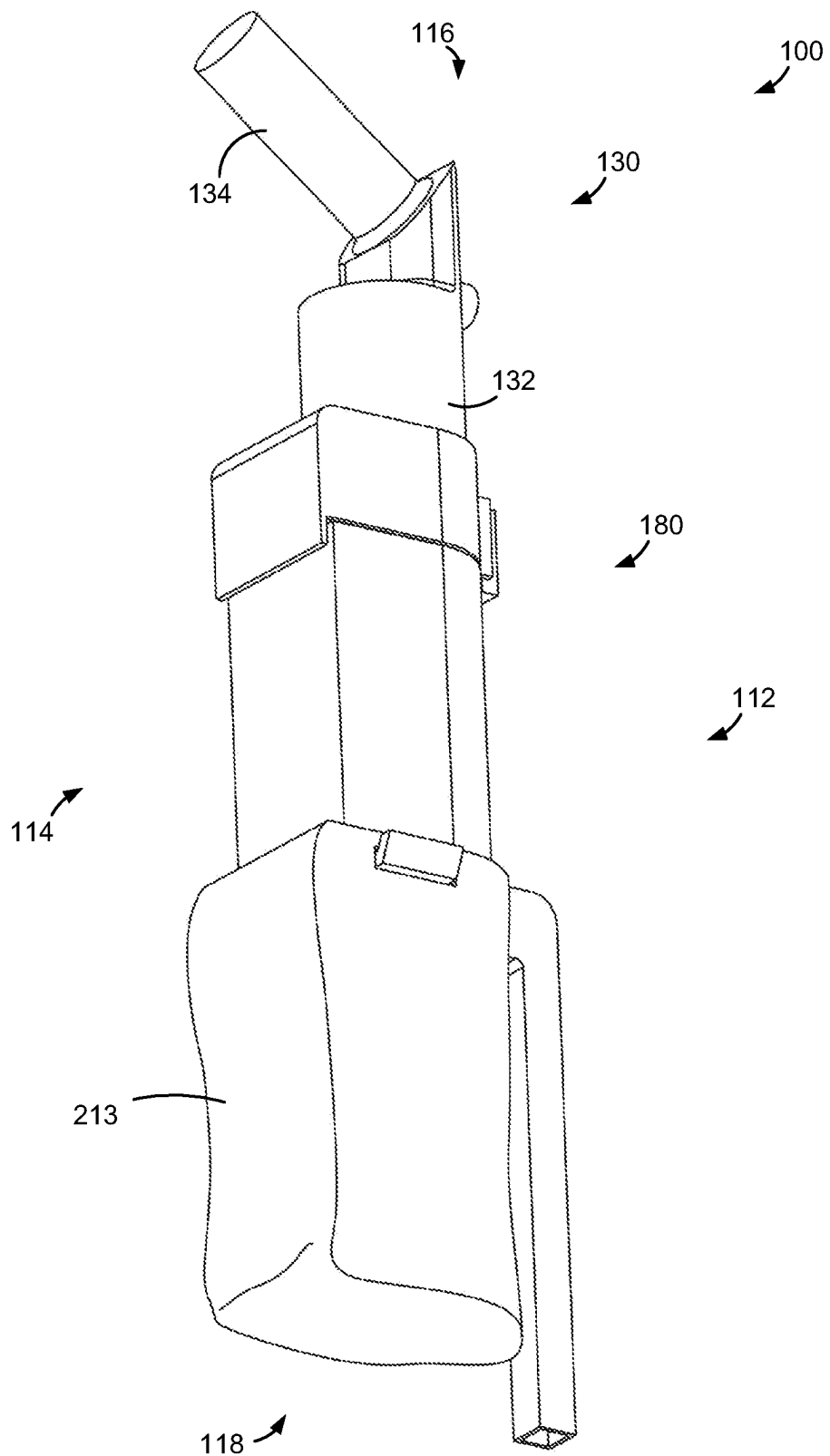
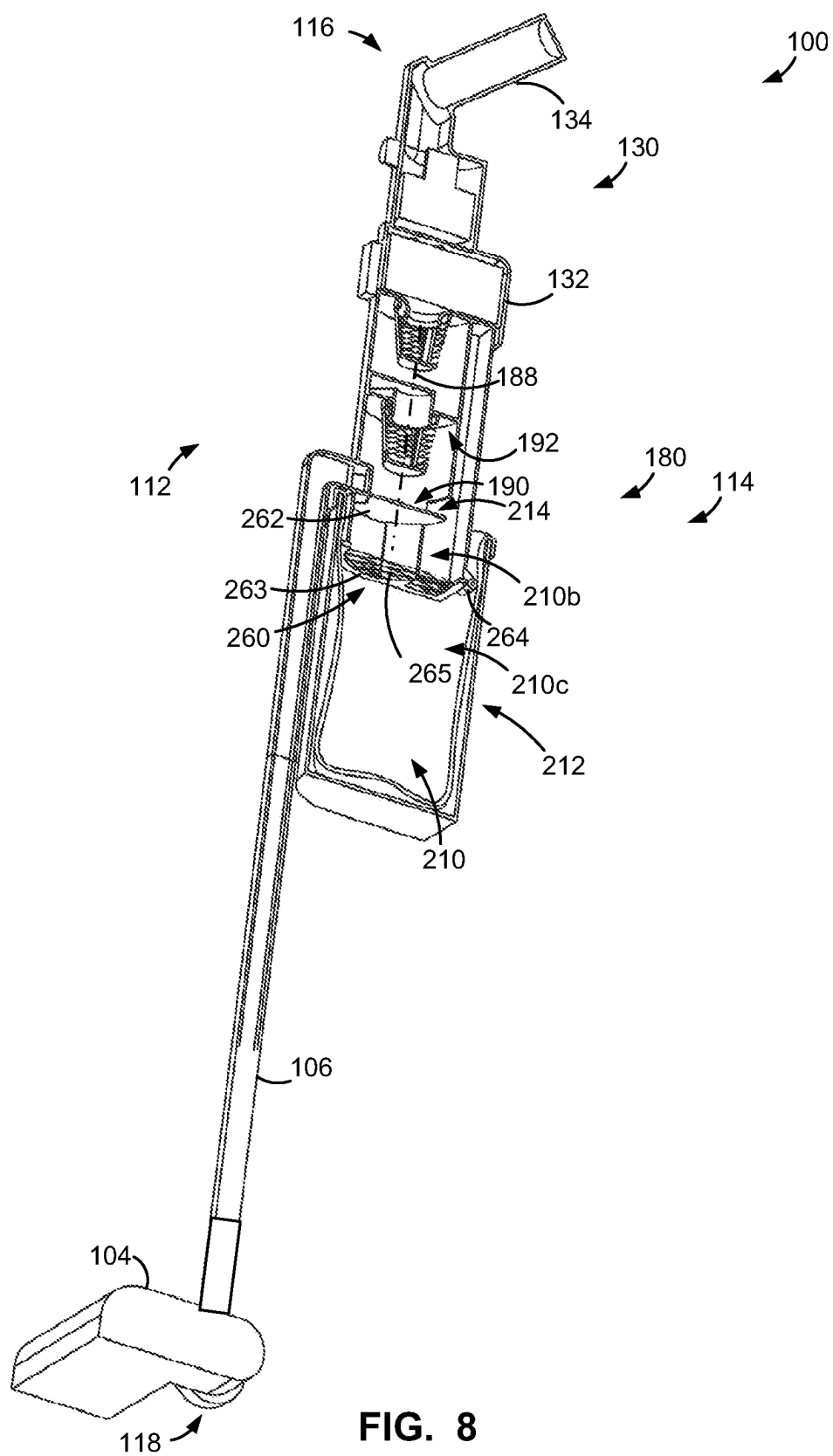


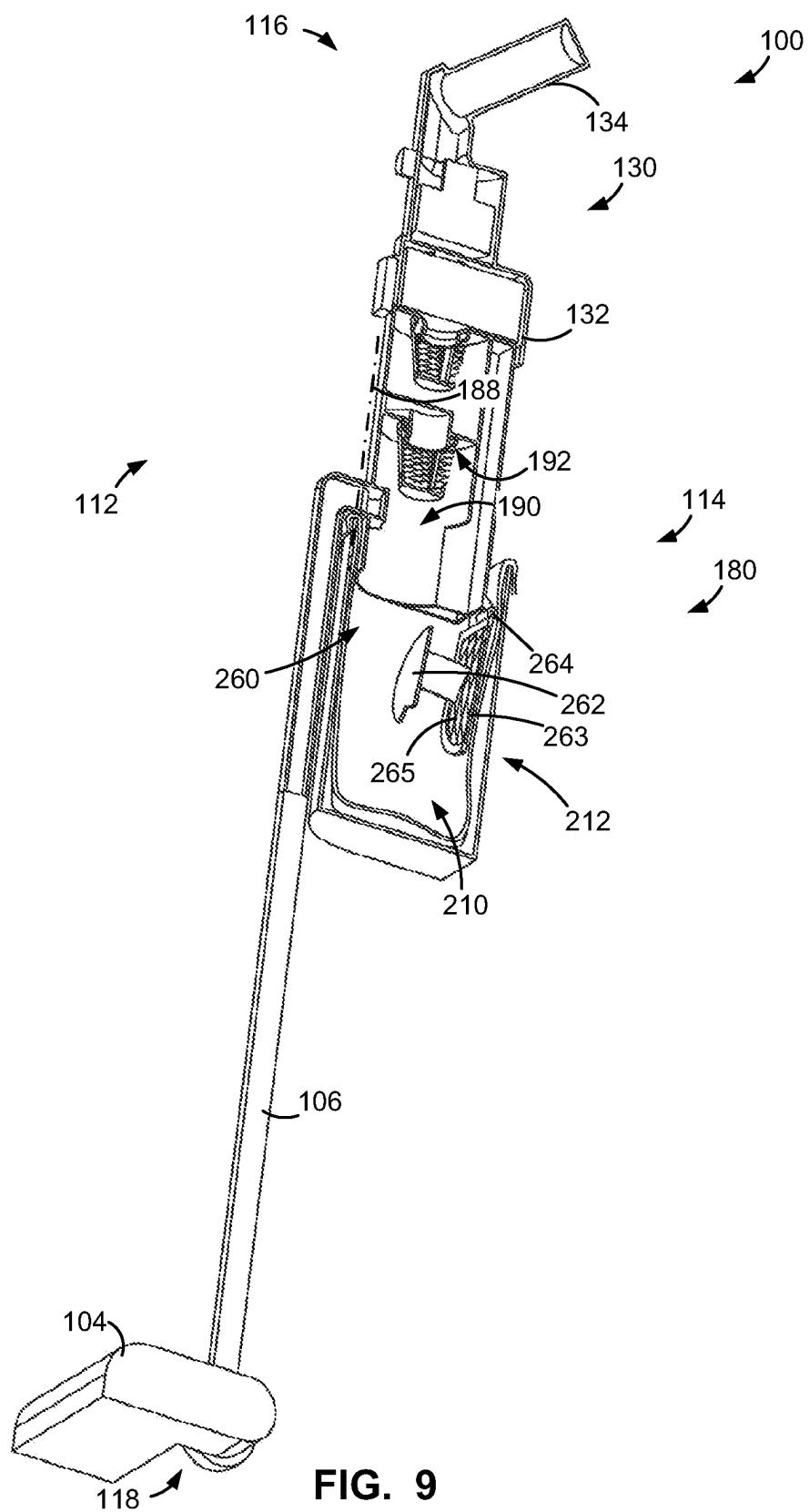
FIG. 6



**FIG. 7**







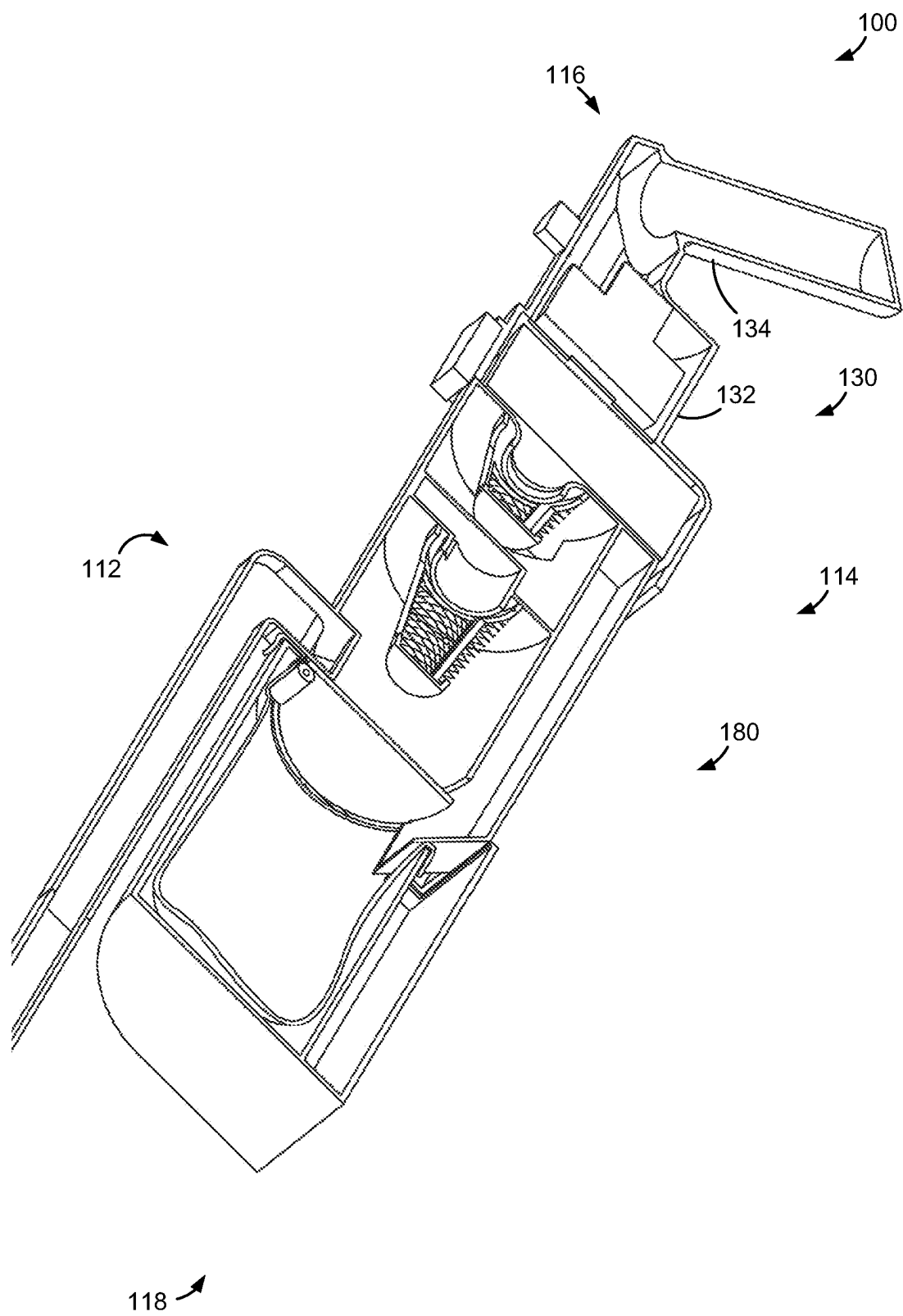


FIG. 10

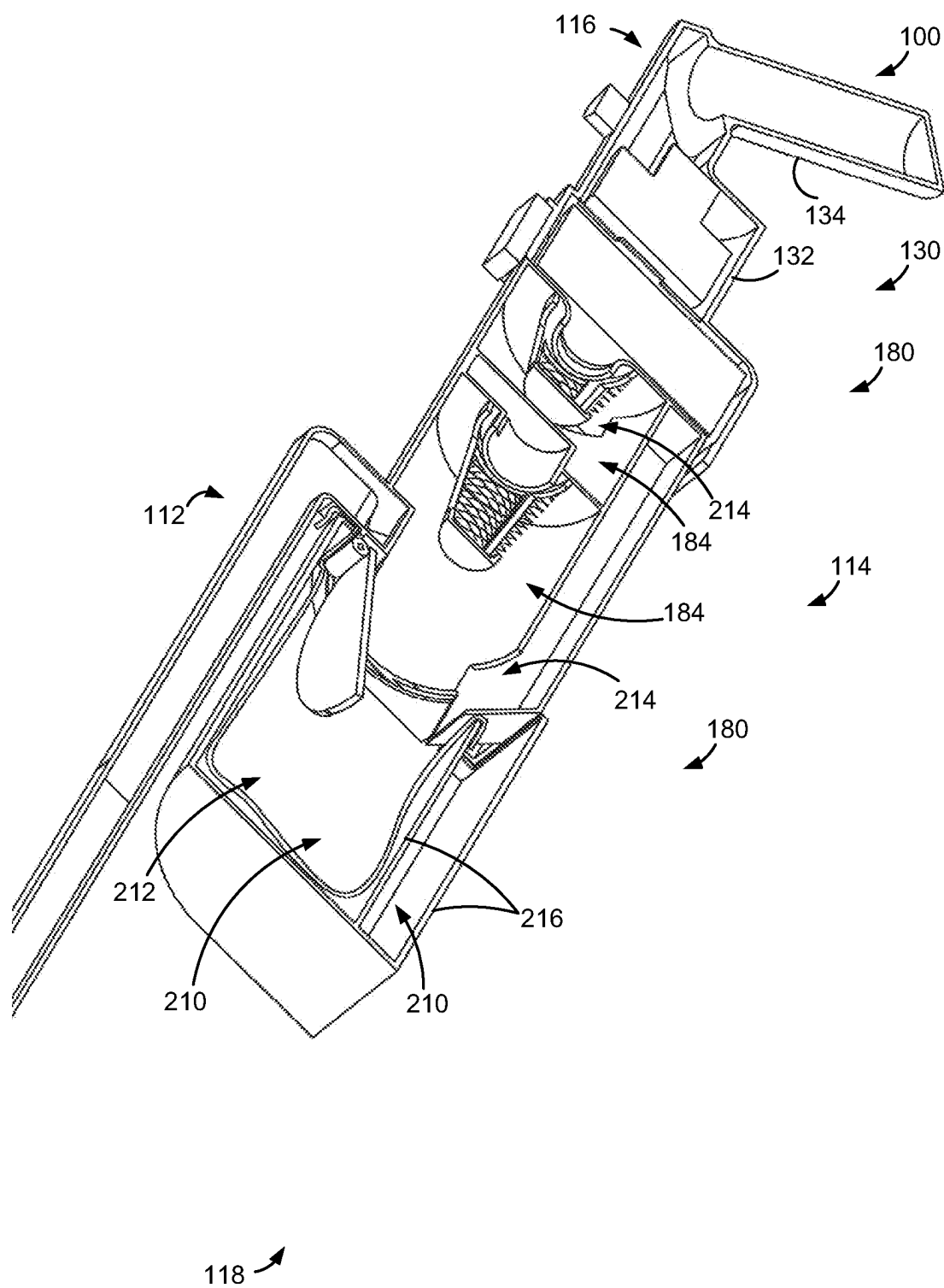


FIG. 11

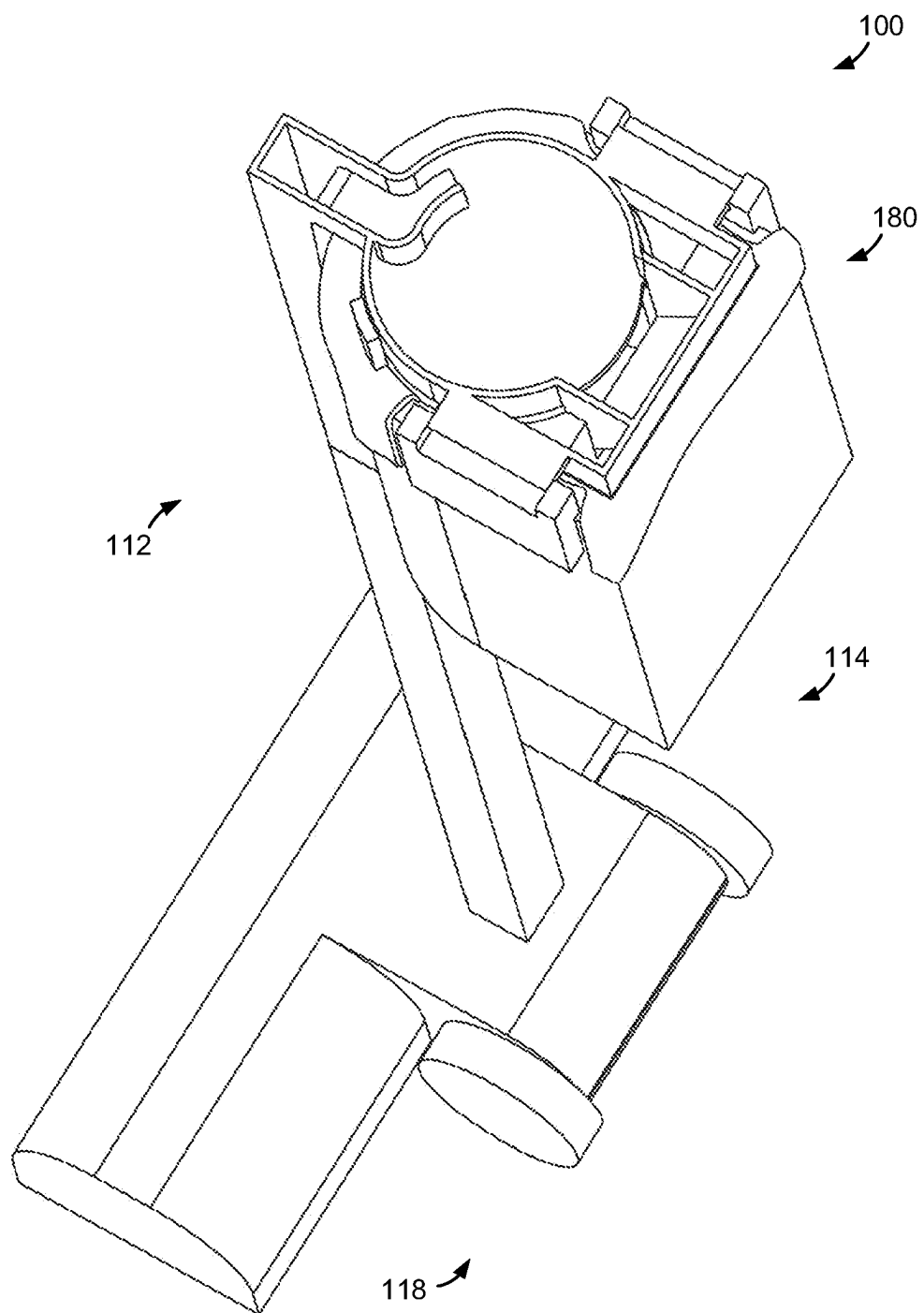


FIG. 12

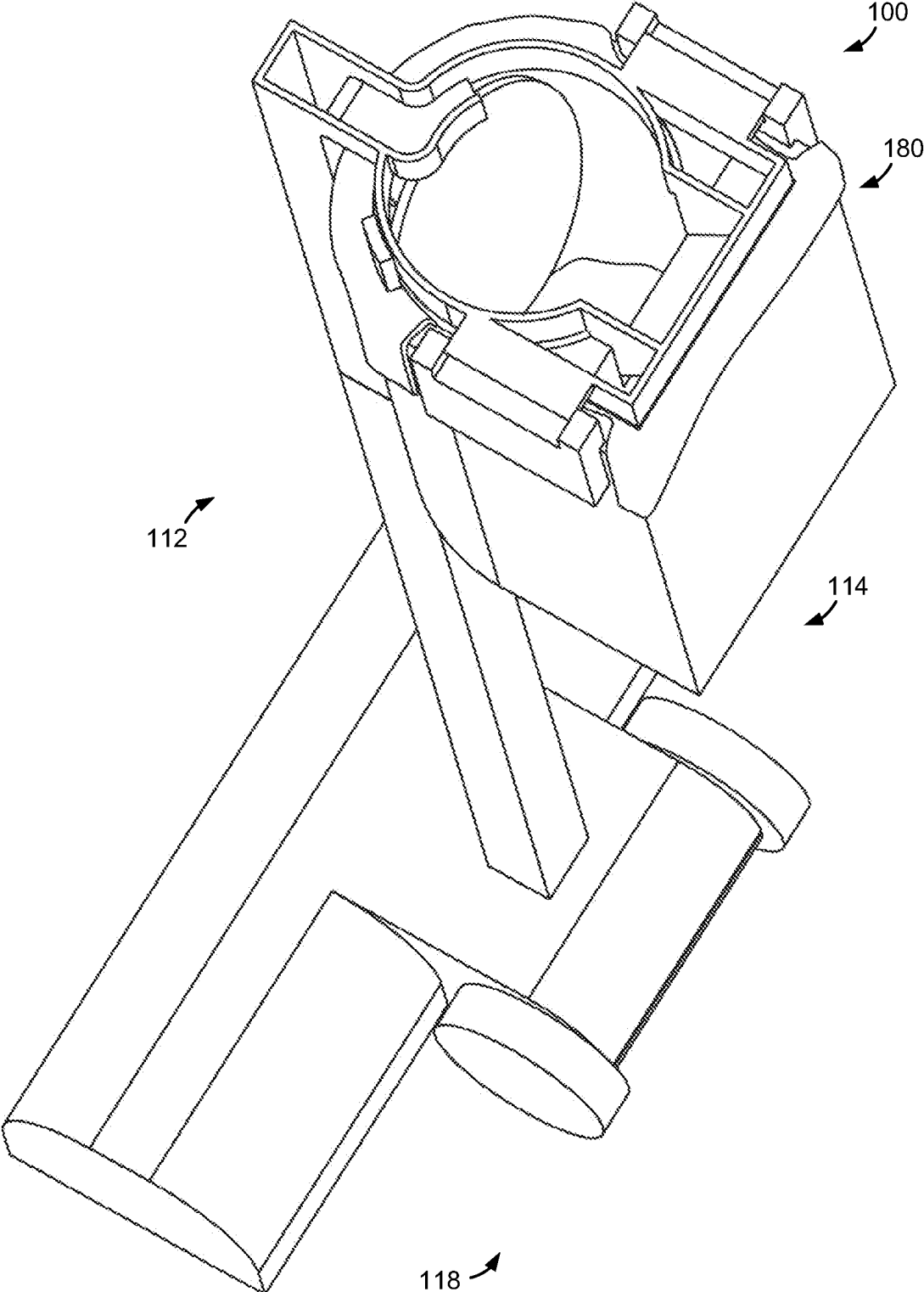
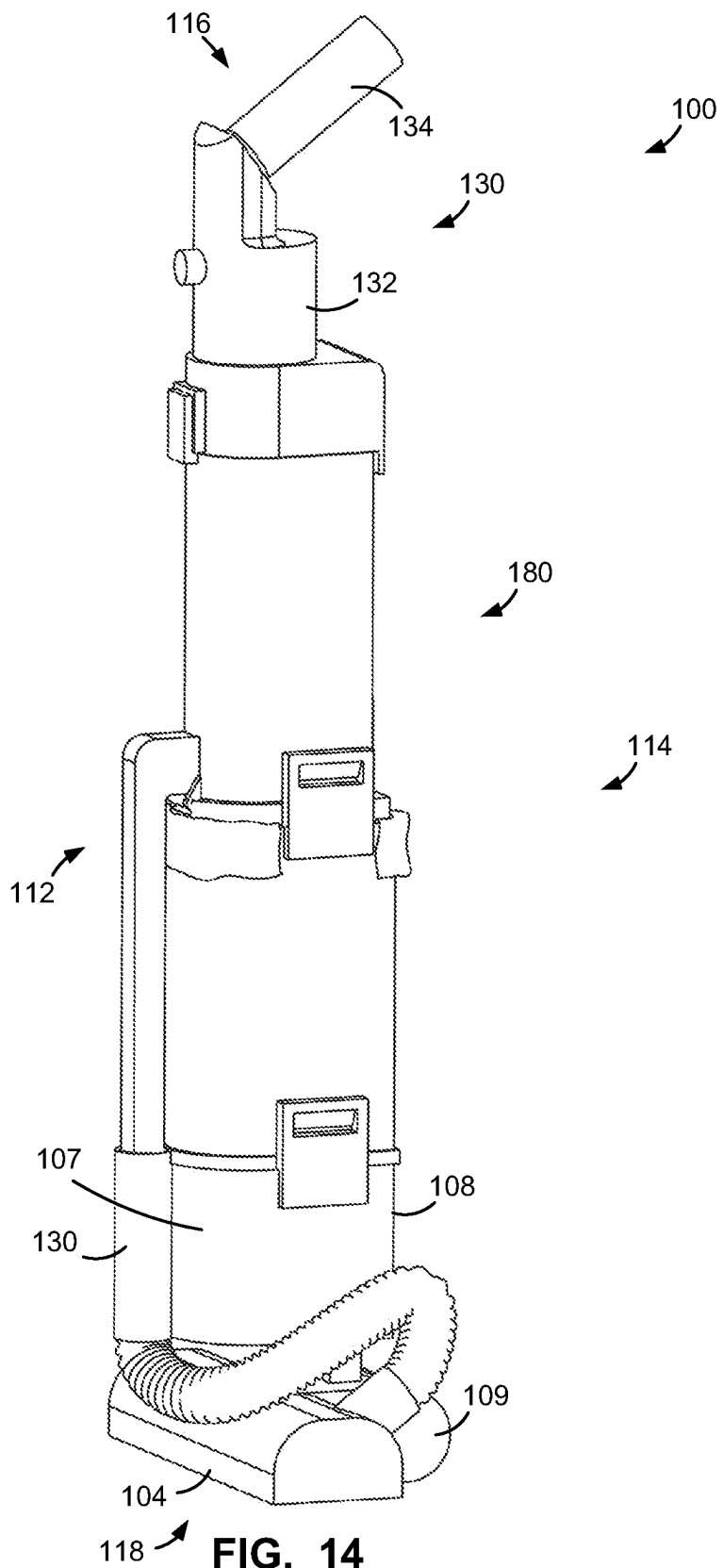


FIG. 13



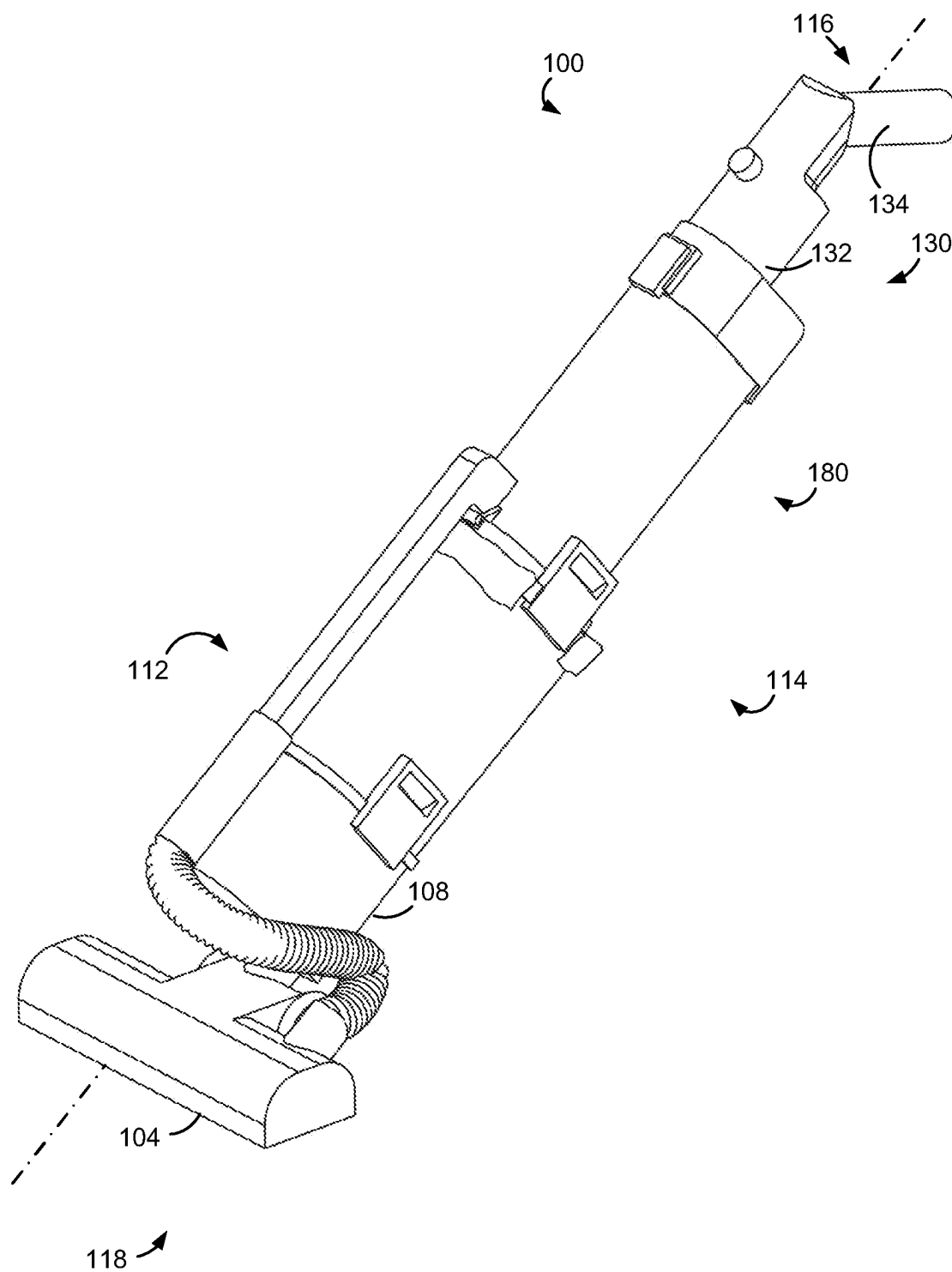


FIG. 15



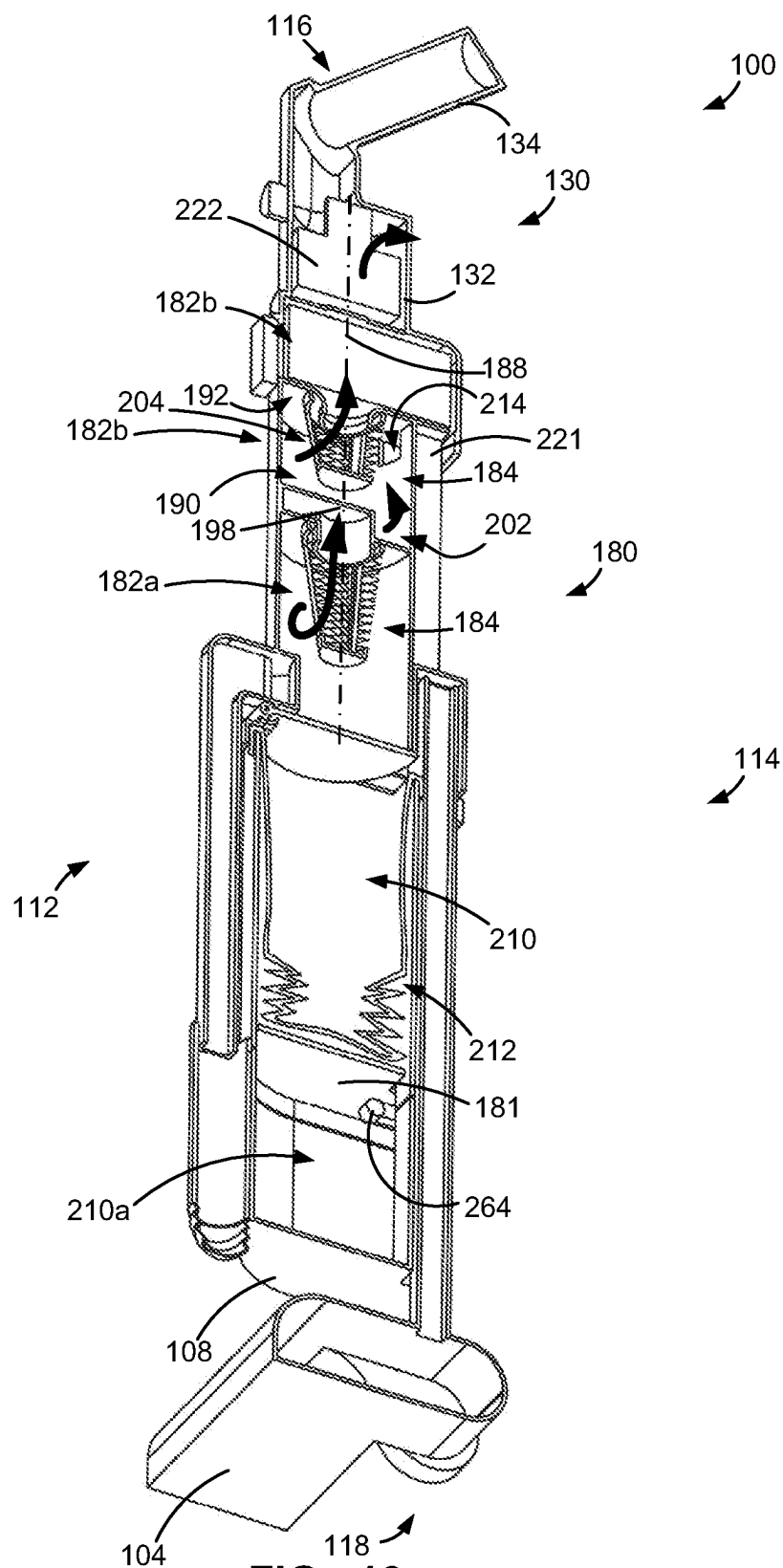
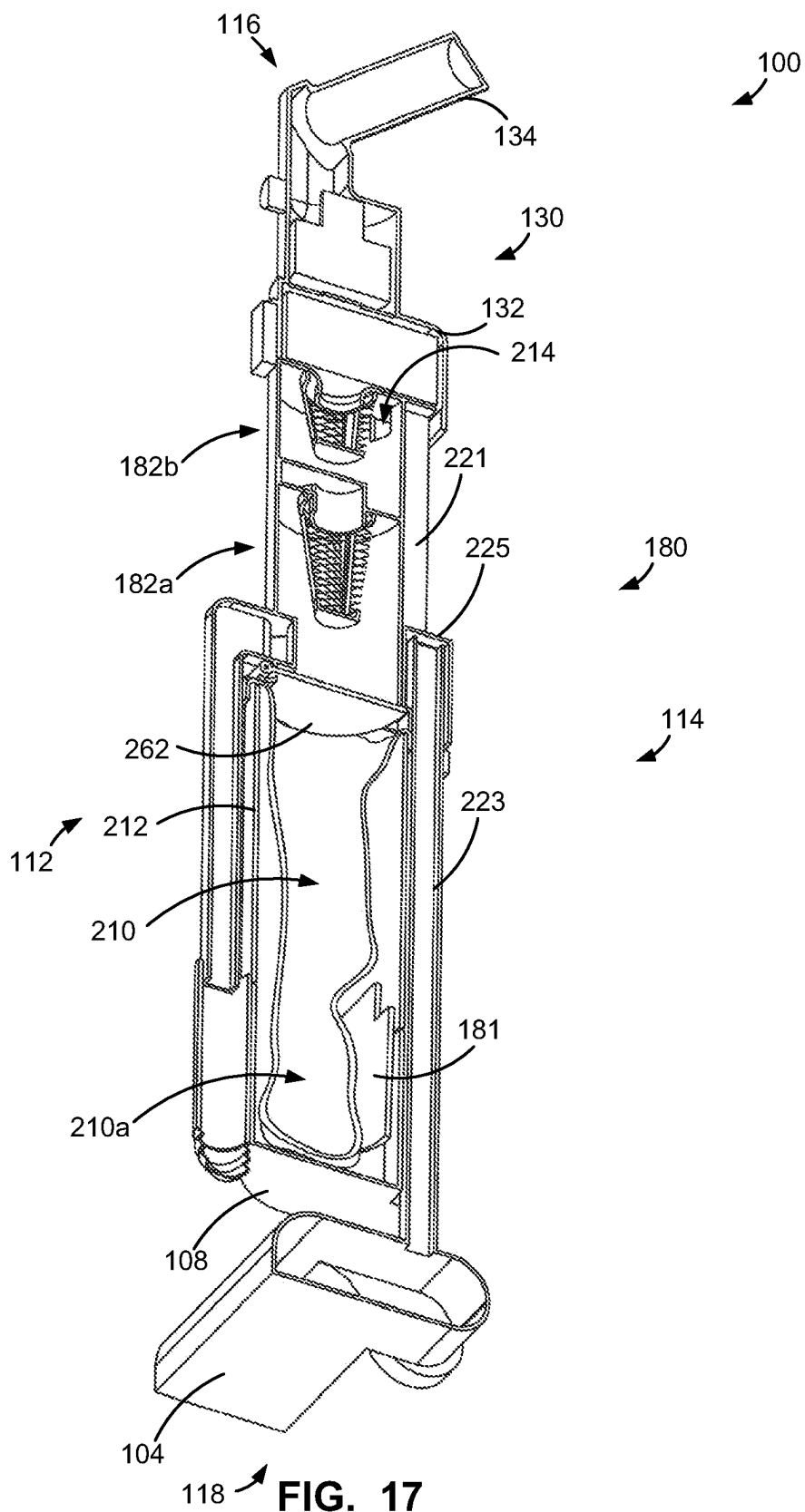


FIG. 16



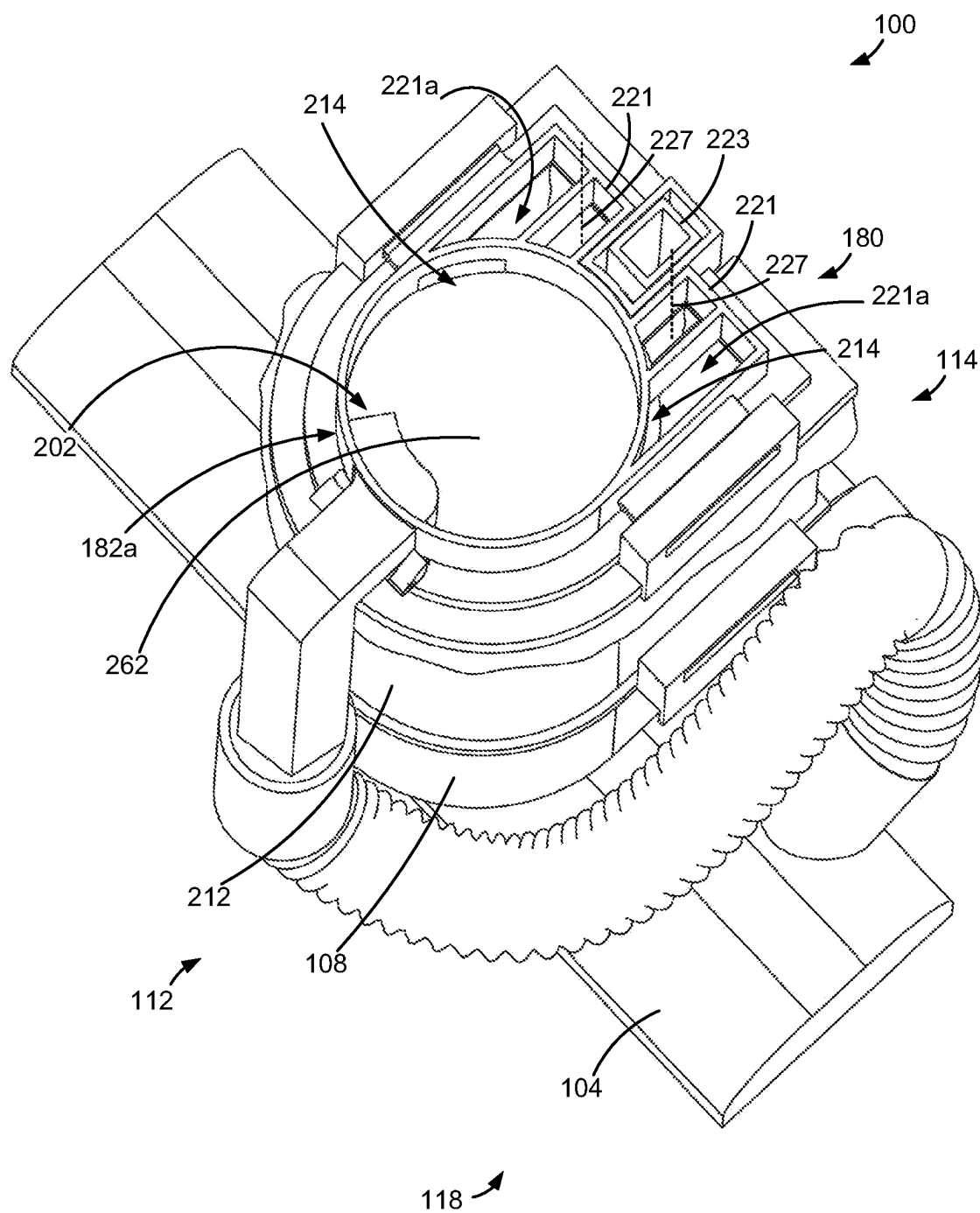


FIG. 18

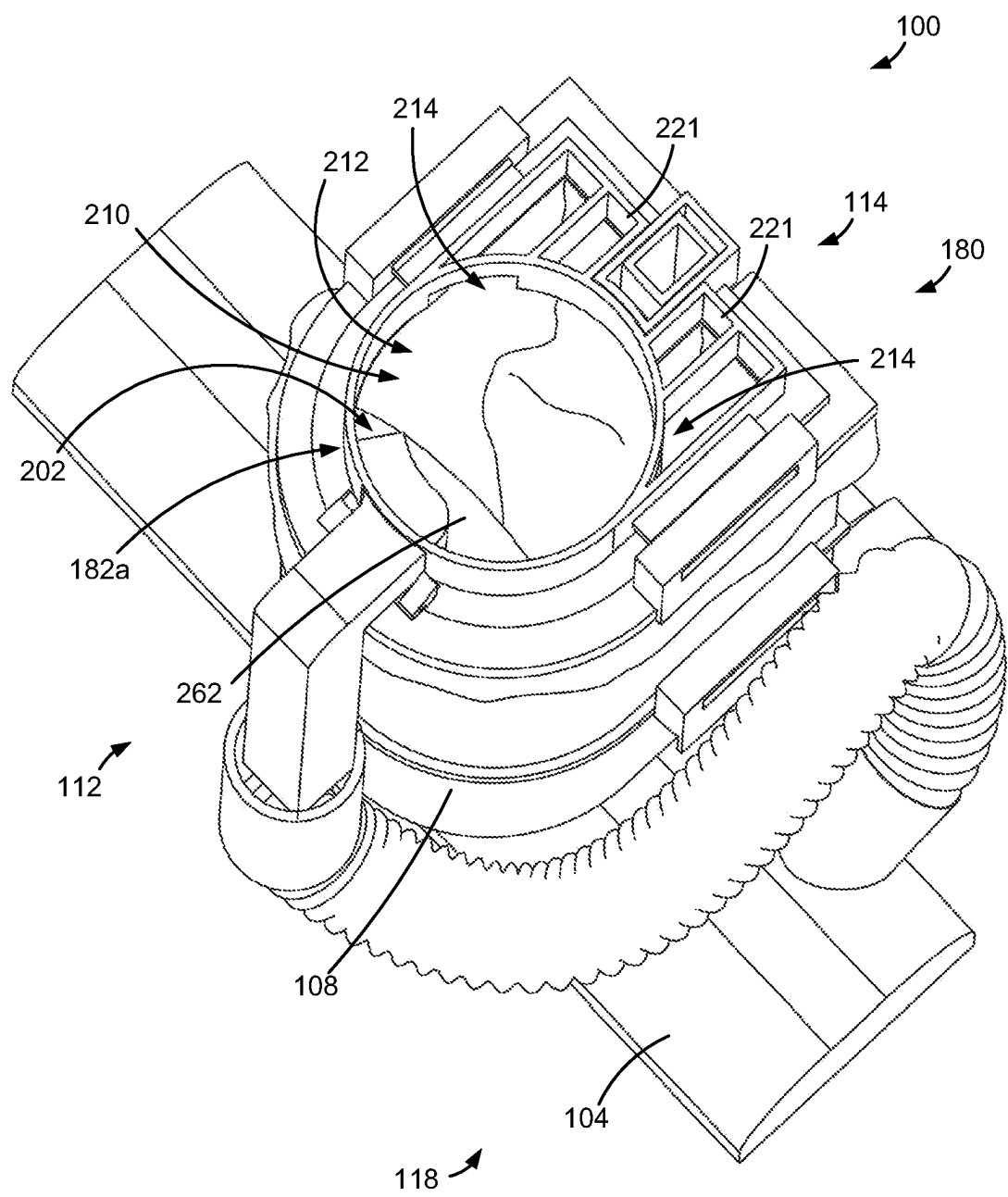
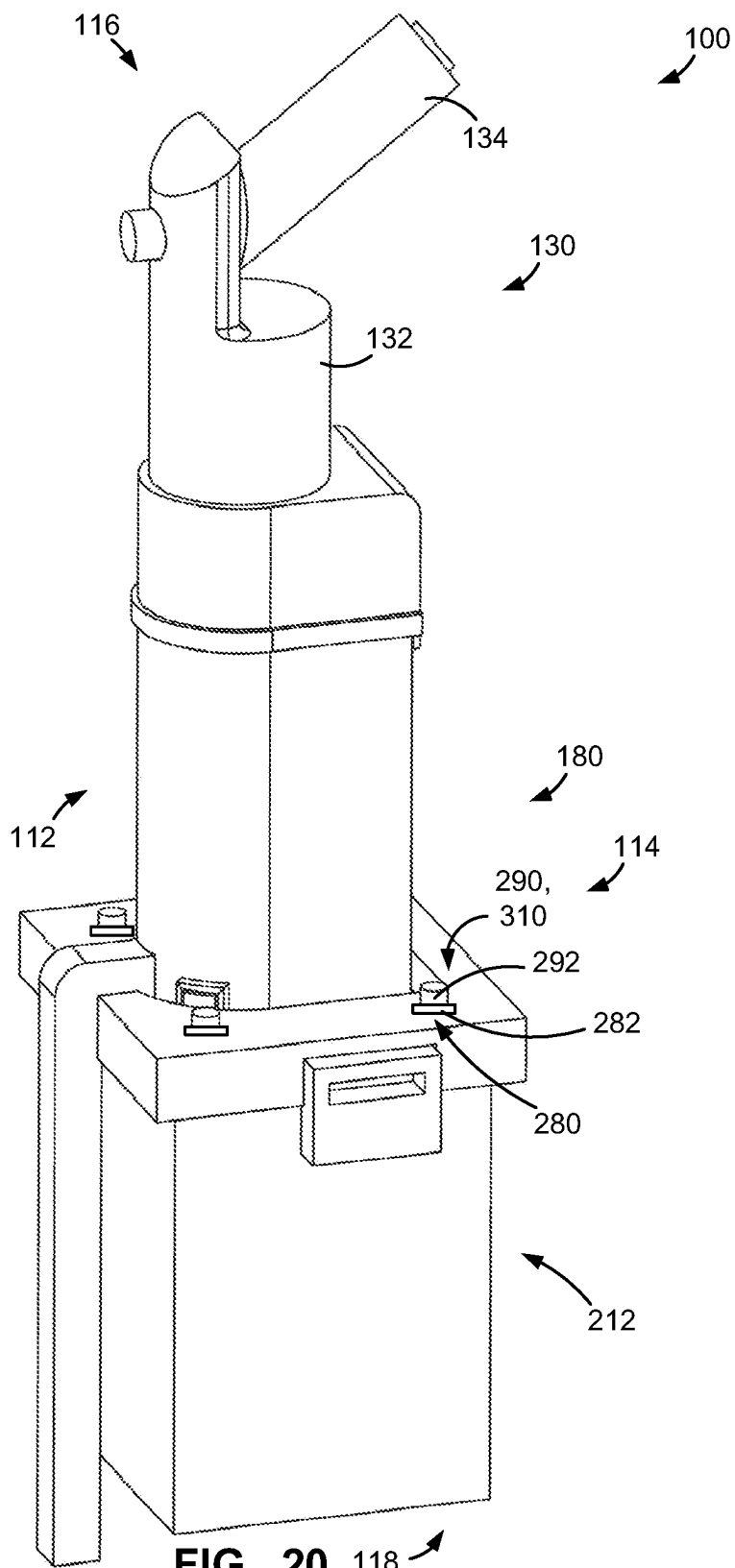
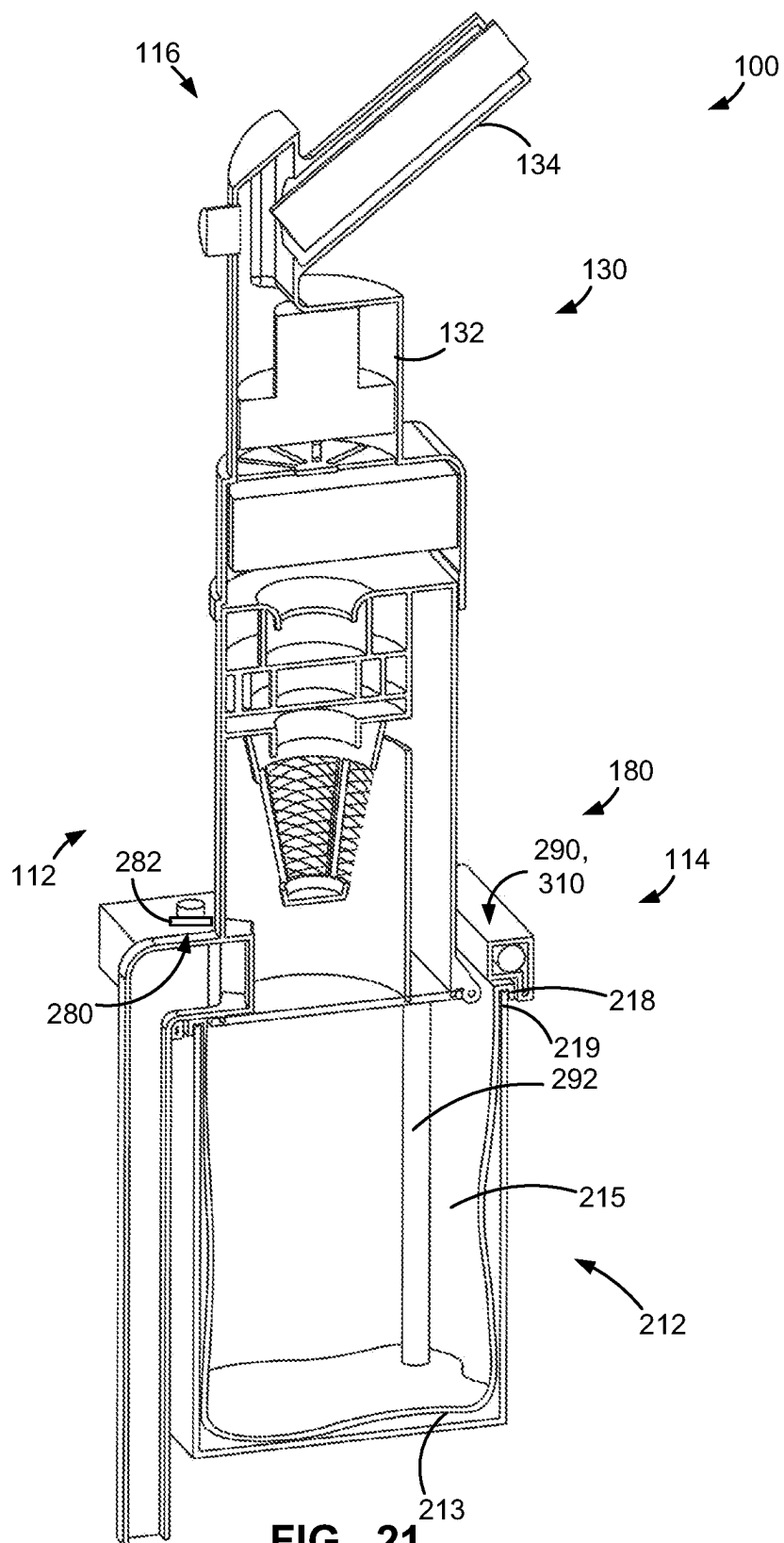
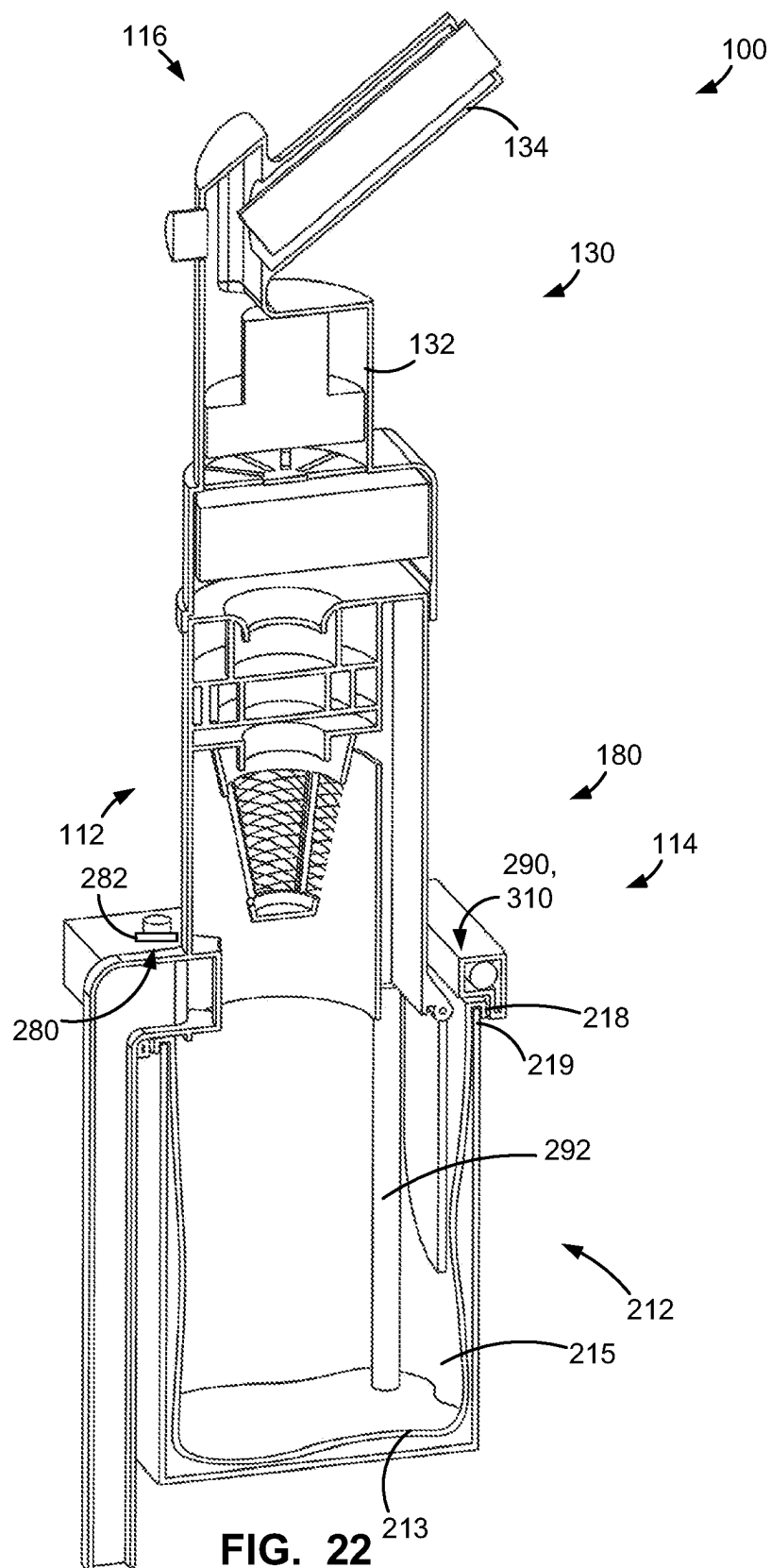


FIG. 19







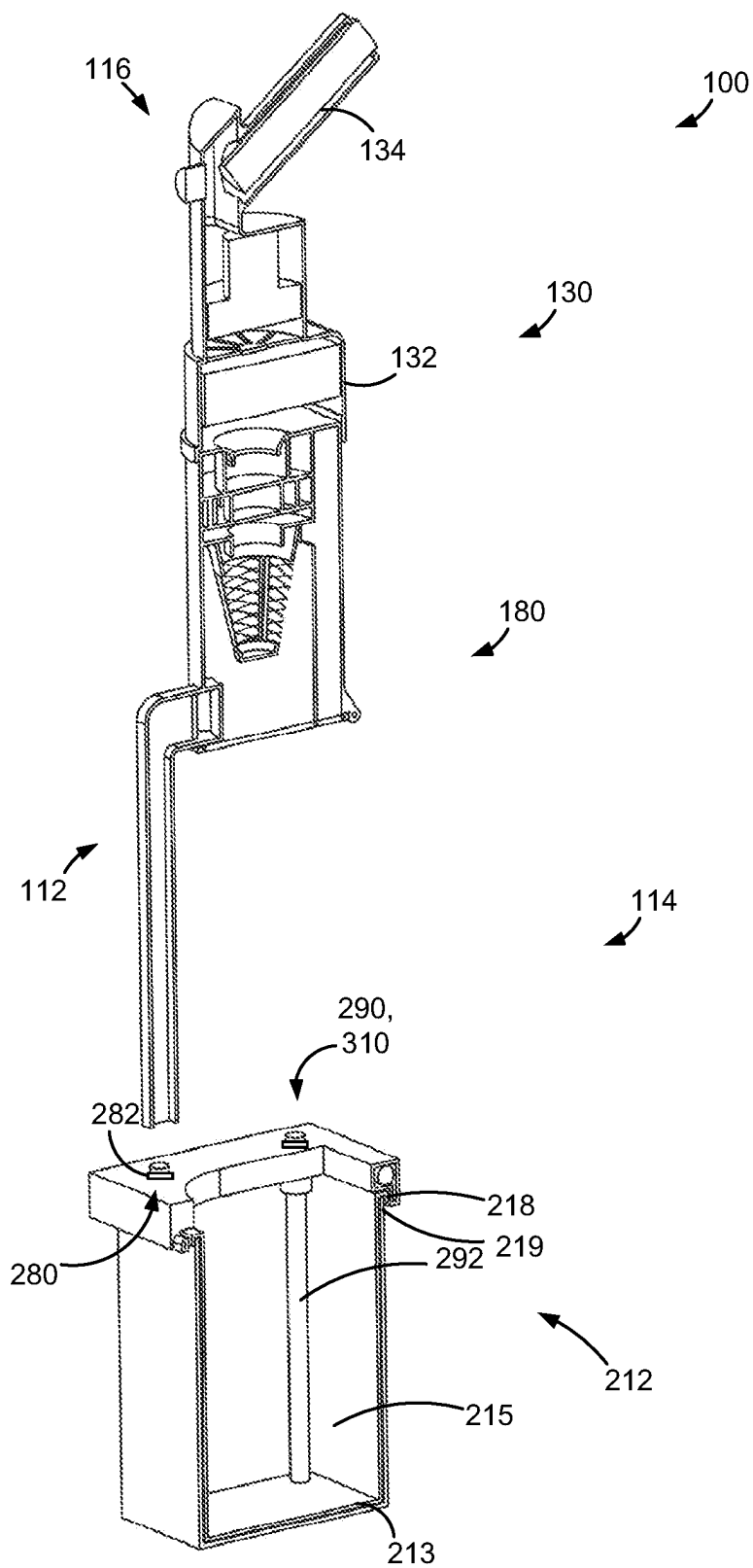
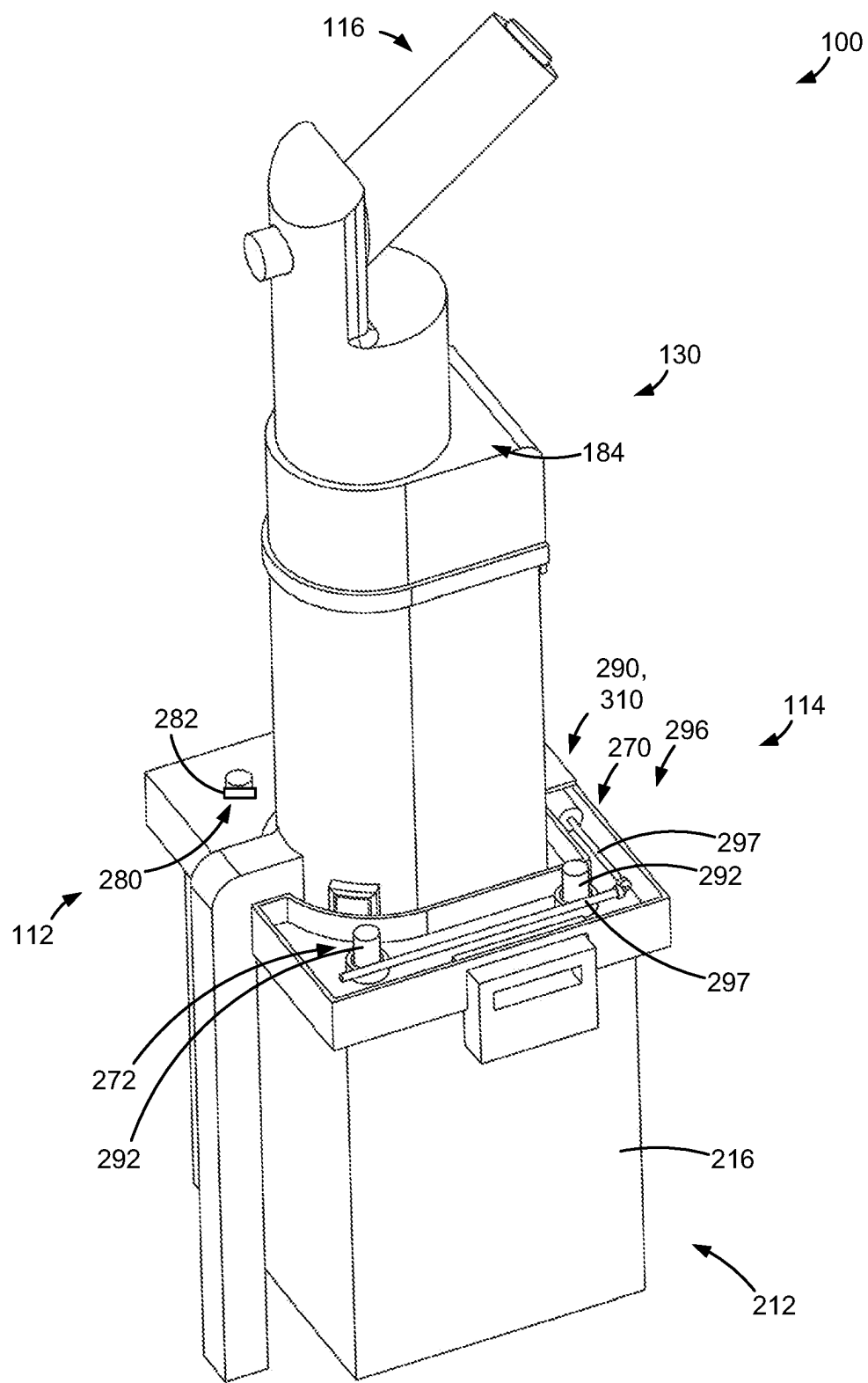


FIG. 23





118 **FIG. 24**

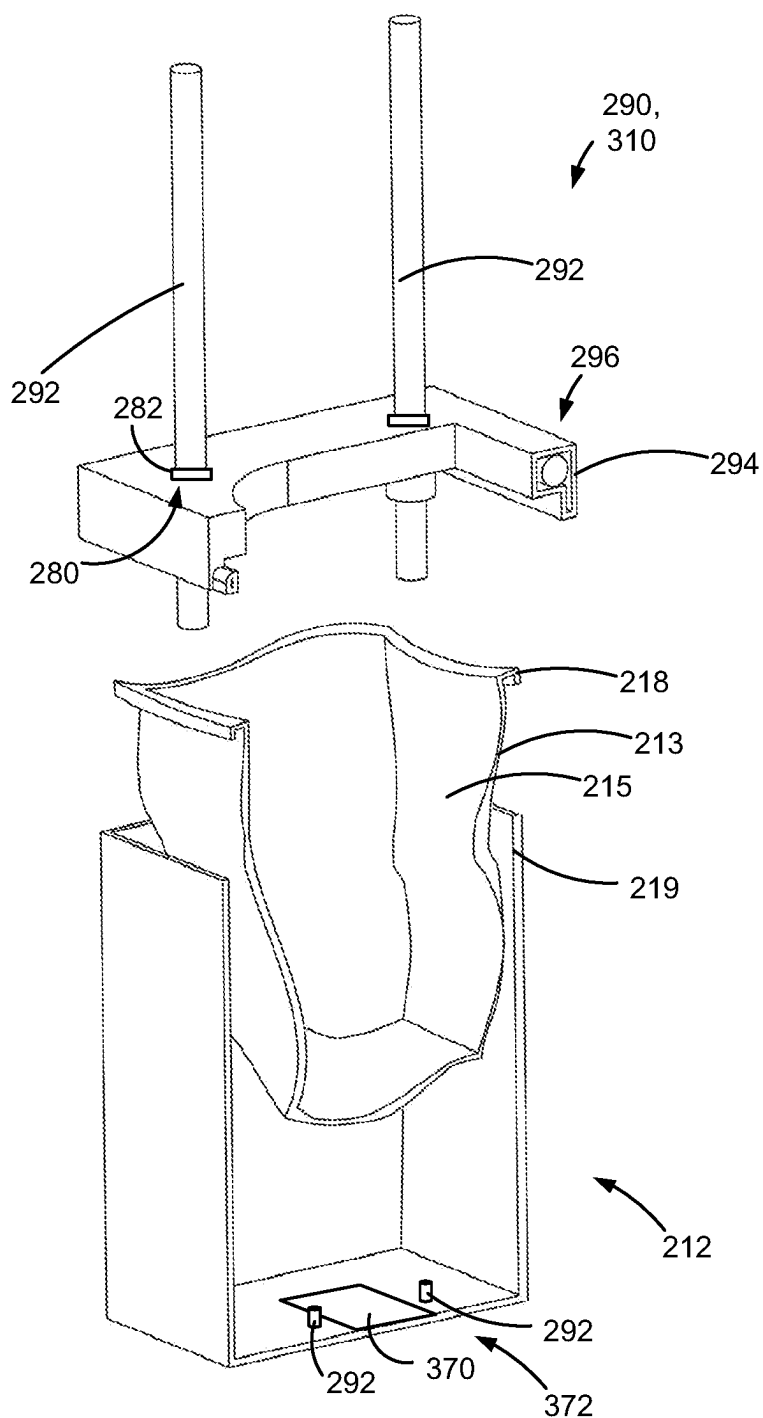


FIG. 25

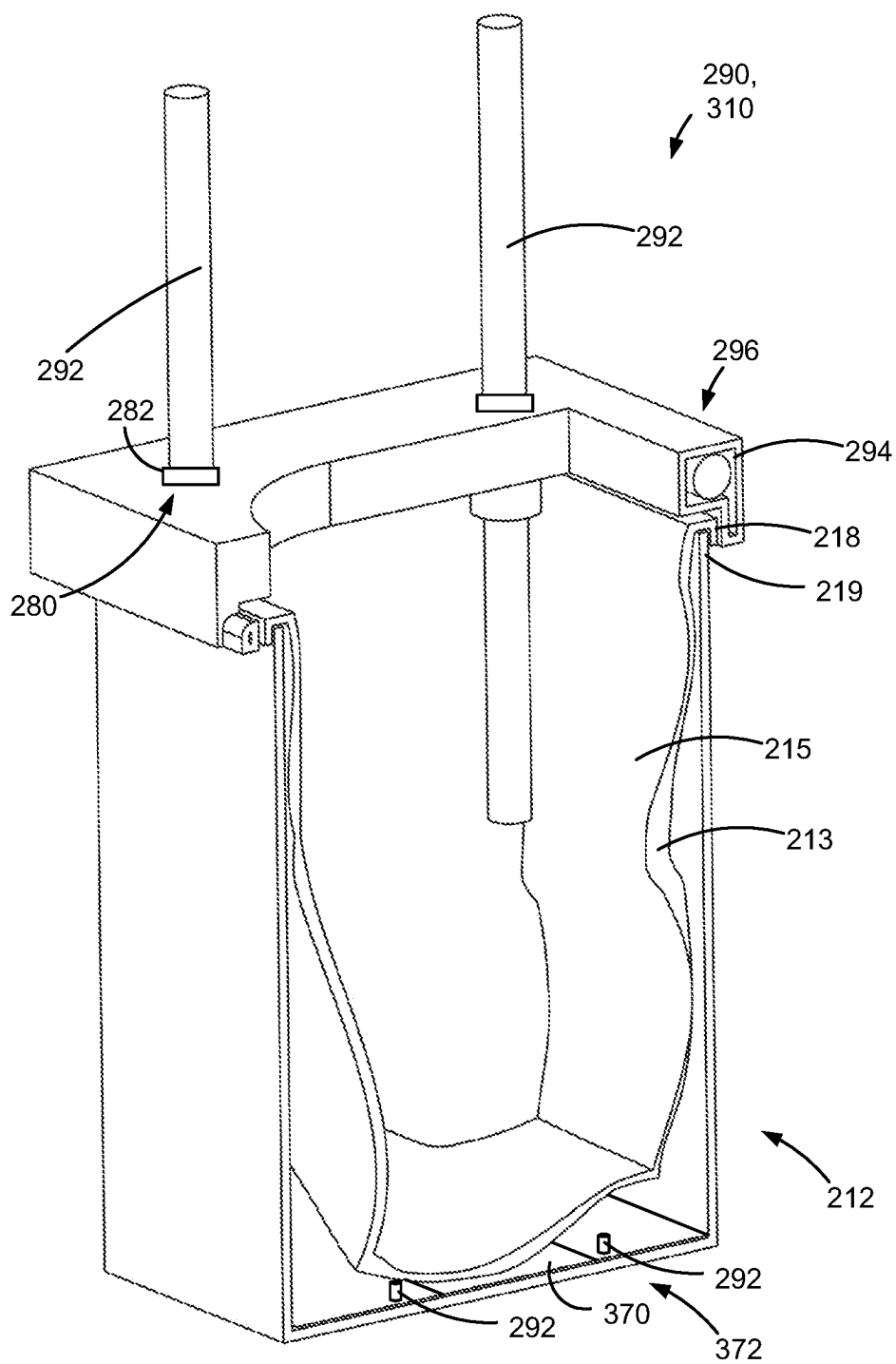


FIG. 26

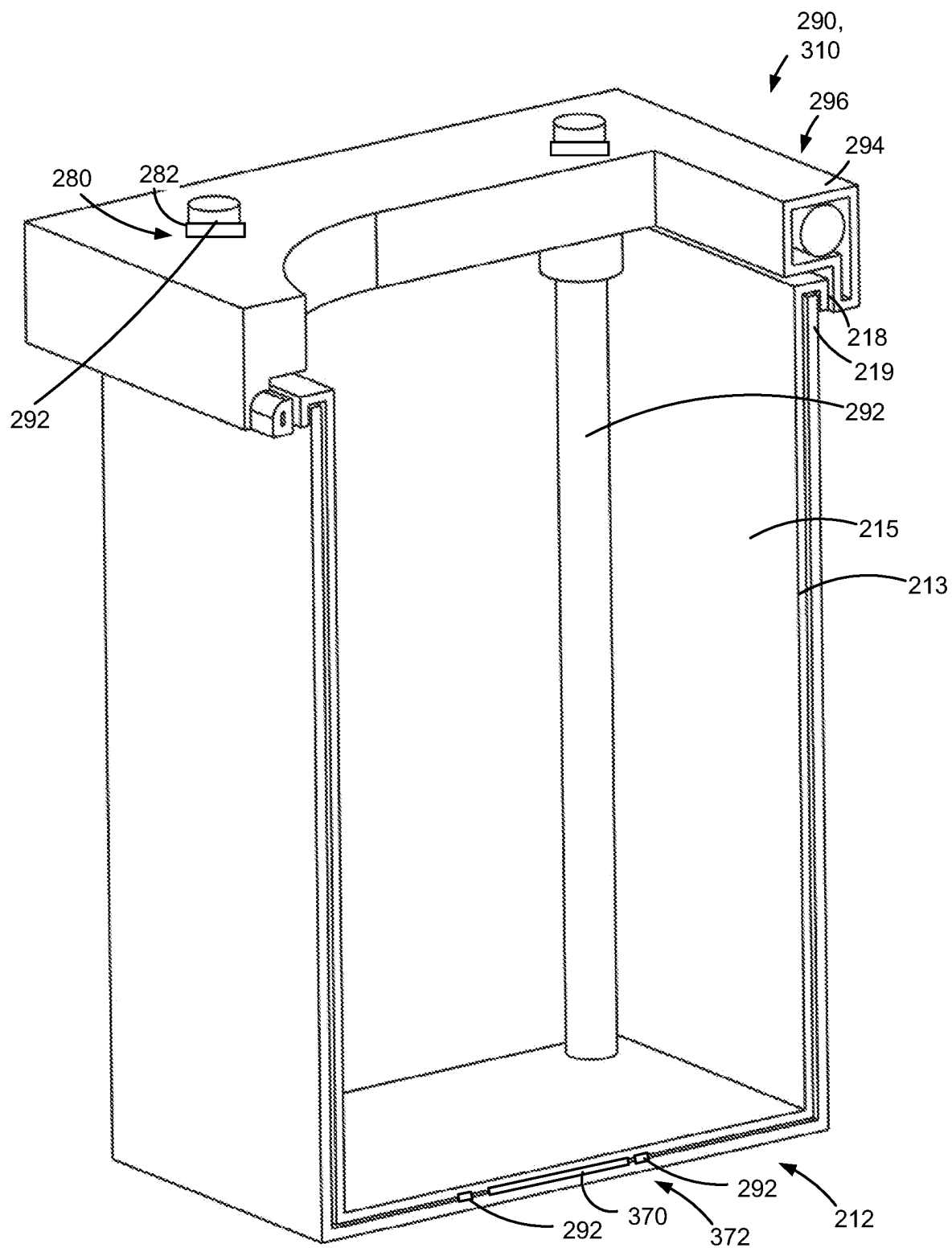


FIG. 27

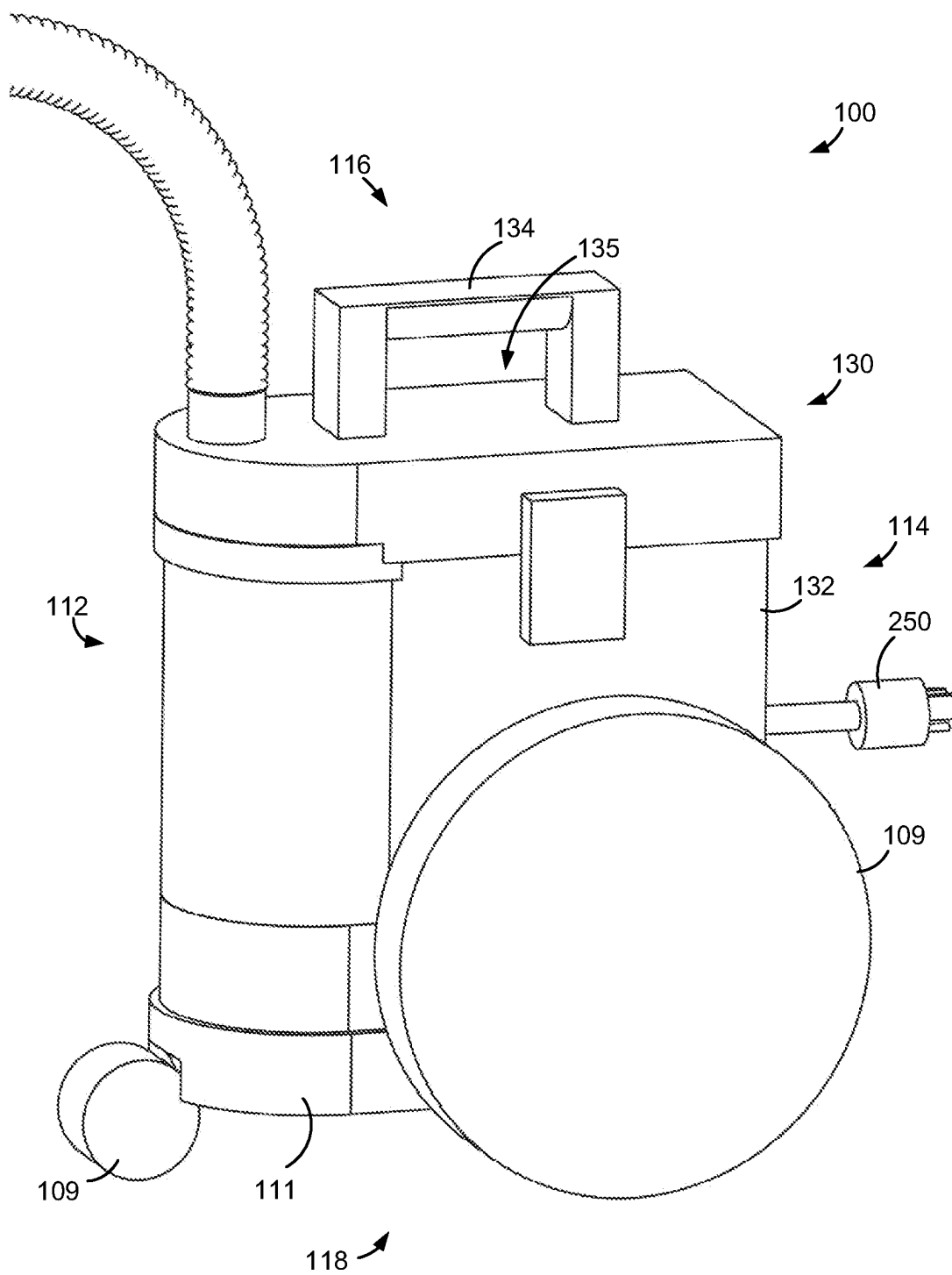


FIG. 28

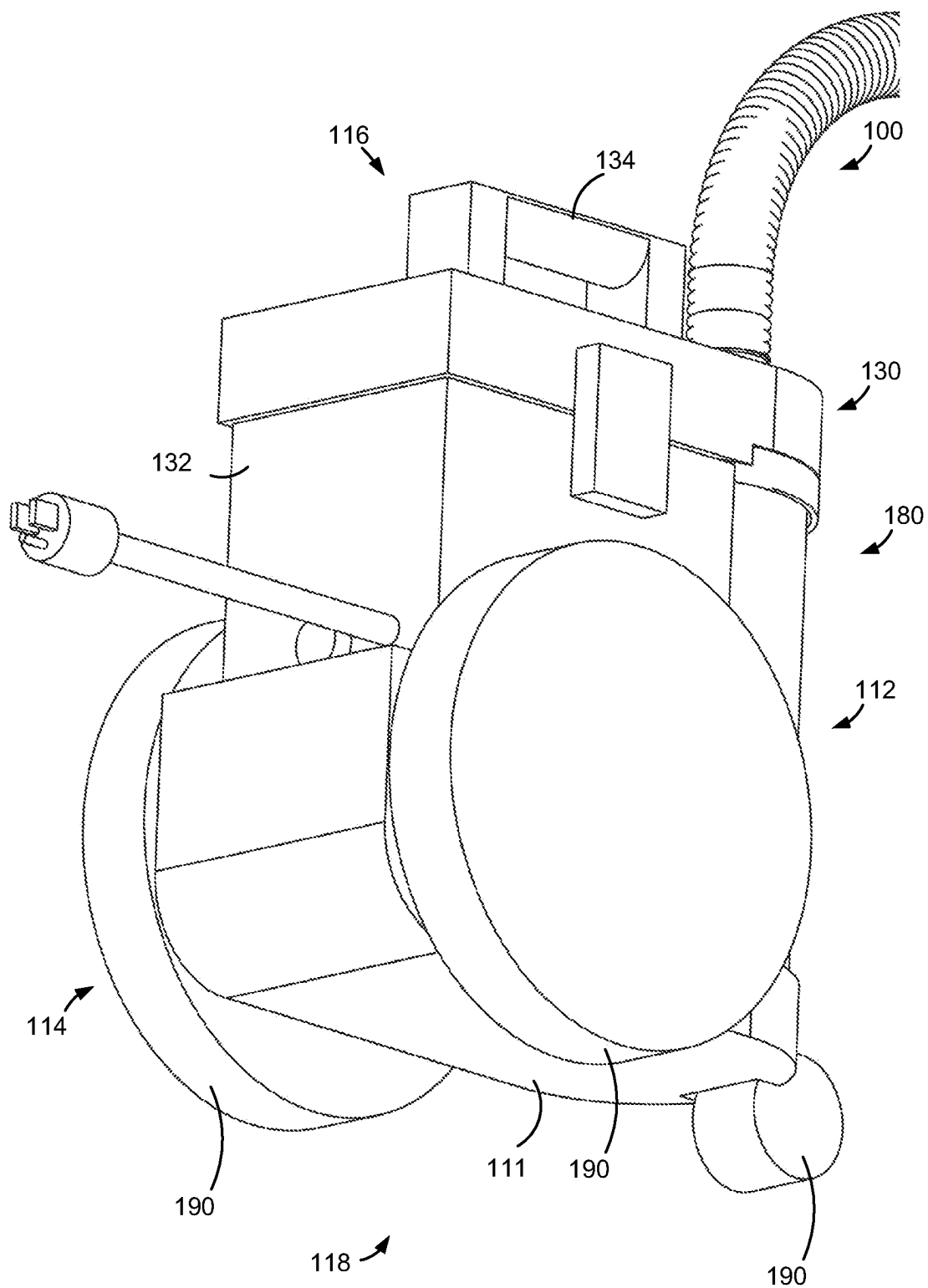


FIG. 29

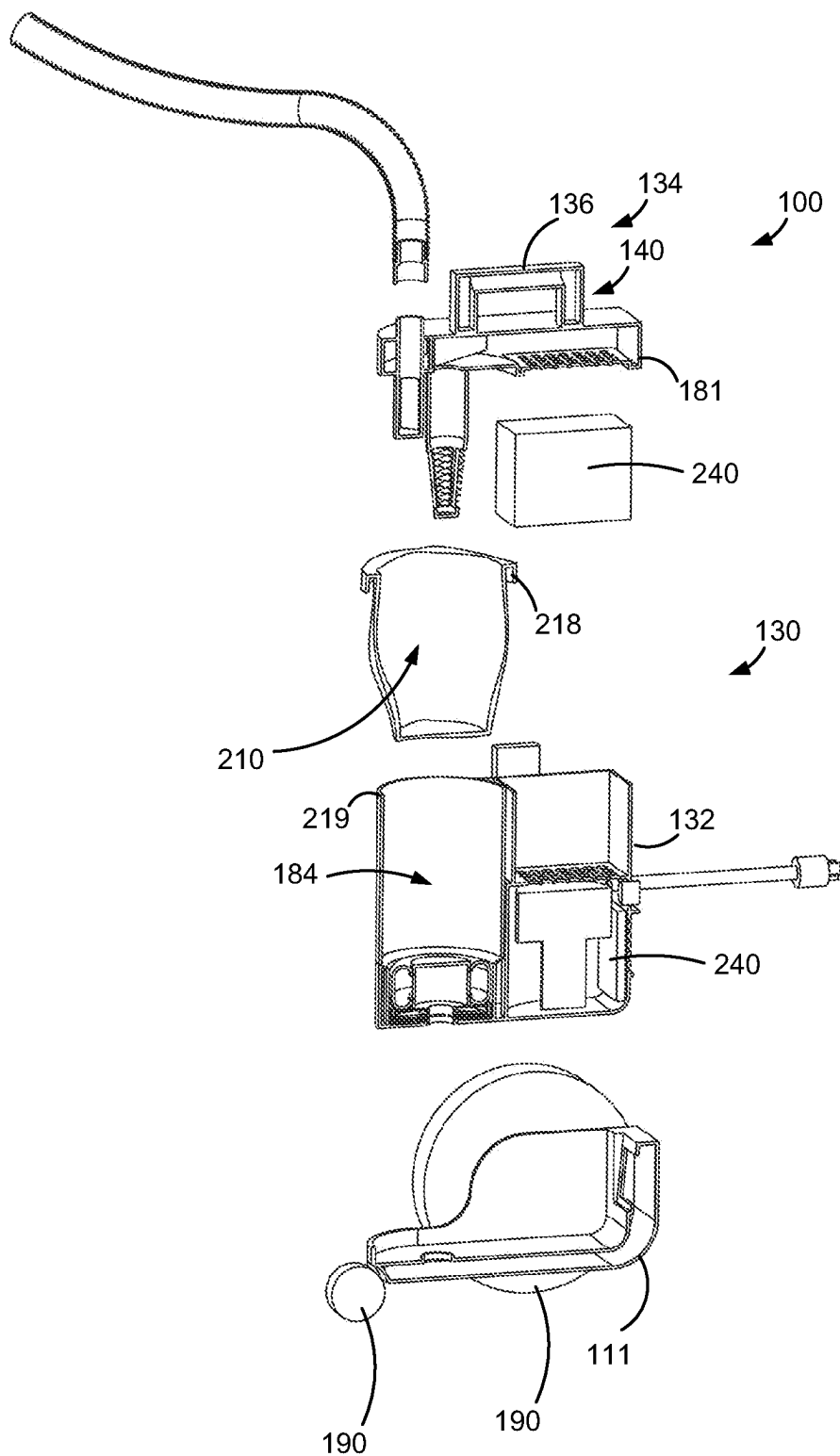
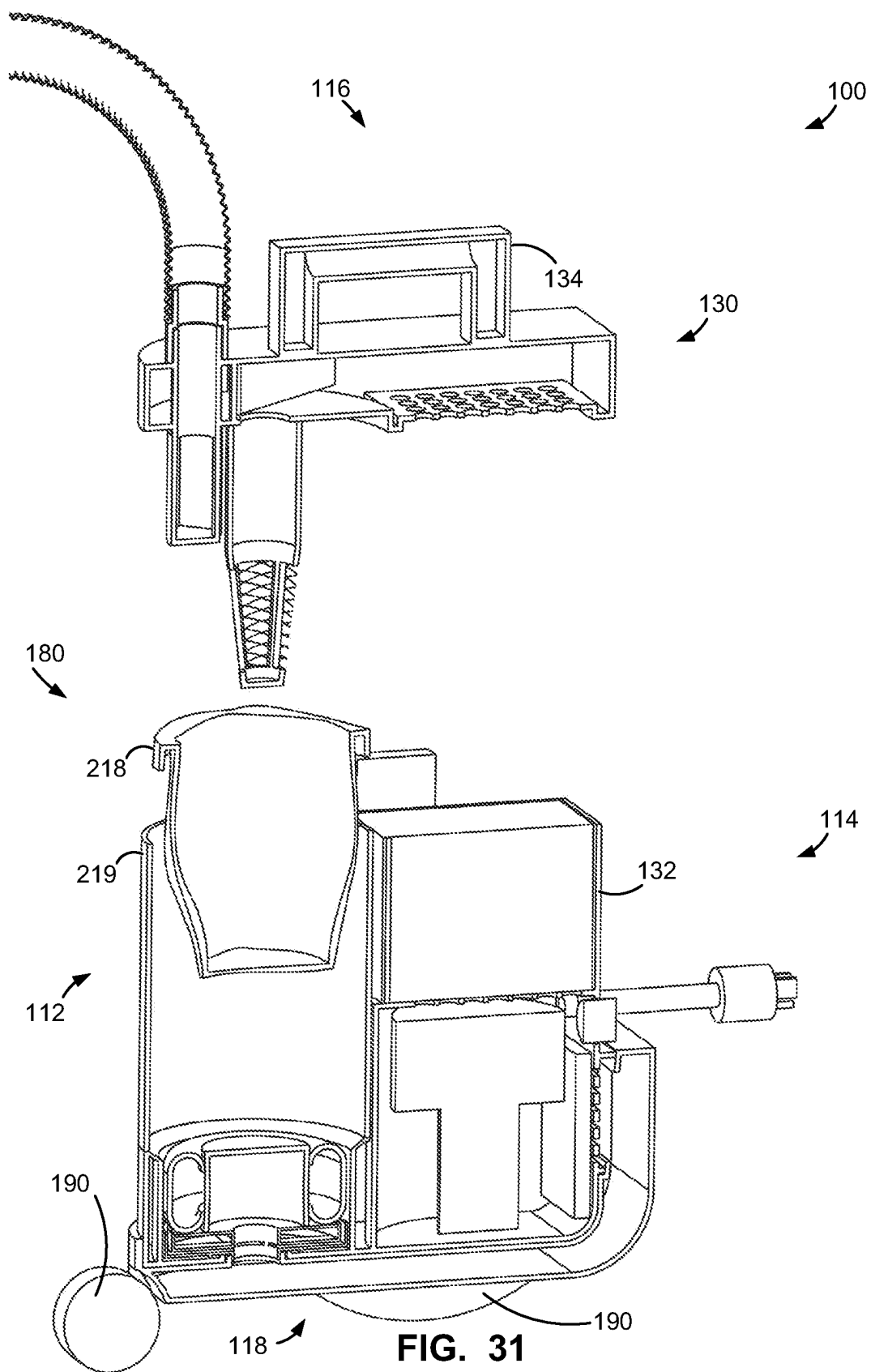


FIG. 30





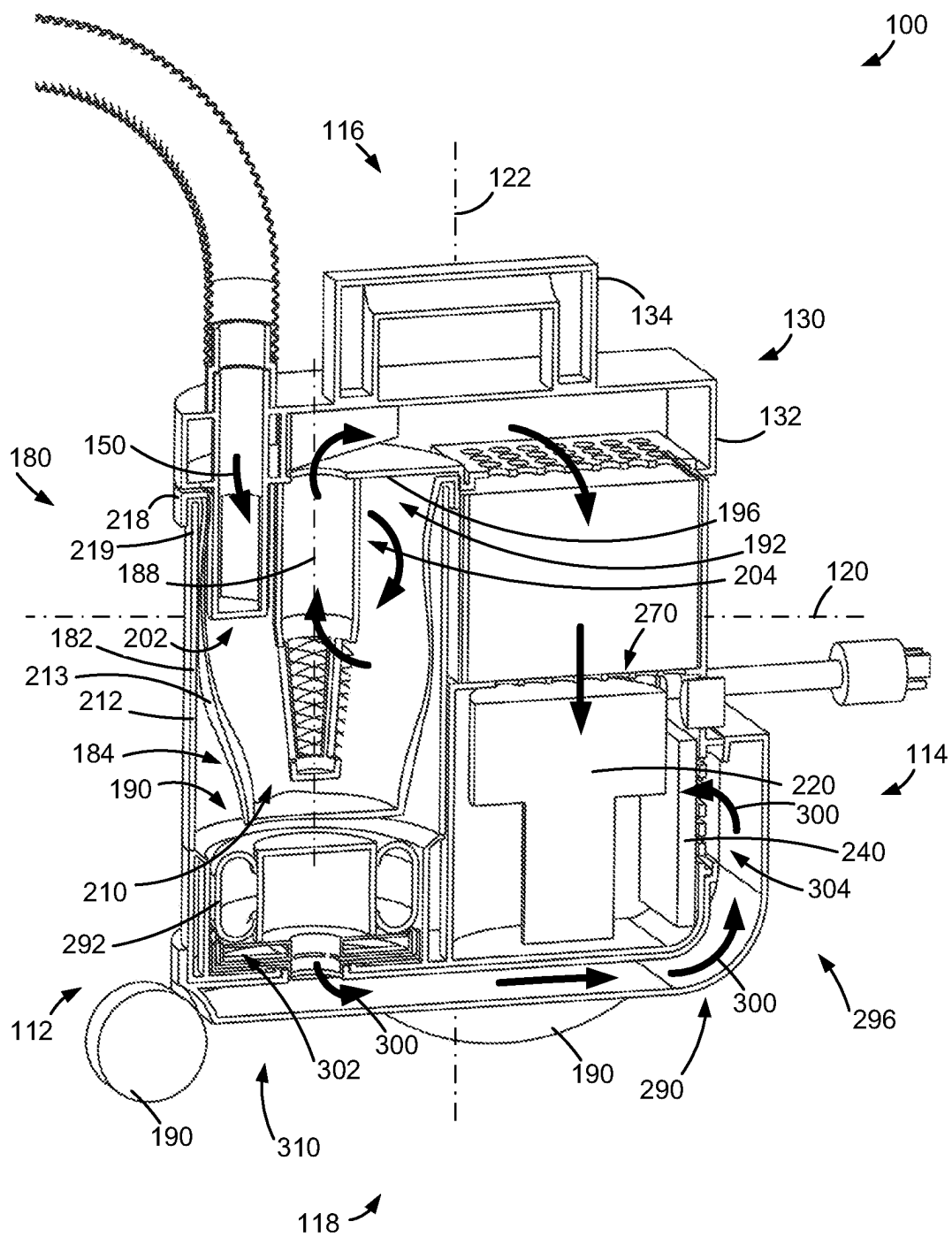


FIG. 32

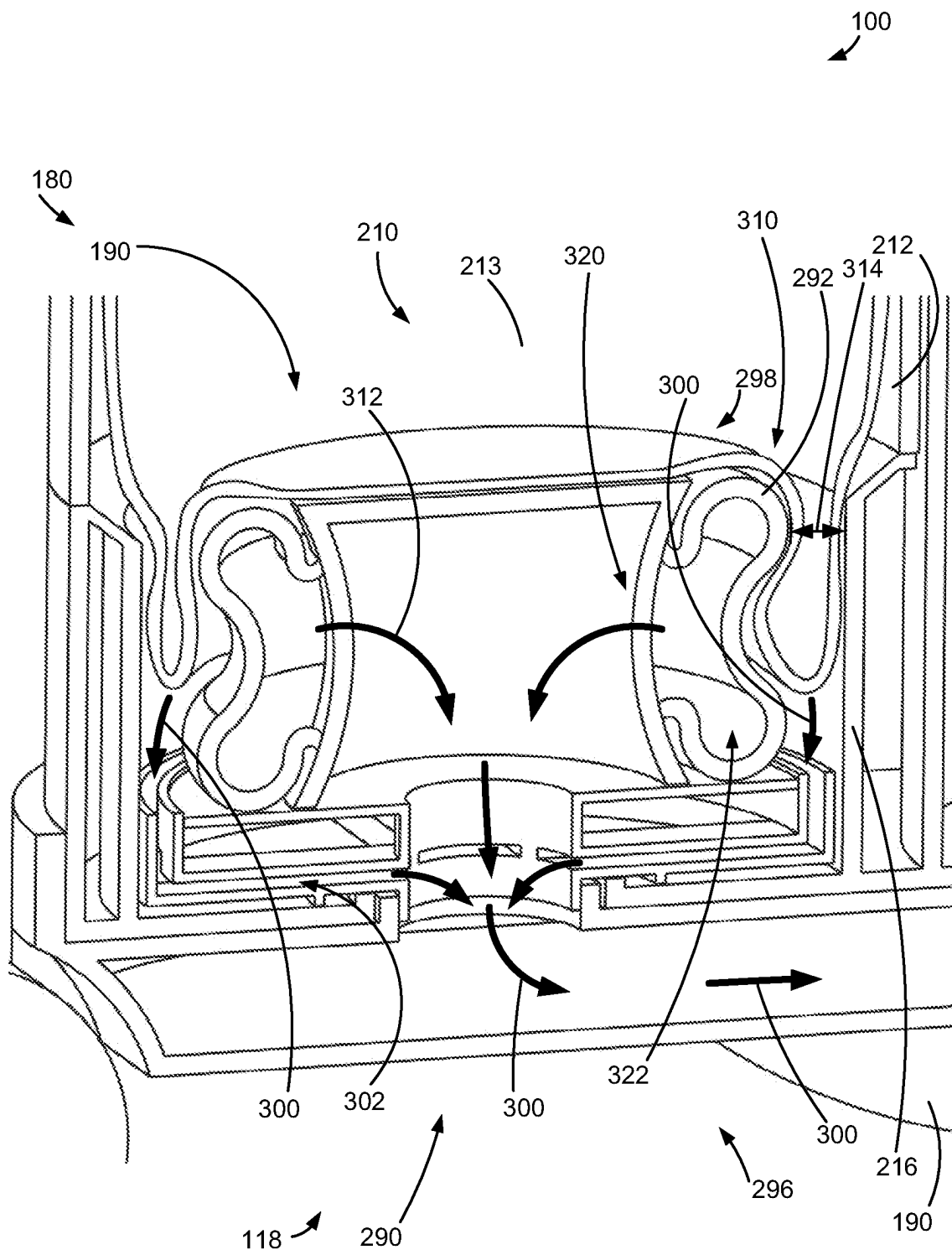


FIG. 33

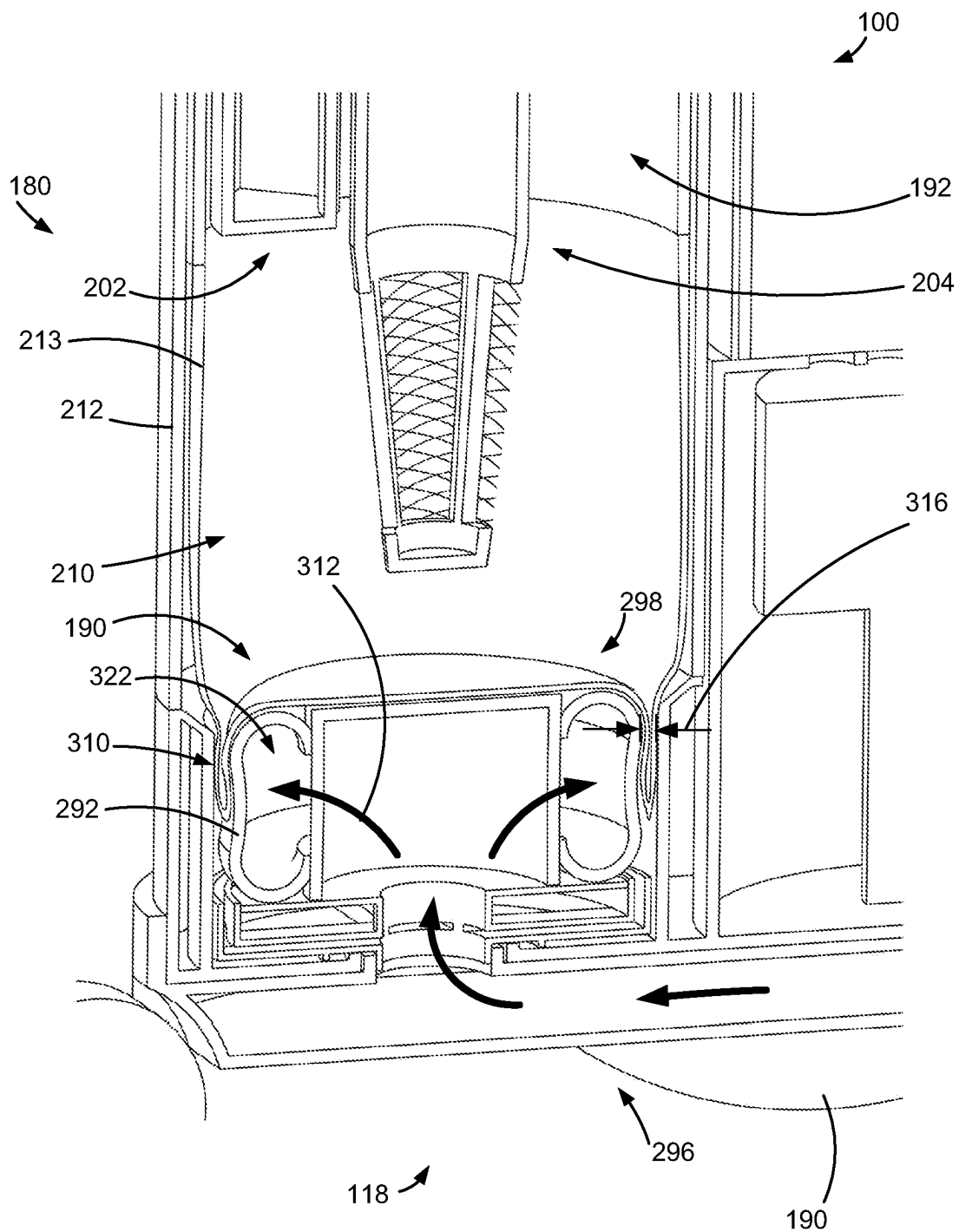


FIG. 34

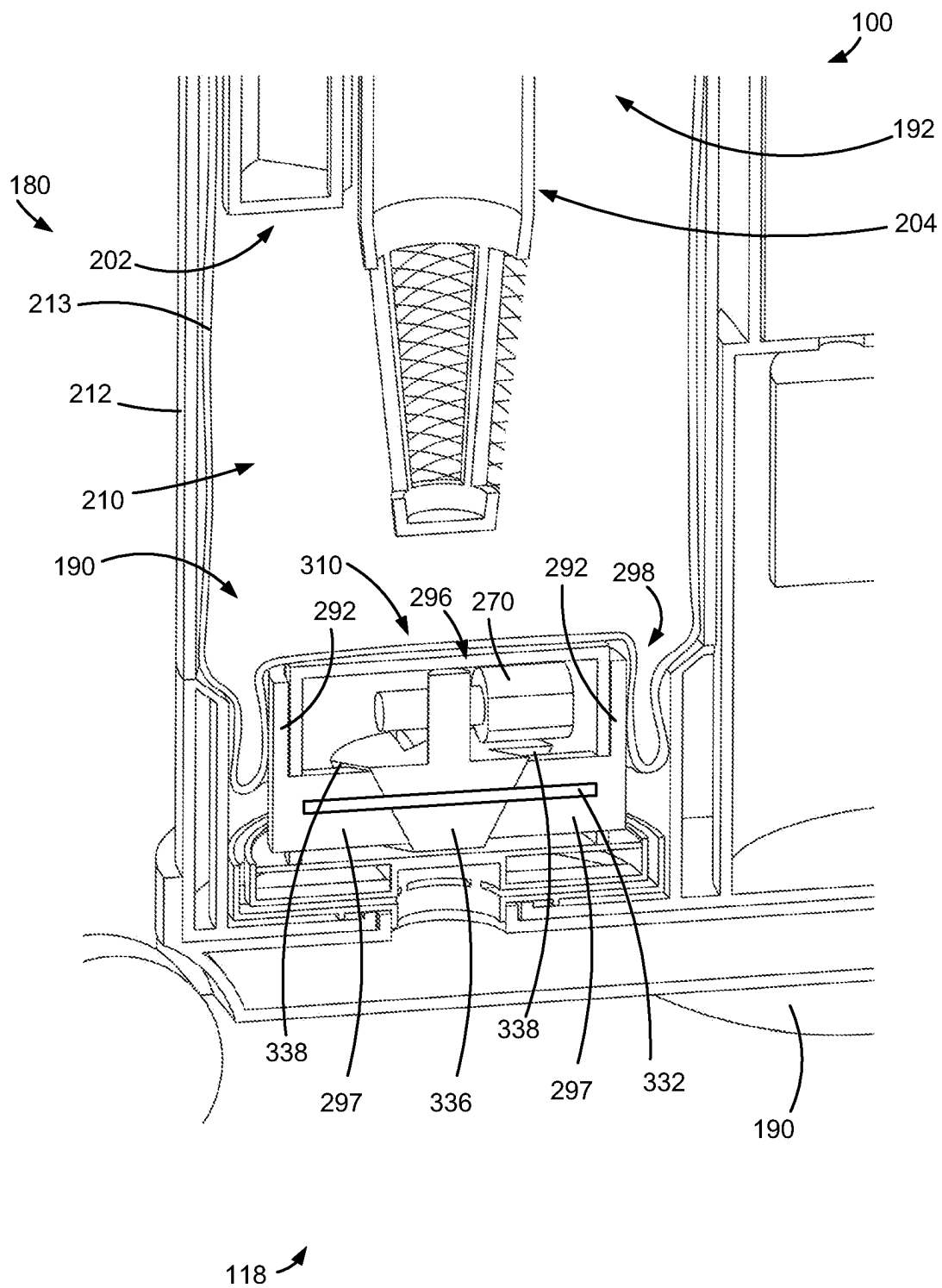
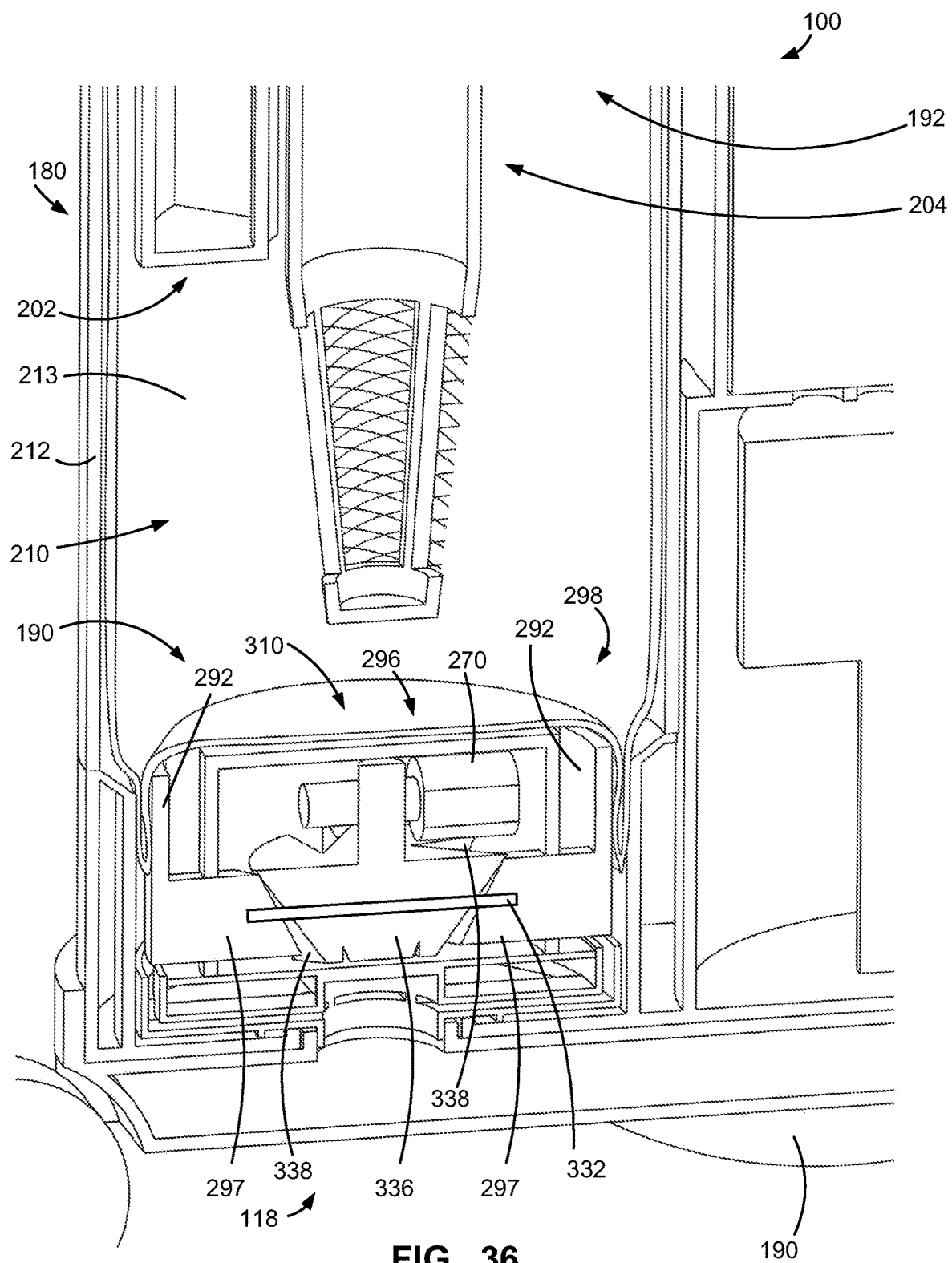


FIG. 35



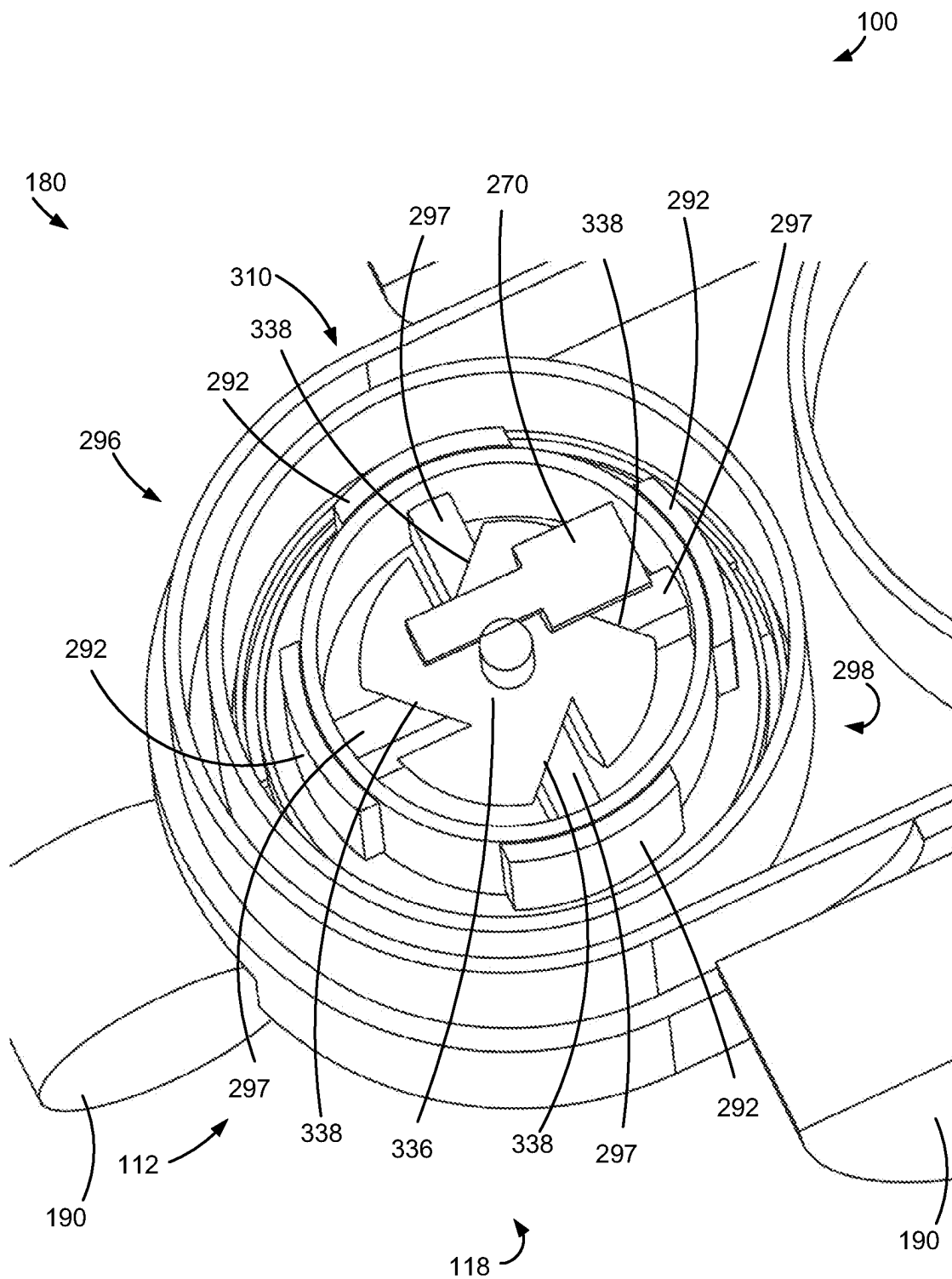
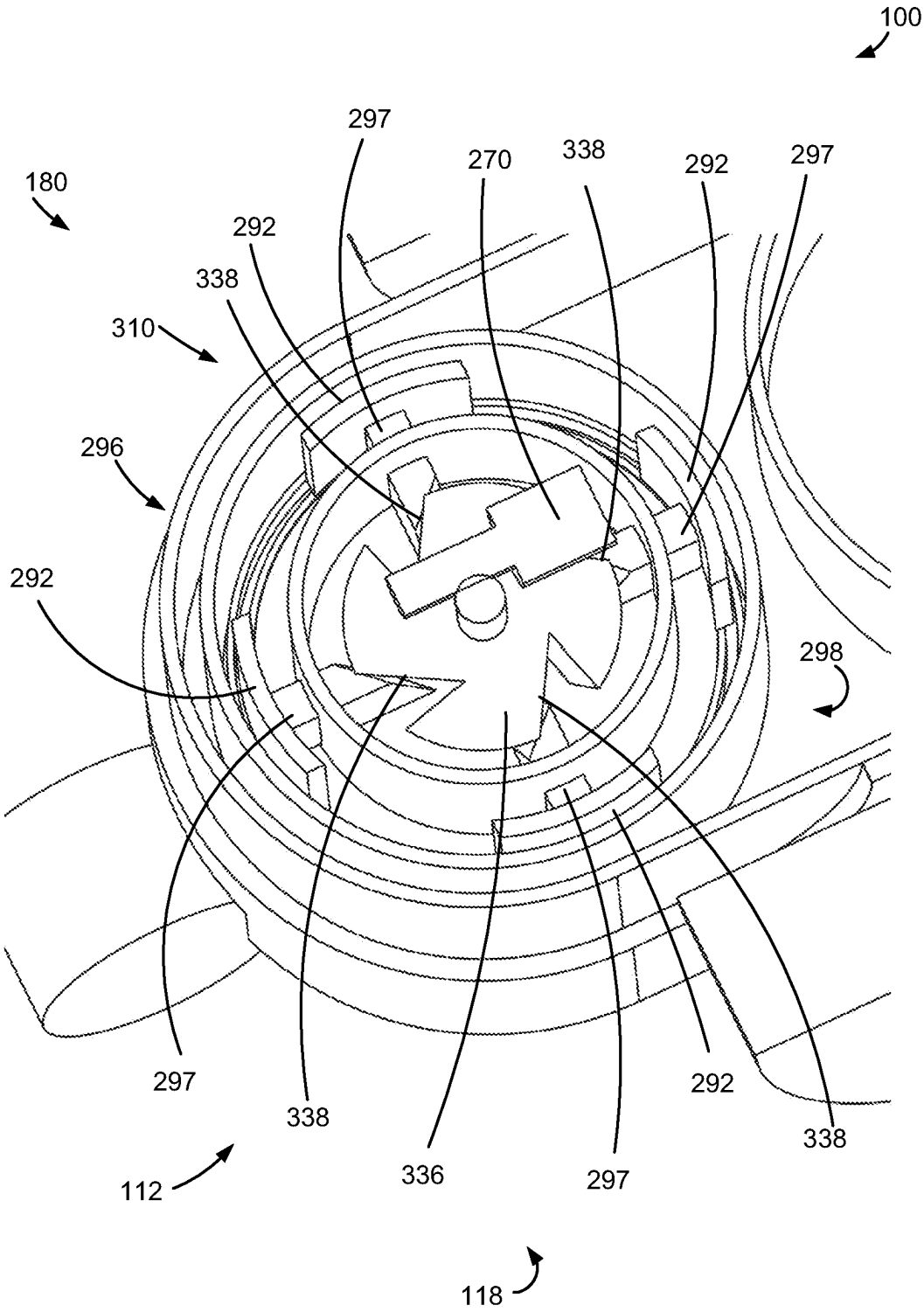


FIG. 37



**FIG. 38**

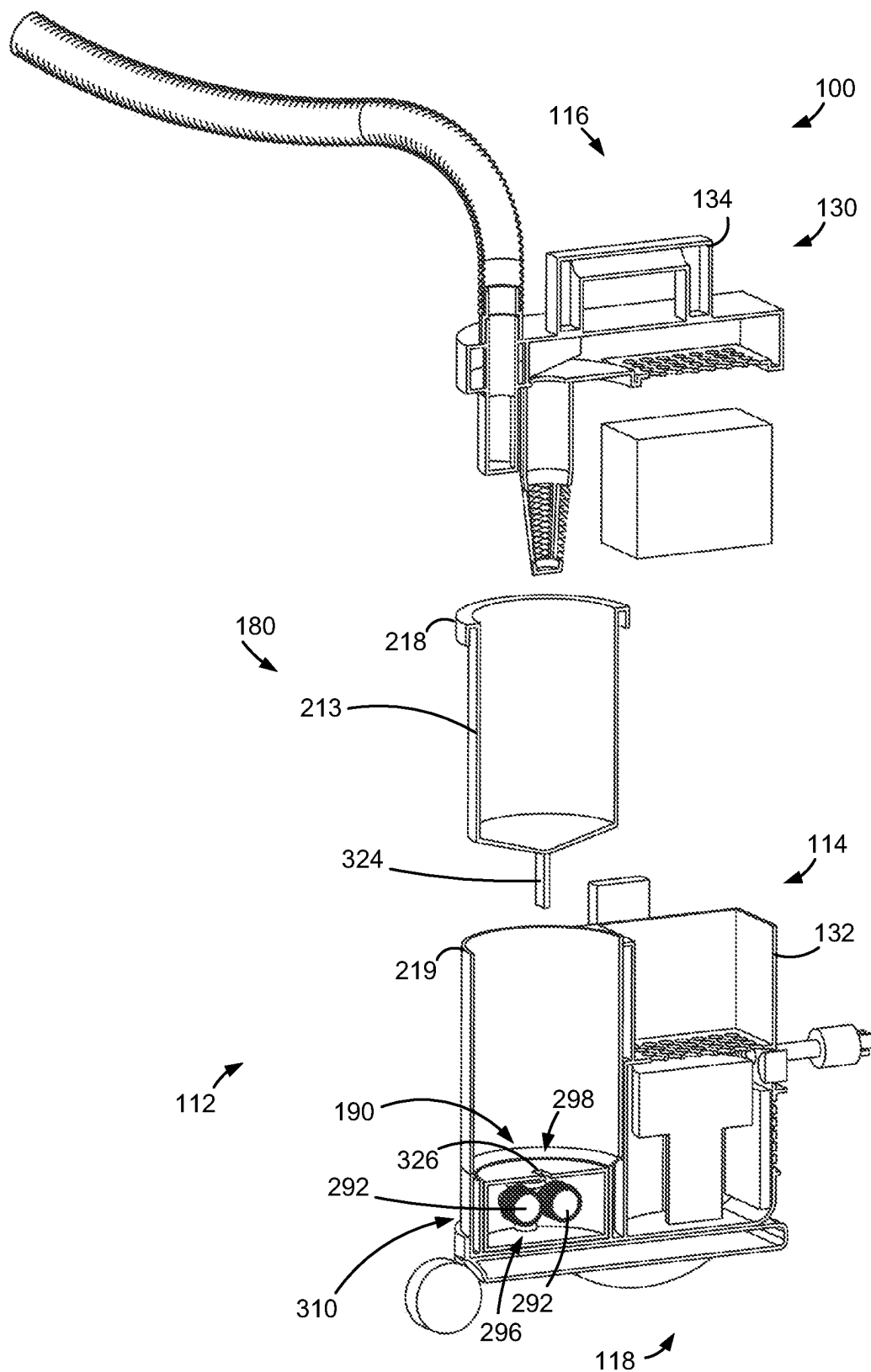
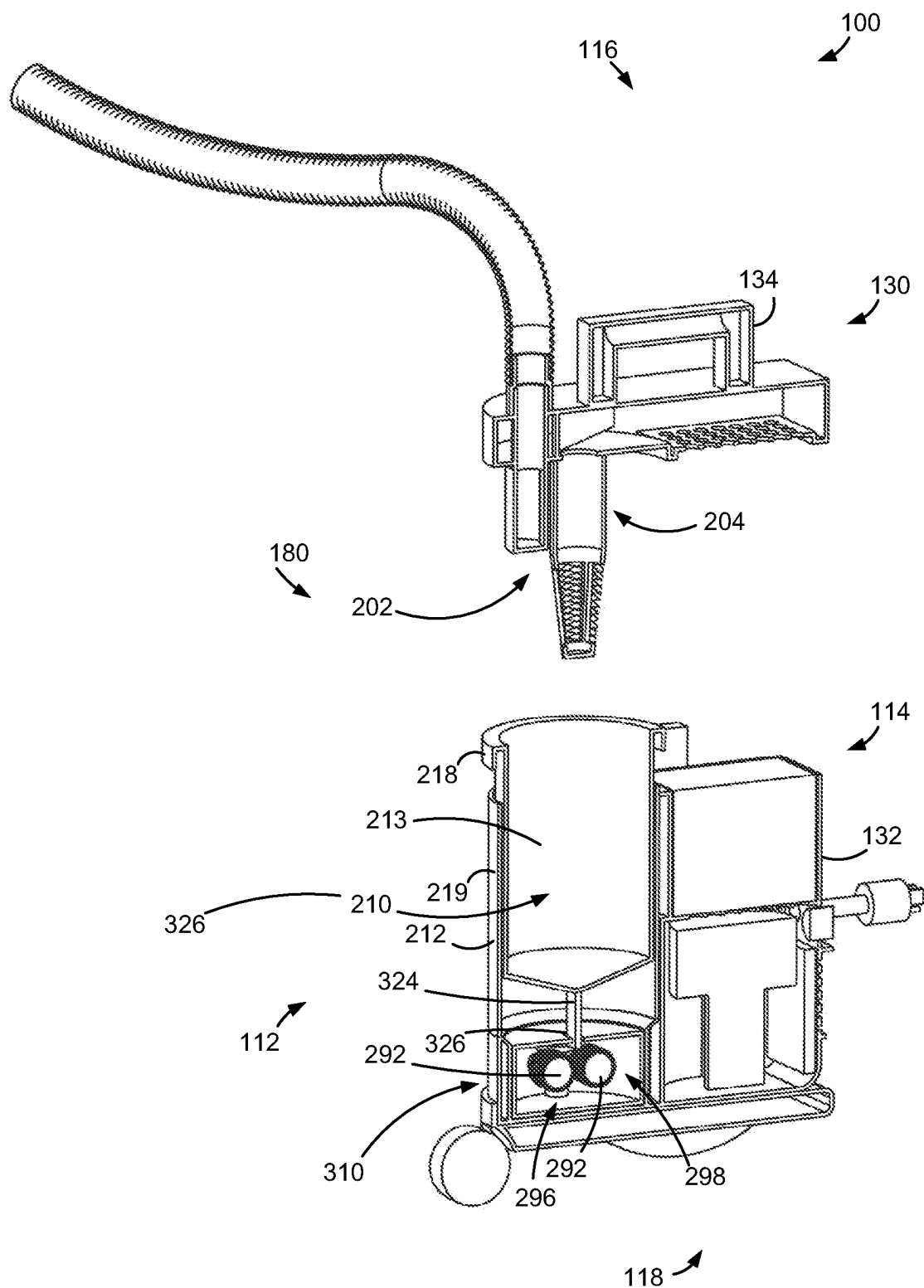
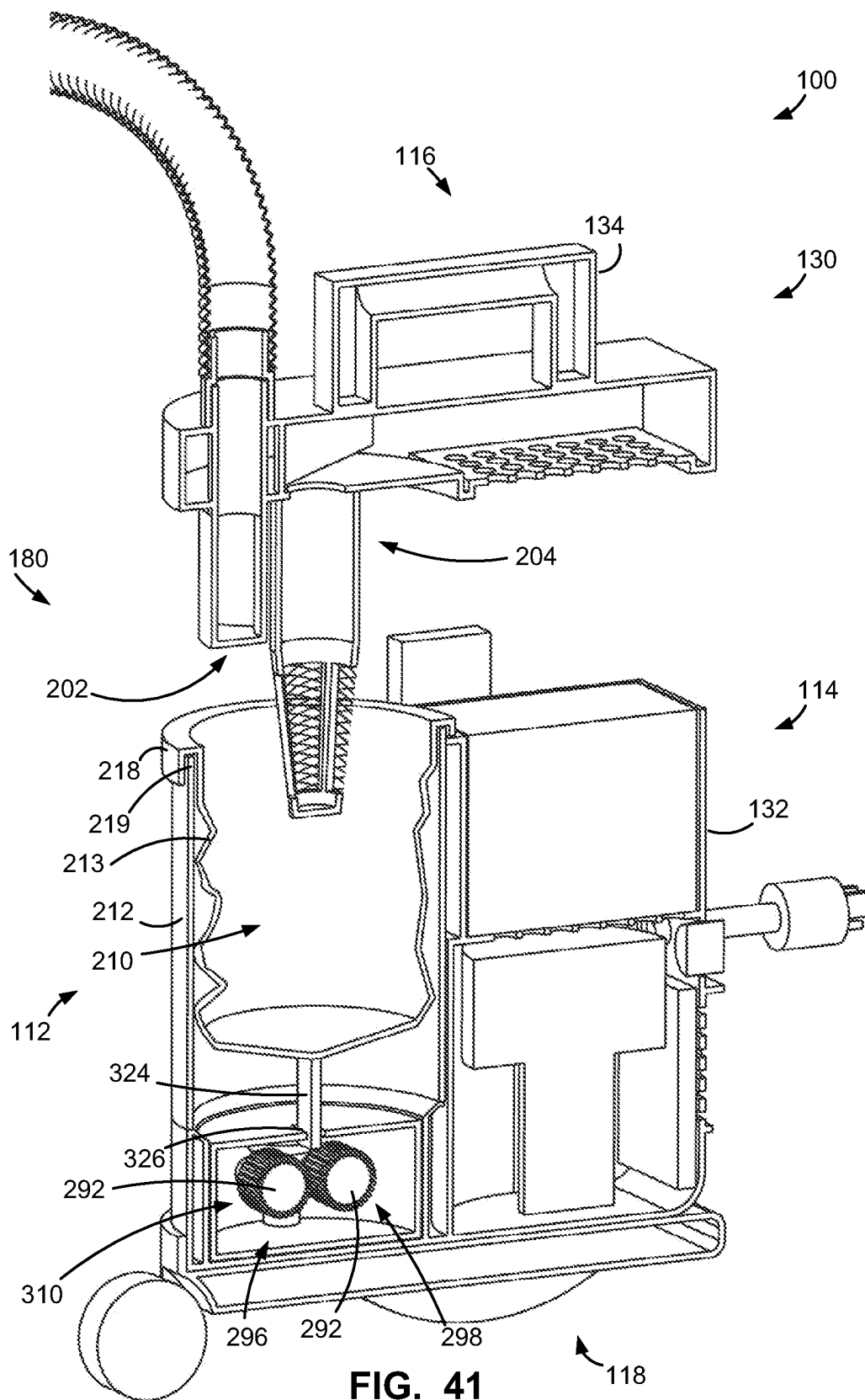


FIG. 39







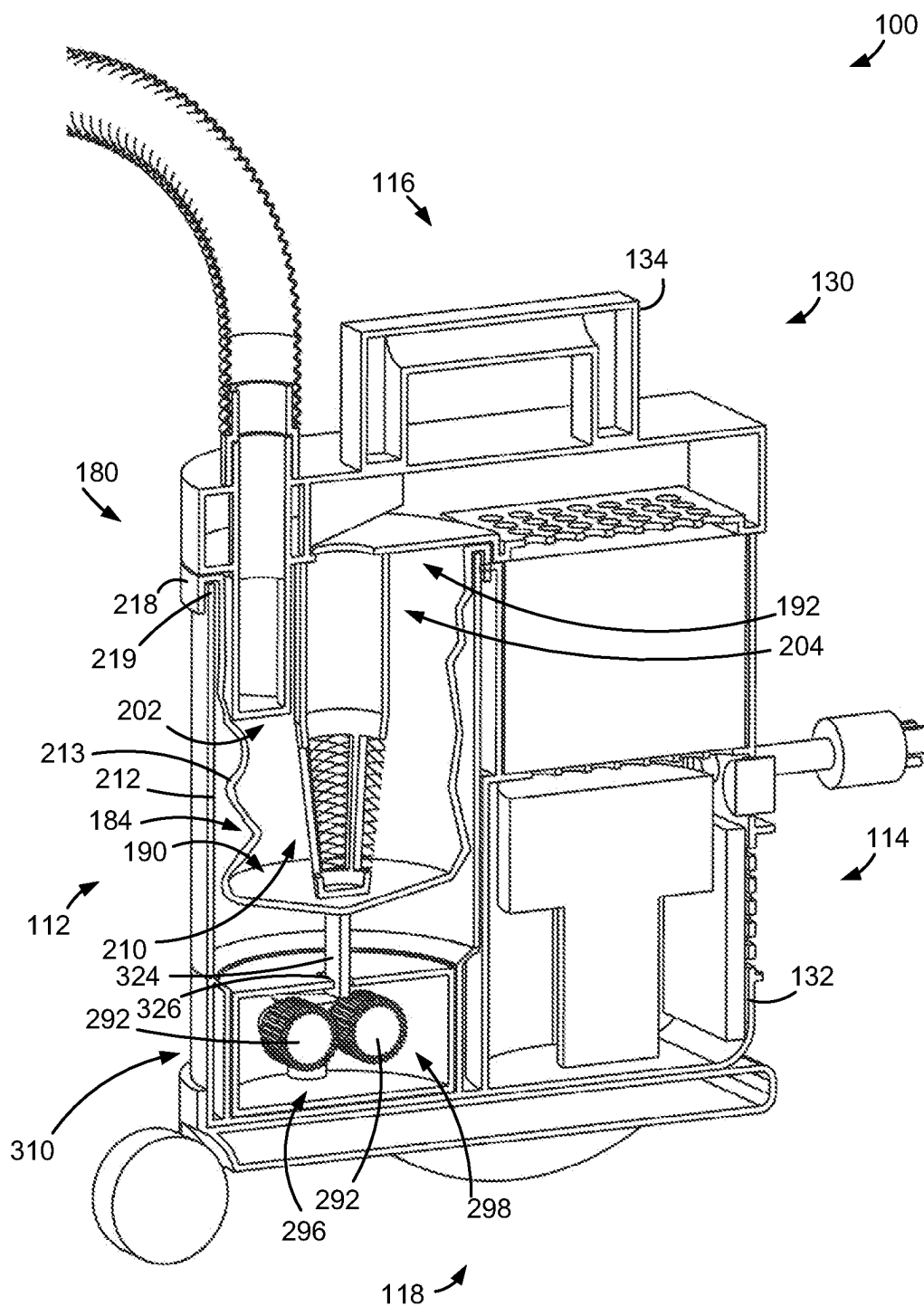


FIG. 42

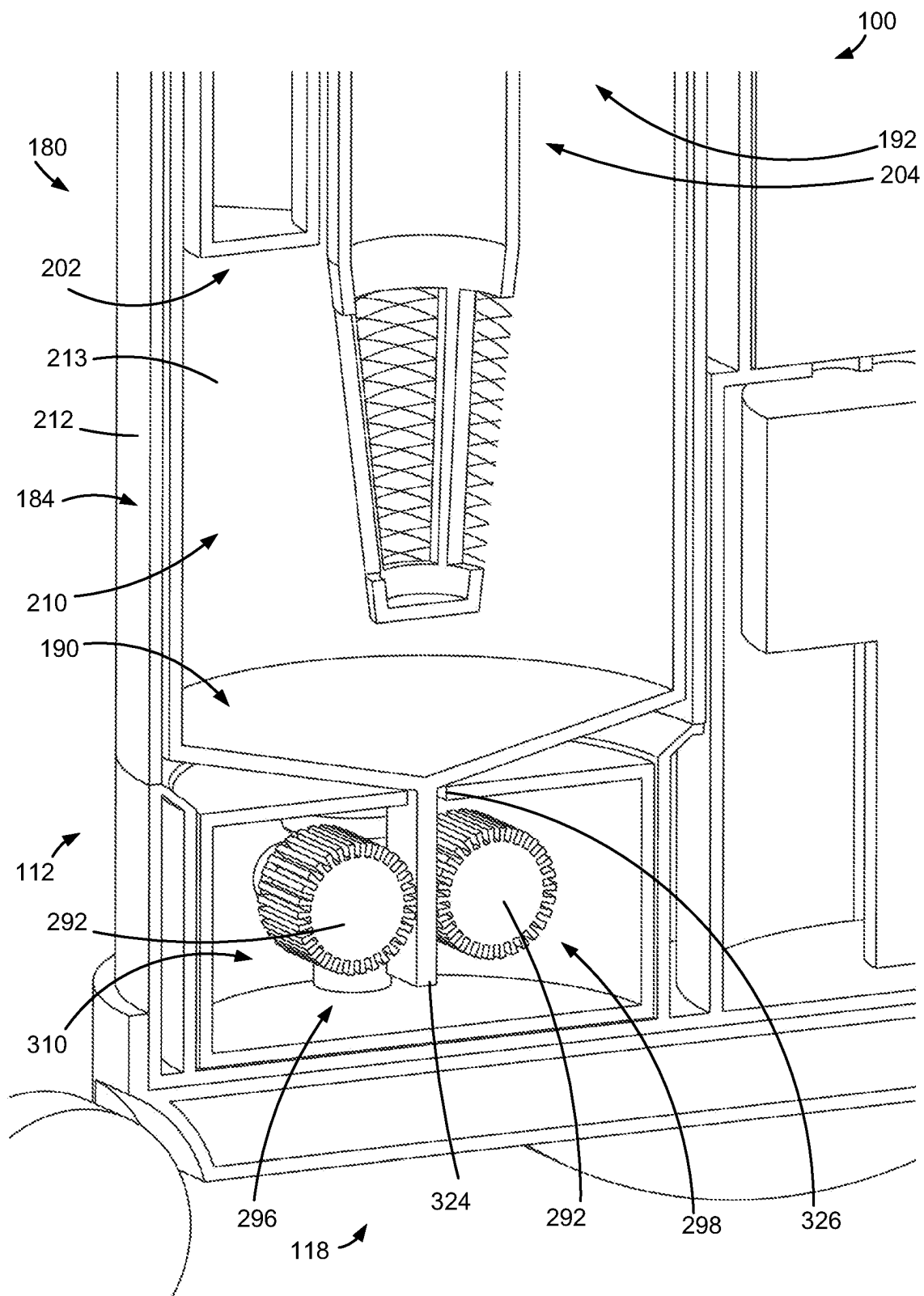
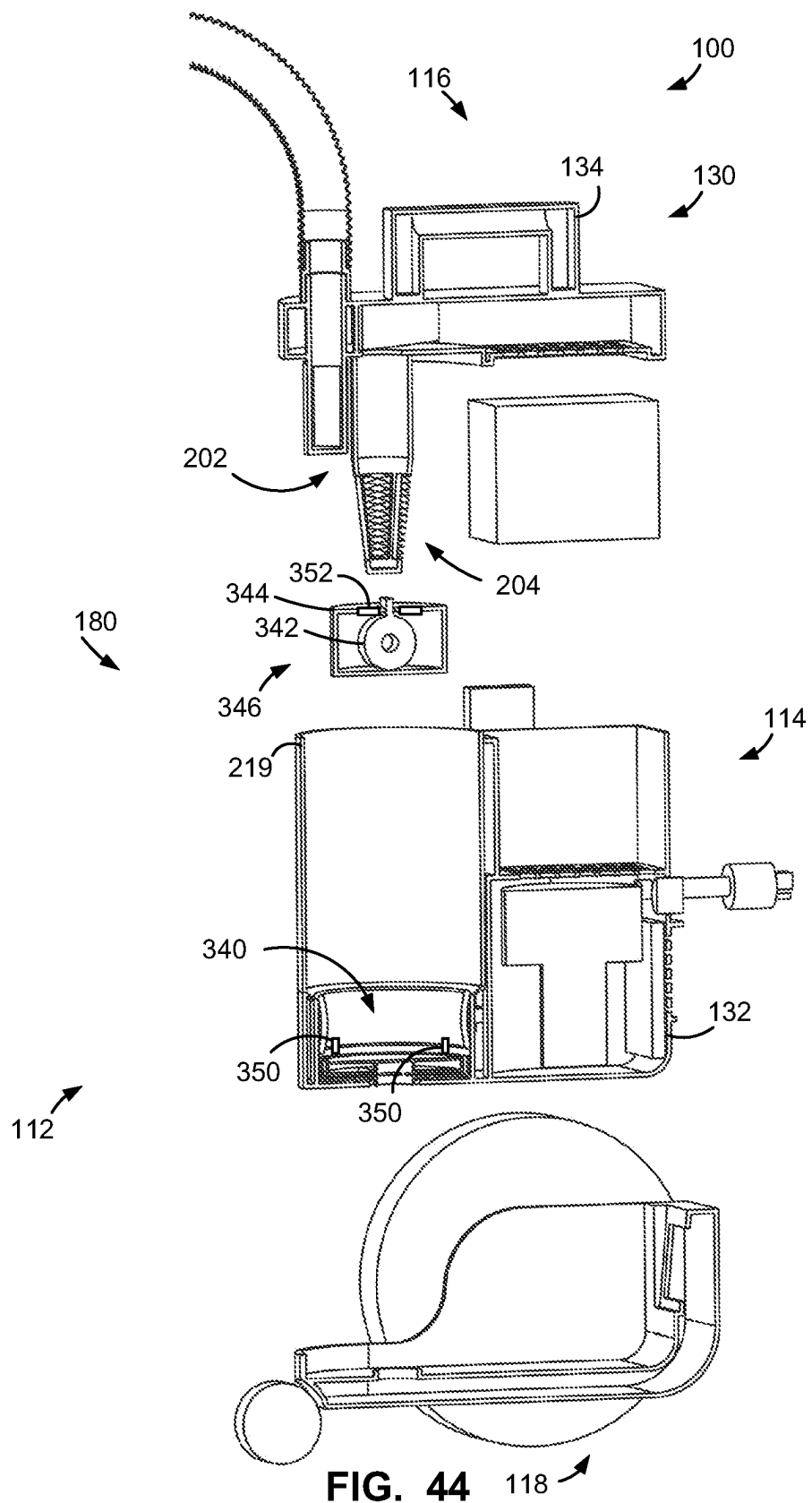


FIG. 43



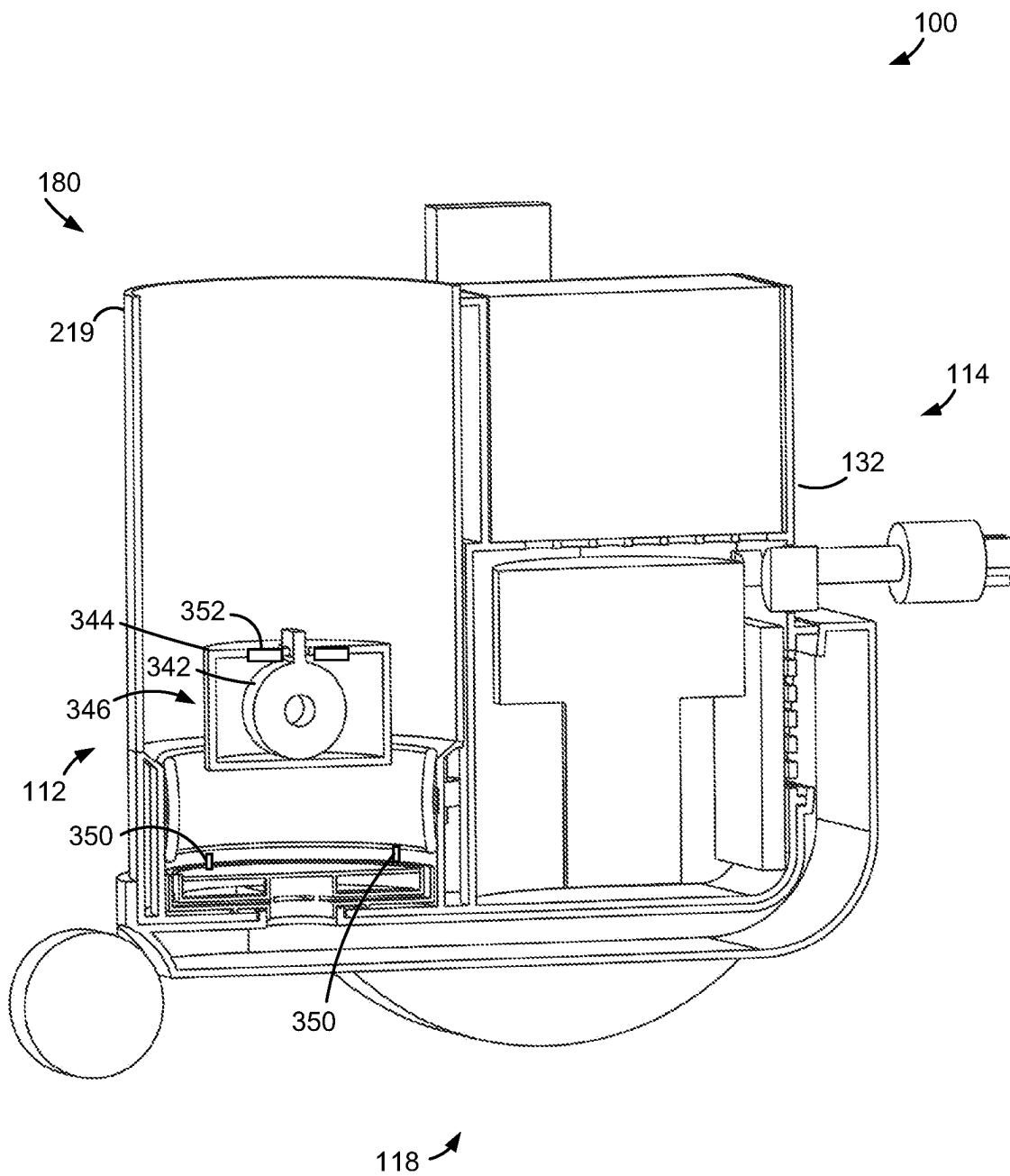


FIG. 45

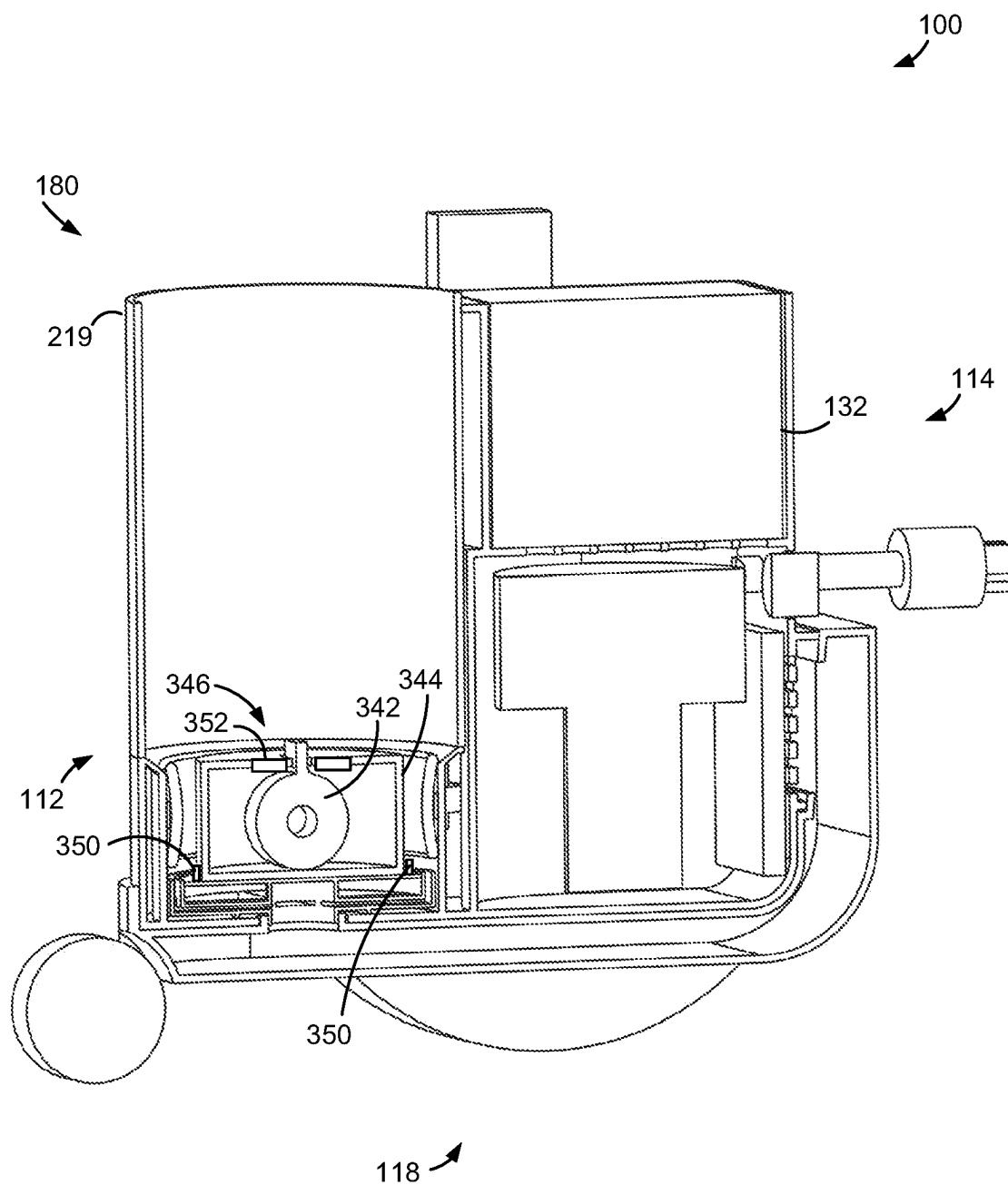


FIG. 46

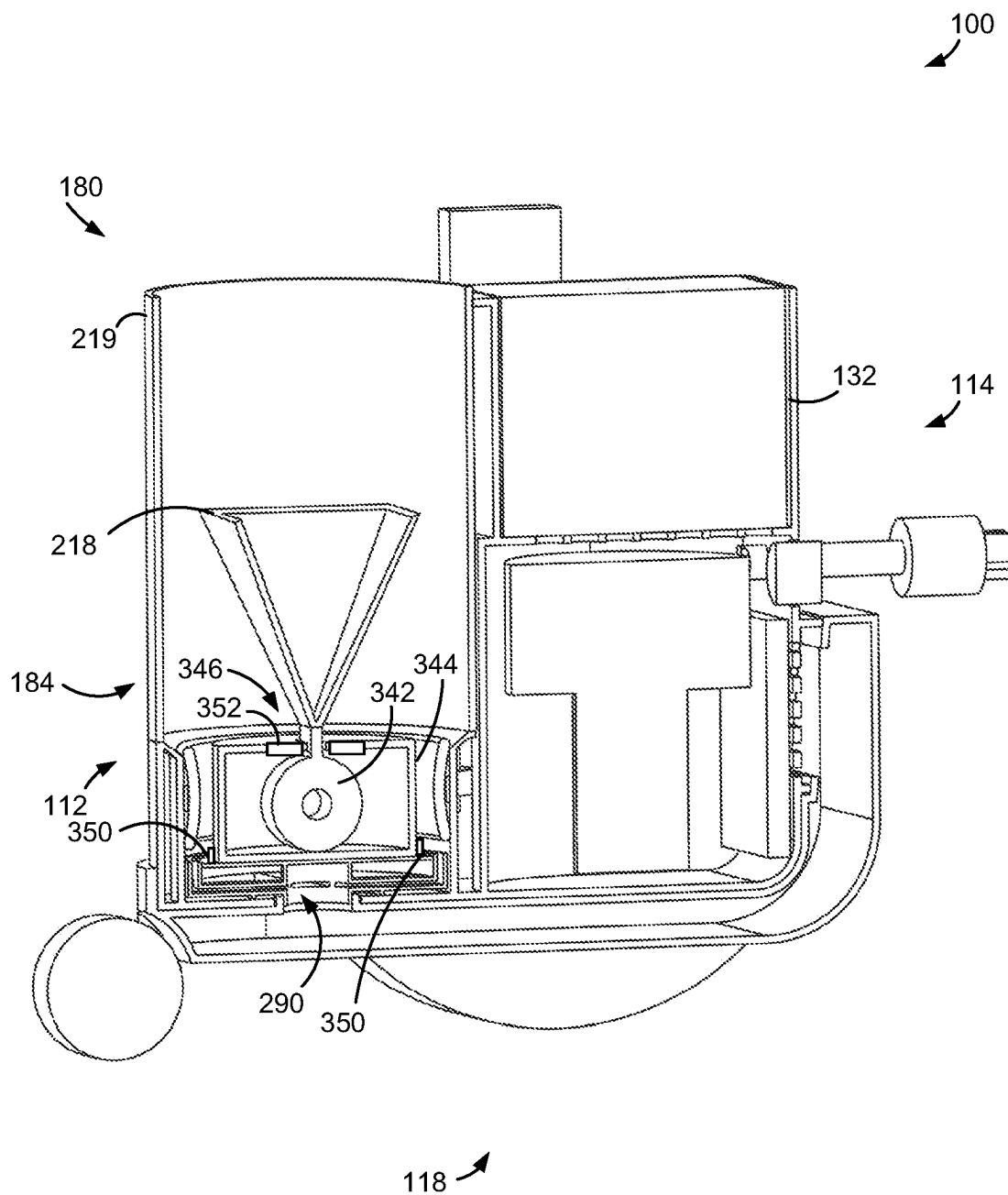


FIG. 47



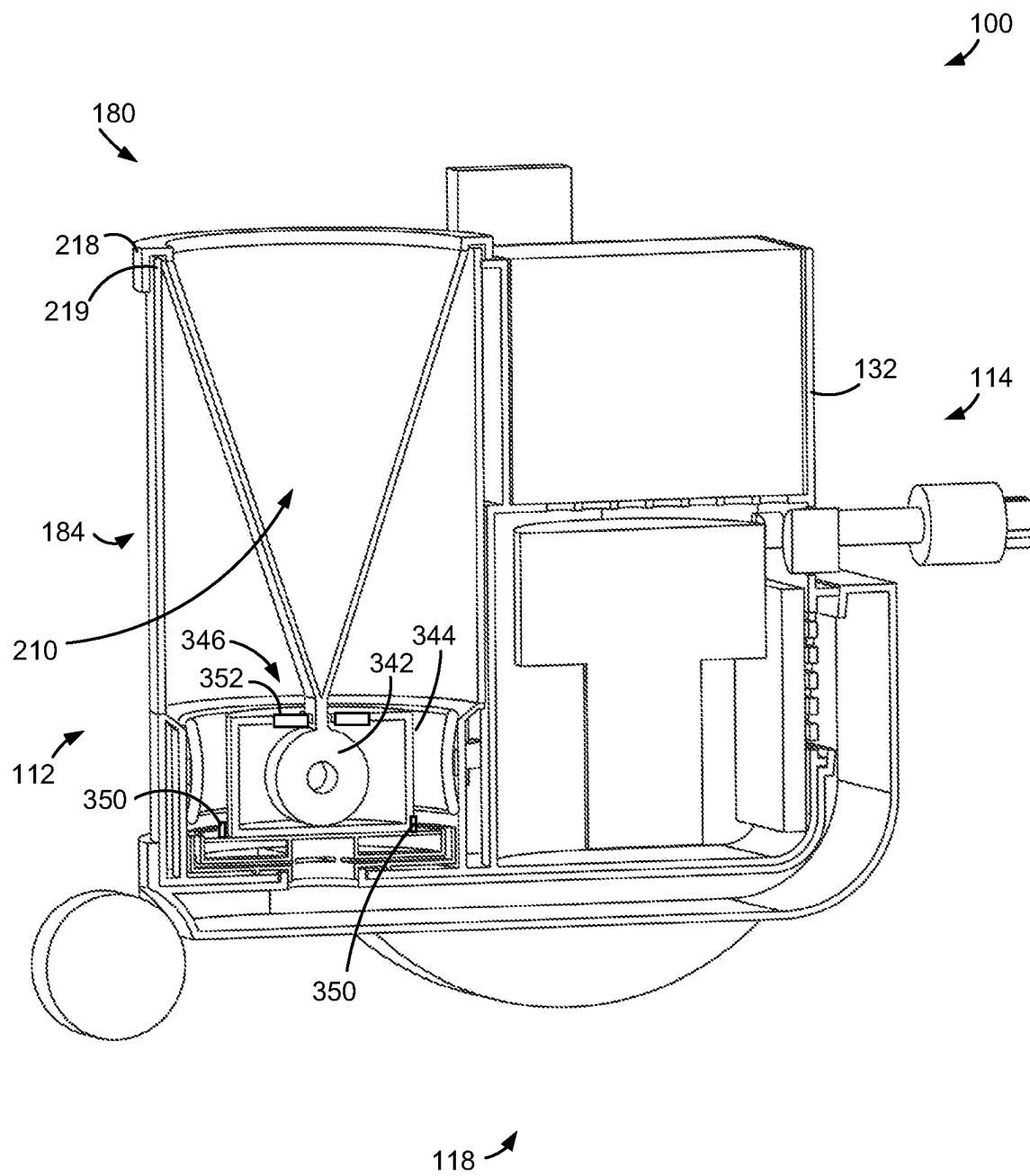


FIG. 48

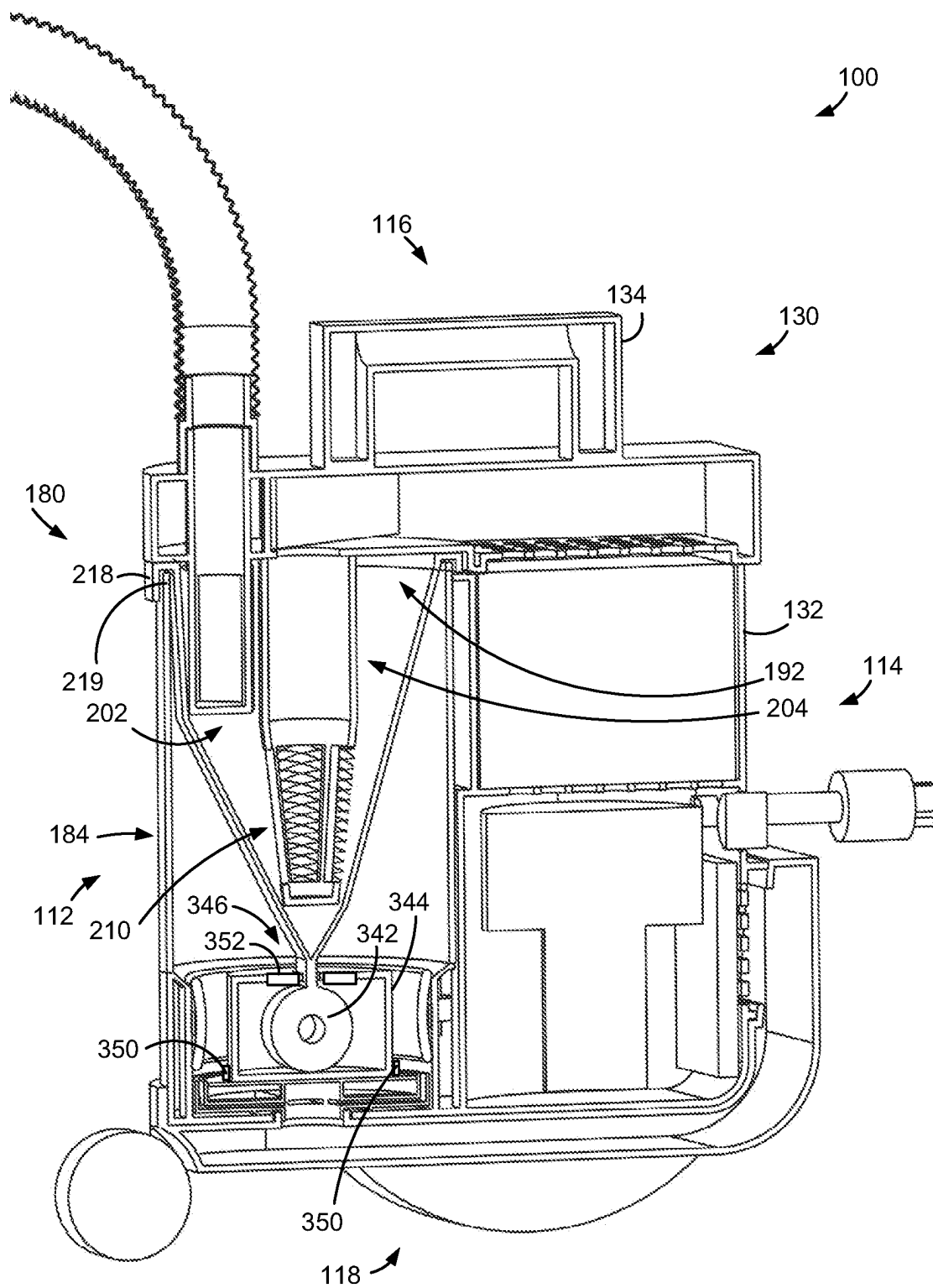
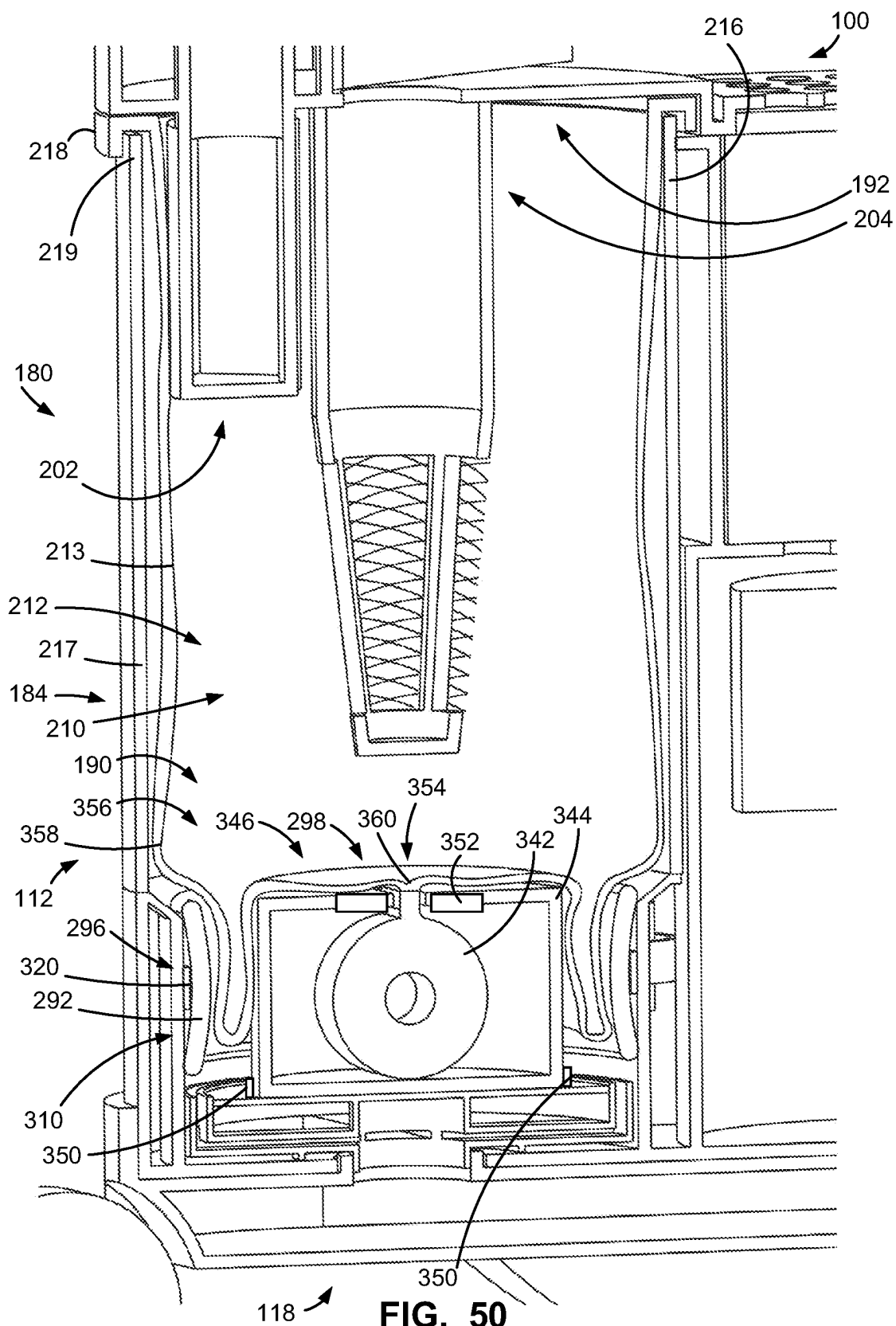
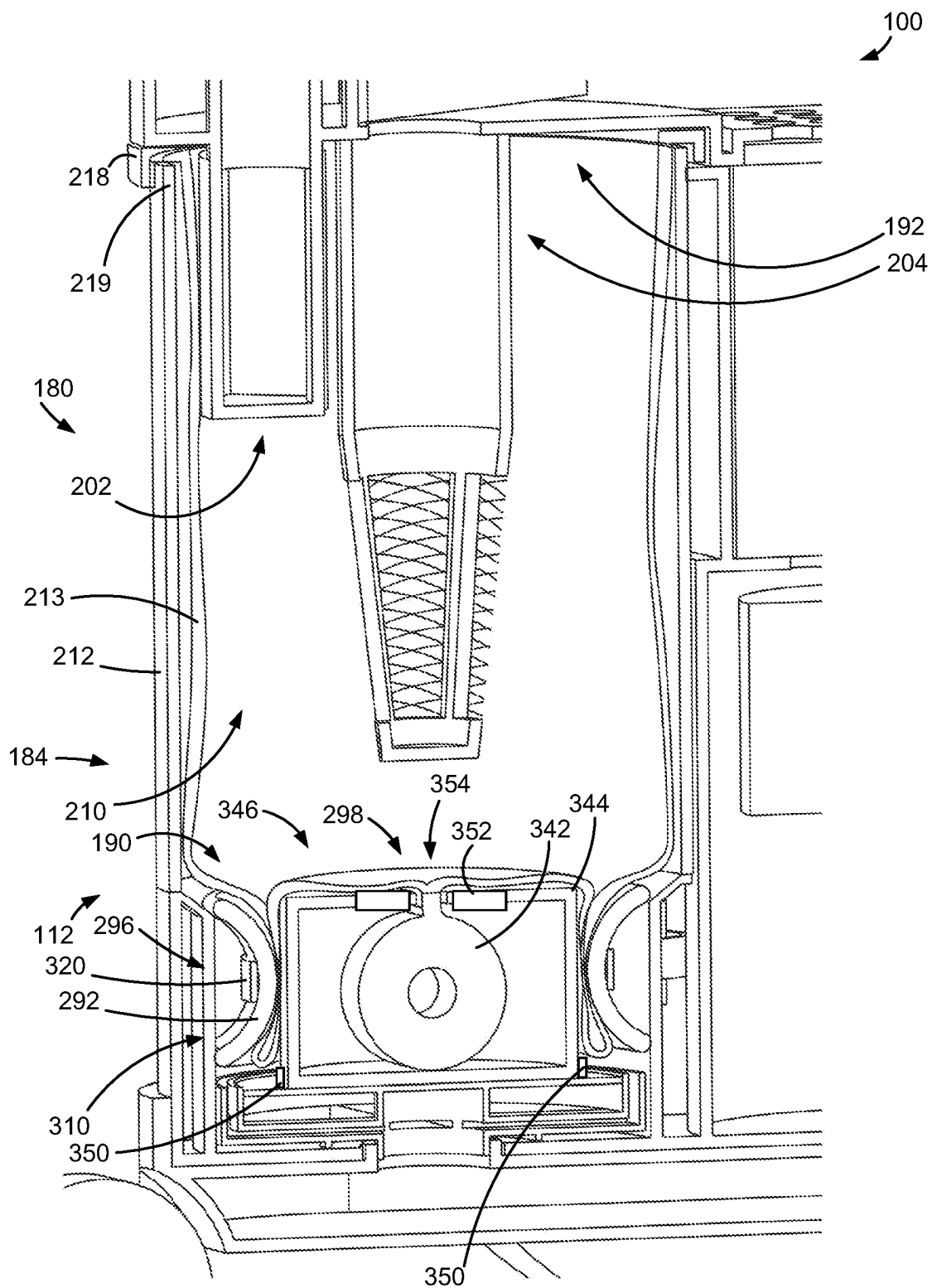


FIG. 49





118 **FIG. 51**

## SURFACE CLEANING APPARATUS

### FIELD

**[0001]** This disclosure relates generally to surface cleaning apparatus and a docking station which has an air treatment chamber such as a cyclone.

### INTRODUCTION

**[0002]** The following is not an admission that anything discussed below is part of the prior art or part of the common general knowledge of a person skilled in the art.

**[0003]** Various types of surface cleaning apparatus are known, including upright surface cleaning apparatus, canister surface cleaning apparatus, stick surface cleaning apparatus, central vacuum systems, and hand carriable surface cleaning apparatus such as hand vacuum cleaners. Further, various designs for cyclonic surface cleaning apparatus, including battery operated cyclonic hand vacuum cleaners are known in the art.

### SUMMARY

**[0004]** The following introduction is provided to introduce the reader to the more detailed discussion to follow. The introduction is not intended to limit or define any claimed or as yet unclaimed invention. One or more inventions may reside in any combination or sub-combination of the elements or process steps disclosed in any part of this document including its claims and figures.

**[0005]** A surface cleaning apparatus or a docking station includes an air flow path between a dirty air inlet and a clean air outlet. An air treatment member, which comprises an air treatment chamber, is provided in the air flow path. A dirt collection region, comprising a region within the air treatment chamber and/or a discrete dirt collection chamber, is also provided. A bag lining the dirt collection region is removably positionable in the region within the air treatment chamber and/or the discrete dirt collection chamber. The liner bag may be removed and disposed of along with contained dirt for ease of disposal.

**[0006]** In accordance with an aspect of this disclosure, which may be used alone or in combination with any one or more other aspects, the air treatment chamber includes a dirt outlet that is upstream of a dirt conduit. The dirt conduit extends from the dirt outlet to a remote dirt collection region. The air treatment chamber may be a second stage air treatment chamber, and the dirt collection region may be remote from the second stage chamber, such as with a first stage air treatment chamber received between the second stage air treatment chamber and a dirt collection chamber. The dirt conduit may extend along the first stage air treatment chamber. The treatment chamber may include two or more dirt outlets, each opening into a dirt conduit. In some examples, the dirt outlets and/or dirt conduits are angularly spaced from one another, leaving a space between for an additional component. For example, a portion of an upright section may be received between the dirt conduits when the surface cleaning apparatus is mounted to the upright section to form an upright vacuum. The additional component may be received snugly between the dirt conduits for a stable connection between the surface cleaning apparatus and the additional component. The additional component may be elongated and fit snugly between the dirt conduits along its length for a stable connection.

**[0007]** In accordance with another aspect of this disclosure, which may be used alone or in combination with any one or more other aspects, the surface cleaning apparatus includes a uniflow cyclone chamber and a discrete dirt collection chamber, and the dirt collection chamber is at an end of the uniflow cyclone chamber. The uniflow cyclone includes an air inlet at one of the first and second ends of the cyclone chamber and an air outlet at the other. Arranging the dirt collection chamber at an end of a uniflow cyclone results in a compact width of the air treatment assembly in combination with the compact size and low back pressure of a uniflow cyclone. The dirt collection chamber is generally at the first end when the apparatus is in an in-use position. The first end may be a lower end and the collection chamber may be below the cyclone such that gravity urges dirt into the collection chamber. The dirt collection chamber is at the same end of the air treatment chamber as the air inlet or the air outlet of the cyclone. In some embodiments, the dirt collection chamber is at the same end as the air inlet of the cyclone.

**[0008]** In accordance with another aspect of this disclosure, which may be used alone or in combination with any one or more other aspects, a porous member is between the air treatment chamber and the dirt collection chamber. The porous member is positioned at the interface between the air treatment chamber and the dirt collection chamber. The dirt outlet from the air treatment chamber may comprise or consist of openings in the porous member. Liquids such as water can pass through the porous member into the dirt collection chamber, while larger debris is blocked by the porous member. The porous member may allow for separate handling of liquids and larger debris. The liquids may pass through the porous member into a first collection region and the larger debris may be directed into a second, separate collection region (e.g., a separate region of a dirt collection chamber or kept in a large debris collection region within the air treatment chamber).

**[0009]** In accordance with another aspect of this disclosure, which may be used alone or in combination with any one or more other aspects, a surface cleaning apparatus includes a bag positioning system to arrange the bag within the dirt collection chamber. The user may be required to arrange a rim of the bag on a rim of the chamber. The bag positioning system repositions the bag within the chamber into an open position to receive dirt. The bag positioning system may reposition the bag by moving a bottom of the bag down into a bottom of the chamber. The bag positioning system may partially or fully open up the bag, such as by extending a bottom of the bag to line the bottom of the dirt collection chamber. The bag positioning system may be a mechanical system and/or a pneumatic system. A mechanical bag positioning system may include one or more arms selectively extendable down into the bag to open up the bag. A pneumatic bag positioning system may include a vacuum line opening from a bottom of the dirt collection chamber to draw the bag down into the chamber.

**[0010]** In accordance with another aspect of this disclosure, which may be used alone or in combination with any one or more other aspects, a surface cleaning apparatus includes a bag holding system. The bag holding system holds the bag in position during a dirt collection operation to prevent the bag from being moved by, e.g., air flow through the surface cleaning apparatus. The bag may be in a chamber that is in the air flow path (e.g., lining a dirt collection region

that is within an air treatment chamber), and the bag holding system prevents the bag from being sucked inwardly and collapsing on itself during a cleaning operation. The bag holding system includes one or more bag securing members. Each bag securing member comprises one or more bag engaging members which are moveable between a retracted bag insertion position in which a bag is positionable in the dirt collection chamber and an extended bag engaging position in which the bag is secured in the dirt collection chamber. The bag holding system and the bag positioning system may be a common system, such as mechanical arms which extend down into the bag to open up the bag and then remain in the bag to keep it open. Alternatively, the surface cleaning apparatus may include a bag holding system which is discrete from the bag positioning system, such as one or more mechanical members which are moveable into an extended bag engaging position after a pneumatic bag positioning system has arranged the bag in a desired position.

**[0011]** In accordance with another aspect of this disclosure, which may be used alone or in combination with any one or more other aspects, the bag engaging member(s) are internal to the bag when in the bag engaging position. In some examples, bag engaging members extend down into the bag to prevent the bottom of the bag from moving upwards towards the rim. The bag engaging members may be or include arms which are mounted to an upper wall of the dirt collection chamber (e.g., a fixed wall or a removable wall such as a lid) and moveable between a retracted position removed from a bottom of a dirt collection chamber and an extended position in which the arms extend into the bottom of the dirt collection chamber. Arms may be retracted by moving the arms relative to a mounting location and/or by moving the mounting location, such as by lifting a lid of the dirt collection region to which the arms are mounted away from the bottom of the dirt collection region to remove the arms from the bottom of the dirt collection region. Alternately, the arms may be telescopic. The bag engaging members internal to the bag which form part of the bag holding system may also form part of the bag positioning system.

**[0012]** In accordance with another aspect of this disclosure, which may be used alone or in combination with any one or more other aspects, the bag engaging members are external of the bag when in the bag engaging position. An external member holding the bag avoids obstructing the interior of the bag and/or separates the external member from the dirt collected in the bag. A moveable bag engaging member outside the bag simplifies protecting the moveable components from contamination by dirt that is within the bag. The bag engaging members may be moveable in any suitable direction, such as generally parallel to an axis of rotation of an air treatment chamber which the bag lines or generally radially relative to the axis of rotation. For example, the bag engaging members may move radially inwardly into the bag engaging position and/or radially outwardly into the bag engaging position. In some examples, the member or members sandwich a portion of the bag to hold it in position. The portion of the bag may be sandwiched between two bag engaging members (e.g., one moving radially inward and one moving radially outward) or between a bag engaging member and a block such as a radially inward block over which the bag is draped or a radially outward block which the bag hangs in front of. For

example, a bag engaging member may move radially outwardly to sandwich the bag against a wall of the dirt collection chamber or a bag engaging member may move radially inwardly to sandwich the bag against a centrally arranged block over which the bag is draped.

**[0013]** In accordance with another aspect of this disclosure, which may be used alone or in combination with any one or more other aspects, a surface cleaning apparatus includes a roller bag engaging member. The roller bag engaging member rolls about an axis of rotation to draw a portion (e.g., a bottom) of the bag past the roller. The portion of the bag drawn past the roller is held against the roller and an adjacent member. Optionally, the bag securing member includes a pair of rollers to draw the portion of the bag between the rollers. A pair of rollers may be arranged adjacent one another and may be rotated in opposite directions to pull the bag through between them. The rollers may rotate downwardly towards one another to pull the bag down between them. In some examples, the roller engages an external surface of the bag.

**[0014]** In accordance with another aspect of this disclosure, which may be used alone or in combination with any one or more other aspects, an adhesive holds a bag in position during a cleaning operation of the surface cleaning apparatus. The adhesive joins a wall of the bag to a surface of the dirt collection chamber during the cleaning operation. The adhesive may be provided on a disposable bag and/or on the wall of the chamber. In some examples, the surface cleaning apparatus includes a mechanical bag engaging member to separate the bag wall and the wall of the dirt collection chamber after the cleaning operation to break the adhesive bond to allow the bag to be removed. The mechanical bag engaging member is moveable between a retracted position and an extended position. For example, the member may move from the retracted position to the extended position to push the bag wall away from the chamber wall to break the adhesive bond. The member may extend out from the chamber wall to push the bag wall away from the chamber wall.

**[0015]** In accordance with another aspect of this disclosure, which may be used alone or in combination with any one or more other aspects, a surface cleaning apparatus is useable with a bag roll. The surface cleaning apparatus may include a roll holder to receive a bag roll. The bag roll may be any suitable roll of material to be used in lining the dirt collection chamber with a bag, such as a roll of preformed bags, a continuous tube to be formed into bags, or one or more rolls of sheeting to be formed into bags. In some examples, a heat sealing member is arranged to engage a portion of the material of the bag roll, e.g., to heat seal a portion of plastic sheeting to another portion of plastic sheeting to form an edge of a bag. A heat sealing member may close a single edge of a bag, e.g., closing the bottom of a bag by sealing across a continuous tube of plastic. A heat sealing member may close multiple edges of a bag, e.g., sealing two sheets of plastic together to form a tube or sealing one edge of a doubled plastic sheet to the opposite edge to form a tube and concurrently or subsequently sealing across the tube to close a bottom of the bag. The bag may be formed prior to a dirt cleaning operation. Alternatively, the bag may be formed after a cleaning operation but before the dirt is emptied from the chamber. A bag may remain attached to a remainder of the bag roll during a cleaning operation. Optionally, an interface between a bag and the remainder of

the bag roll includes a line of weakness for easier separation of the bag from the remainder of the roll of bags. Alternatively, the interface between a bag and the remainder of the bag roll is unweakened. A bag may be cut free from the remainder of the bag roll following a cleaning operation, e.g., automatically upon sensing that the bag is full or in response to a user action such as a button push. The bag roll may form part of a bag holding system, with the remainder of a roll of bag material holding a bag in position during operation. Alternatively, or additionally, a discrete bag holding system may be used to hold the bag, which may be removably positionable in the container.

[0016] It will be appreciated by a person skilled in the art that an apparatus or method disclosed herein may embody any one or more of the features contained herein and that the features may be used in any particular combination or sub-combination.

[0017] These and other aspects and features of various embodiments will be described in greater detail below.

#### BRIEF DESCRIPTION OF THE DRAWINGS

[0018] For a better understanding of the described embodiments and to show more clearly how they may be carried into effect, reference will now be made, by way of example, to the accompanying drawings in which:

[0019] FIG. 1 is rear perspective view of an exemplary surface cleaning apparatus;

[0020] FIG. 2 is an exploded perspective view from the side and the front of the surface cleaning apparatus of FIG. 1;

[0021] FIG. 3 is a perspective view from the side and the front of the surface cleaning of FIG. 1 mounted to base to form a stick vacuum cleaner;

[0022] FIG. 4 is a bottom perspective vertical cross sectional view of the stick vacuum of FIG. 3 with a floor of a first stage cyclone closed;

[0023] FIG. 4A is an enlarged bottom perspective vertical cross sectional view of a portion of another surface cleaning apparatus with a porous separation member;

[0024] FIG. 5 is a bottom perspective vertical cross sectional view of the stick vacuum of FIG. 3 with the floor of the first stage cyclone opened;

[0025] FIG. 6 is a perspective view from the side and the front of another exemplary surface cleaning apparatus;

[0026] FIG. 7 is a rear perspective view of the surface cleaning apparatus of FIG. 6;

[0027] FIG. 8 is a bottom perspective vertical cross sectional view of another exemplary surface cleaning apparatus with a floor of a first stage cyclone closed;

[0028] FIG. 9 is a bottom perspective vertical cross sectional view of the surface cleaning apparatus of FIG. 8 with the floor of the first stage cyclone opened;

[0029] FIG. 10 is a side perspective vertical cross sectional view of another exemplary surface cleaning apparatus with a floor of a first stage cyclone closed;

[0030] FIG. 11 is a side perspective vertical cross sectional view of the surface cleaning apparatus of FIG. 10 with the floor of the first stage cyclone opened;

[0031] FIG. 12 is a top perspective horizontal cross sectional view of the surface cleaning apparatus of FIG. 10 with the floor of the first stage cyclone closed;

[0032] FIG. 13 is a top perspective horizontal cross sectional view of the surface cleaning apparatus of FIG. 10 with the floor of the first stage cyclone opened;

[0033] FIG. 14 is a perspective view from the side and the front of another exemplary surface cleaning apparatus in a storage position;

[0034] FIG. 15 is a perspective view from the side and the front of the surface cleaning apparatus of FIG. 14 in a reclined position;

[0035] FIG. 16 is a bottom perspective vertical cross sectional view of the surface cleaning apparatus of FIG. 14 with a floor of a dirt collection region closed;

[0036] FIG. 17 is a bottom perspective vertical cross sectional view of the surface cleaning apparatus of FIG. 14 with the floor of the dirt collection region opened;

[0037] FIG. 18 is a top perspective horizontal cross sectional view of the surface cleaning apparatus of FIG. 14 with a floor of a first stage cyclone closed;

[0038] FIG. 19 is a top perspective horizontal cross sectional view of the surface cleaning apparatus of FIG. 14 with the floor of the first stage cyclone opened;

[0039] FIG. 20 is a perspective view from the side and the front of another exemplary surface cleaning apparatus;

[0040] FIG. 21 is a side perspective vertical cross sectional view of the surface cleaning apparatus of FIG. 20 with a floor of a first stage cyclone closed;

[0041] FIG. 22 is a side perspective vertical cross sectional view of the surface cleaning apparatus of FIG. 20 with the floor of the first stage cyclone opened;

[0042] FIG. 23 is a side perspective exploded vertical cross sectional view of the surface cleaning apparatus of FIG. 20;

[0043] FIG. 24 is a perspective cut view from the side and the front cut away view of the surface cleaning apparatus of FIG. 20;

[0044] FIG. 25 is a front perspective exploded vertical cross sectional view of a portion of an air treatment assembly of the surface cleaning apparatus of FIG. 20;

[0045] FIG. 26 is a front perspective vertical cross sectional view of the portion of the air treatment assembly of FIG. 25 with a bag positioning system in a partially retracted position;

[0046] FIG. 27 is a front perspective vertical cross sectional view of the portion of the air treatment assembly of FIG. 25 with the bag positioning system in an extended position;

[0047] FIG. 28 is a perspective view from the side and the front of another exemplary surface cleaning apparatus;

[0048] FIG. 29 is a rear perspective view of the surface cleaning apparatus of FIG. 28;

[0049] FIG. 30 is a side perspective exploded vertical cross sectional view of the surface cleaning apparatus of FIG. 28;

[0050] FIG. 31 is a side perspective vertical cross sectional view of the surface cleaning apparatus of FIG. 28 with a lid removed and a bag partially inserted;

[0051] FIG. 32 is a side perspective vertical cross sectional view of the surface cleaning apparatus of FIG. 28 with the lid closed and the bag inserted;

[0052] FIG. 33 is a side perspective vertical cross sectional view of a portion of the surface cleaning apparatus of FIG. 28 with a bag engaging member in a retracted position;

[0053] FIG. 34 is a side perspective vertical cross sectional view of the portion of the surface cleaning apparatus of FIG. 33 with the bag engaging member in an extended position;

[0054] FIG. 35 is a side perspective vertical cross sectional view of a portion of another exemplary surface cleaning apparatus, with a bag engaging member in a retracted position;

[0055] FIG. 36 is a side perspective vertical cross sectional view of the portion of the surface cleaning apparatus of FIG. 35, with the bag engaging member in an extended position;

[0056] FIG. 37 is a top perspective horizontal cross sectional view of the portion of the surface cleaning apparatus of FIG. 35, with the bag engaging member in the retracted position;

[0057] FIG. 38 is a top perspective horizontal cross sectional view of the portion of the surface cleaning apparatus of FIG. 35, with the bag engaging member in the extended position;

[0058] FIG. 39 is a side perspective exploded vertical cross sectional view of another exemplary surface cleaning apparatus;

[0059] FIG. 40 is a side perspective vertical cross sectional view of the surface cleaning apparatus of FIG. 39 with a lid removed and a bag partially inserted;

[0060] FIG. 41 is a side perspective vertical cross sectional view of the surface cleaning apparatus of FIG. 39 with the lid removed and the bag inserted;

[0061] FIG. 42 is a side perspective vertical cross sectional view of the surface cleaning apparatus of FIG. 39 with the lid closed and the bag inserted;

[0062] FIG. 43 is a side perspective vertical cross sectional view of a portion of the surface cleaning apparatus of FIG. 39 with the bag held by bag engaging rollers;

[0063] FIG. 44 is a side perspective exploded vertical cross sectional view of another exemplary surface cleaning apparatus;

[0064] FIG. 45 is a side perspective vertical cross sectional view of the surface cleaning apparatus of FIG. 44 with a lid removed and a bag roll partially inserted;

[0065] FIG. 46 is a side perspective vertical cross sectional view of the surface cleaning apparatus of FIG. 44 with the lid removed and the bag roll inserted;

[0066] FIG. 47 is a side perspective vertical cross sectional view of the surface cleaning apparatus of FIG. 44 with the bag roll inserted and partially drawn up;

[0067] FIG. 48 is a side perspective vertical cross sectional view of the surface cleaning apparatus of FIG. 44 with the bag roll inserted and drawn up;

[0068] FIG. 49 is a side perspective vertical cross sectional view of the surface cleaning apparatus of FIG. 44 with the bag roll inserted and drawn up and the lid closed;

[0069] FIG. 50 is a side perspective vertical cross sectional view of a portion of the surface cleaning apparatus of FIG. 44 with a bottom of the bag opened up and a bag engaging member in a retracted position; and,

[0070] FIG. 51 is a side perspective vertical cross sectional view of the portion of the surface cleaning apparatus of FIG. 50 with the bag engaging member in an extended position.

[0071] The drawings included herewith are for illustrating various examples of articles, methods, and apparatuses of the teaching of the present specification and are not intended to limit the scope of what is taught in any way.

#### DESCRIPTION OF EXAMPLE EMBODIMENTS

[0072] Various apparatuses, methods and compositions are described below to provide an example of an embodiment of each claimed invention. No embodiment described below limits any claimed invention and any claimed invention may cover apparatuses and methods that differ from those described below. The claimed inventions are not limited to apparatuses, methods and compositions having all of the features of any one apparatus, method or composition described below or to features common to multiple or all of the apparatuses, methods or compositions described below. It is possible that an apparatus, method or composition described below is not an embodiment of any claimed invention. Any invention disclosed in an apparatus, method or composition described below that is not claimed in this document may be the subject matter of another protective instrument, for example, a continuing patent application, and the applicant(s), inventor(s) and/or owner(s) do not intend to abandon, disclaim, or dedicate to the public any such invention by its disclosure in this document.

[0073] The terms “an embodiment,” “embodiment,” “embodiments,” “the embodiment,” “the embodiments,” “one or more embodiments,” “some embodiments,” and “one embodiment” mean “one or more (but not all) embodiments of the present invention(s),” unless expressly specified otherwise.

[0074] The terms “including,” “comprising” and variations thereof mean “including but not limited to,” unless expressly specified otherwise. A listing of items does not imply that any or all of the items are mutually exclusive, unless expressly specified otherwise. The terms “a,” “an” and “the” mean “one or more,” unless expressly specified otherwise.

[0075] As used herein and in the claims, two or more parts are said to be “coupled,” “connected,” “attached,” or “fastened” where the parts are joined or operate together either directly or indirectly (i.e., through one or more intermediate parts), so long as a link occurs. As used herein and in the claims, two or more parts are said to be “directly coupled,” “directly connected,” “directly attached,” or “directly fastened” where the parts are connected in physical contact with each other. None of the terms “coupled,” “connected,” “attached,” and “fastened” distinguish the manner in which two or more parts are joined together.

[0076] Furthermore, it will be appreciated that for simplicity and clarity of illustration, where considered appropriate, reference numerals may be repeated among the figures to indicate corresponding or analogous elements. In addition, numerous specific details are set forth in order to provide a thorough understanding of the example embodiments described herein. However, it will be understood by those of ordinary skill in the art that the example embodiments described herein may be practiced without these specific details. In other instances, well-known methods, procedures, and components have not been described in detail so as not to obscure the example embodiments described herein. Also, the description is not to be considered as limiting the scope of the example embodiments described herein.

#### General Description of a Surface Cleaning Apparatus

[0077] Referring to FIG. 1, an exemplary embodiment of a surface cleaning apparatus is shown generally as 100.



[0078] The exemplary surface cleaning apparatus 100 of FIGS. 1 and 2 is a hand-held surface cleaning apparatus, which is commonly referred to as a “hand vacuum cleaner” or a “handvac”. As used herein, a hand-held surface cleaning apparatus or hand vacuum cleaner or handvac is a vacuum cleaner that can be operated generally one-handedly to clean a surface while its weight is held by the same one hand. For example, the carry handle and a clean air inlet may be rigidly coupled to each other (directly or indirectly) so as to move as one while maintaining a constant orientation relative to each other. This is contrasted with upright and canister surface cleaning apparatus, the weight of which is supported by a surface (e.g., a floor below) during use.

[0079] It will be appreciated that any one or more of the features of the surface cleaning apparatus 100 set out herein may also or alternately be used in any type of surface cleaning apparatus, such as an upright surface cleaning apparatus, a stick vac, an extractor, a wet-dry vacuum, or the like, or in a docking station for a surface cleaning apparatus. It will also be appreciated that a surface cleaning apparatus or docking station may use any configuration of the operating components and the airflow paths exemplified herein.

[0080] Optionally, a hand surface cleaning apparatus 100 may be removably mountable on a base so as to form, for example, an upright vacuum cleaner, a canister vacuum cleaner, a stick vacuum cleaner or stick vac, a wet-dry vacuum cleaner and the like. For example, the exemplary surface cleaning apparatus of FIGS. 3 to 5 is a stick vacuum. The stick vacuum includes a surface cleaning head 104 and an elongated wand 106 that can be removably connectable to the surface cleaning apparatus 100. As another example, the exemplary surface cleaning apparatus of FIGS. 14-19 is an upright vacuum cleaner. An upright vacuum cleaner includes the surface cleaning head 104 and an upright section 108 moveably mounted to the surface cleaning head 104. An upright section includes a housing to provide or receive a dirt collection region. The exemplary upright section 108 of FIGS. 14-19 includes a housing 107 to provide or receive a dirt collection region to provide additional dirt collection capacity to the surface cleaning apparatus beyond what is available in the hand vacuum, as described further elsewhere herein. In some examples, a surface cleaning apparatus 100 that includes a base with a floor cleaning head 104 is moveable between an upright storage position (e.g., FIG. 14) and a reclined operating or in-use position (e.g., FIG. 15).

[0081] Referring to FIGS. 3 to 5, the exemplary surface cleaning apparatus 100 includes wheels 109. Wheels 109 are included for supporting one or more piece of the surface cleaning apparatus 100 on a floor or other support surface. Wheels may be mounted to any suitable piece of the apparatus 100. The exemplary apparatus 100 of FIGS. 3 to 5 includes wheels 109 on the floor cleaning head 104. In some examples, wheels 109 are on a main body (e.g., carriage 111 of the main body 130 of the exemplary canister surface cleaning apparatus 100 of FIG. 28) of the apparatus 100. A wheeled support reduces the weight that must be supported by a user, which may be particularly helpful when the vacuum is a large-capacity vacuum that can contain a substantial weight of dirt (e.g., dust or liquid), such as more than 3 kg, more than 5 kg, more than 10 kg, or more than 15 kg of dirt.

[0082] Referring again to FIG. 1, the surface cleaning apparatus 100 has an apparatus front end 112, an apparatus

rear end 114, an apparatus upper end 116 and an apparatus lower end 118. An apparatus longitudinal axis 120 extends between the apparatus front end 112 and the apparatus rear end 114 in a forward/rearward direction. An apparatus vertical axis 122 extends between the apparatus upper end 106 and apparatus lower end 118 in an upward/downward direction. The apparatus vertical axis 122 is perpendicular to the apparatus longitudinal axis 120. An apparatus transverse axis 124 is perpendicular to each of the apparatus vertical axis 122 and the apparatus longitudinal axis 120 and extends laterally through the right and left lateral sides of apparatus 100.

[0083] In an in-use position for cleaning a surface such as a floor, the upper end 116 is generally above the lower end 118. In some examples, the transverse axis 124 extends generally horizontally in the in-use position. In some examples, the longitudinal axis 120 extends generally horizontally in the in-use position (e.g., for a robot vacuum or a canister vacuum). Referring to FIGS. 1 and 2, in the in-use position of FIG. 1 the transverse axis 124 of the exemplary surface cleaning apparatus 100 extends generally horizontally, the upper end 116 is generally at an elevation above the lower end 118, and the longitudinal axis 120 extends at an angle 113 to the horizontal 115 (i.e., a reclined use position).

[0084] Referring still to FIG. 1, the surface cleaning apparatus 100 includes a main body 130. The main body 130 includes a main body housing 132 and optionally a carry handle 134. The main body housing 132 may be any suitable main body housing. The main body housing 132 houses one or more of a pre-motor filter, an air moving member (e.g., a suction motor), and a post motor filter. As exemplified in FIG. 2, the main body housing 132 may be selectively openable to provide access to one or more housed components.

[0085] The handle 134 is configured to for use in moving the apparatus 100. In some examples, handle 134 is a drive handle drivingly (e.g., rigidly) connected to a nozzle or floor cleaning head to move the nozzle or floor cleaning head to clean a surface. In some examples, a handle is a carry handle to carry the apparatus 100, such as during a cleaning operation or to and from a storage location. The handle 134 may be any suitable handle such as a recessed handle formed by providing a recess in a surface of the main body housing 132 or a projecting handle extending out from the main body housing 132.

[0086] Referring to FIG. 1, one or both ends of handle 134 of the main body 130 are secured to the main body housing 132, directly or indirectly (e.g., via struts extending from the hand grip 136). The exemplary handle 134 of FIG. 1 is secured to the main body housing 132 at one end. The exemplary handle 134 of FIG. 28 is secured to the main body housing 132 at each end. In some examples, a handle 134 extends out from the main body housing 132 so as to provide a hand grip area 135 between the handle 134 and a surface of the main body housing 132. Optionally, the hand grip area 135 is encircled by the handle and the main body. As exemplified in FIG. 28, the handle 134 provides an enclosed hand grip area 135 between the handle 134 and an upper surface of the main body housing 132.

[0087] Referring to FIG. 1, the carry handle 134 includes a hand grip portion 136 extending generally along a hand grip axis 138. The hand grip axis 138 may extend in any suitable direction in use, such as generally vertically (e.g., a

pistol grip handle) or generally horizontally (e.g., an under-hand carry handle, such as the handle **134** of FIG. **28**).

**[0088]** In some examples, the handle **134** is secured to the upper end **116** of the apparatus **100**. In some examples, when the apparatus **100** is in an in-use position part or all of the hand grip portion **136** overlies part or all of the main body housing **132**, the suction motor, and/or the air treatment assembly. Arranging the handle **134** above a heavy and/or bulky component of the surface cleaning apparatus **100** may result in a more desirable hand-feel of the surface cleaning apparatus **100**. Referring to FIG. **28**, in the illustrated in-use position part of the exemplary handle **134** overlies part of the air treatment assembly and part of the handle **134** overlies part of the suction motor.

**[0089]** The handle **134** may be mounted to or formed on one or more of an openable and/or removeable apparatus wall, a suction motor housing, a pre-motor filter housing, a post-motor filter housing, and the air treatment assembly. The exemplary handle of FIG. **30** is mounted to an upper surface of the main body **132** by upwardly extending struts that extend between the upper surface (at the location of the carry handle base end **140**) and the hand grip portion **136**. The exemplary upper surface of the main body **132** of FIG. **30** is a surface of an openable wall. The exemplary upper surface of the main body **132** of FIG. **30** is a surface of a removeable door **181** (a lid in the exemplary embodiment) of the main body housing **132**.

**[0090]** It will be appreciated that the main body housing **132** and/or carry handle **134** may be in other configurations, shapes, and/or positions in other embodiments.

**[0091]** As exemplified in FIG. **4**, an apparatus air flow path **150** extends from an apparatus dirty air inlet **152** to an apparatus clean air outlet **154**. As exemplified, the apparatus dirty air inlet **152** may be provided at the apparatus front end **112** of a surface cleaning head. Arranging the dirty air inlet at the front end **112** may help with using the dirty air inlet as a surface cleaning nozzle. It will be appreciated that the dirty air inlet **152** may be located anywhere known in the vacuum cleaner art. The inlet may be directed in any suitable direction, such as generally downwardly as exemplified in FIG. **4** or generally forwardly or upwardly or rearwardly. The apparatus dirty air inlet **152** may be provided at an inlet end of an apparatus inlet conduit **160**. The apparatus inlet conduit **160** extends from an apparatus inlet conduit inlet end **162** to an apparatus inlet conduit outlet end **164**. The apparatus inlet conduit outlet end **164** may open into an internal chamber of the surface cleaning apparatus **100** (e.g., an air treatment chamber).

**[0092]** As exemplified in FIG. **4** the apparatus inlet conduit **160** has an inlet conduit longitudinal axis **166** along a longest dimension of the inlet conduit **160** and extending between the apparatus inlet conduit inlet end **162** and the apparatus inlet conduit outlet end **164**. Optionally, the inlet conduit **160** is a generally upwardly extending conduit, as exemplified in FIG. **4**. In the exemplary embodiment of FIG. **4**, the inlet conduit longitudinal axis **166** extends between the apparatus lower end **118** and the apparatus upper end **116**, and, as exemplified, may be generally vertical when the apparatus upper end **116** is above the apparatus lower end **118**, e.g., in a storage position. Alternatively, the inlet conduit axis may extend between the apparatus forward end **112** and the apparatus rear end **114**, and may be generally horizontal when the upper end **116** is above the lower end

**118**. The inlet conduit longitudinal axis **166** may be generally perpendicular to the apparatus longitudinal axis **120**.

**[0093]** As exemplified in FIG. **1**, the apparatus inlet conduit **160** may form a nozzle **168** of the surface cleaning apparatus **100** (e.g., a nozzle for a hand vac). Alternatively, or additionally, the inlet conduit **160** may be connectable or directly connectable to an accessory, such as the wand **106** illustrated in FIG. **3**, a flexible conduit (e.g., a wet/dry vacuum hose with or without a rigid conduit at an upstream end thereof), a crevice tool, a mini brush, and the like. The accessory may be coupled to the surface cleaning apparatus **100** such that the accessory is in air flow communication with the apparatus dirty air inlet **152** (e.g., in air flow communication with the apparatus inlet conduit **160**). For example, the accessory may be or include a conduit (e.g., wand **106**), and the conduit of the accessory may be received within the apparatus inlet conduit **160** or may receive the apparatus inlet conduit **160** within the accessory conduit. Optionally, one or more releasable fasteners may be used to couple the accessory to the surface cleaning apparatus **100**, such as clips or magnets. Alternatively, or additionally, the accessory may be held in air flow communication with the dirty air inlet via a friction fit (e.g., between an outer diameter of an accessory conduit and an inner diameter of the apparatus inlet conduit **160**, or vice versa). Alternately, it will be understood that the apparatus inlet conduit **160** may be slideably receivable in an accessory conduit.

**[0094]** It will also be appreciated that, in some embodiments, the surface cleaning apparatus **100** may not include an apparatus inlet conduit **160**, and the apparatus dirty air inlet **152** may instead open directly into a downstream chamber (e.g., an air treatment chamber) rather than being at an upstream end of a conduit. However, an apparatus inlet conduit **160** allows, e.g., a nozzle **168** to be formed for application to a surface that is to be cleaned (e.g., projecting outward for application to the surface), provides directionality to an air flow entering the surface cleaning apparatus **100**, and/or provides a convenient attachment interface for an accessory. Accordingly, any inlet conduit may be used.

**[0095]** The apparatus clean air outlet **154** may be provided at any location, such as at the apparatus upper end **116** (see FIG. **4**) or the lower end (See FIG. **42**). The apparatus clean air outlet **154** may be provided at the opposite end of the surface cleaning apparatus **100** from the apparatus dirty air inlet **152**. It will be appreciated that the apparatus dirty air inlet **152** and/or the apparatus clean air outlet **154** may each be provided at different locations and/or be of different configurations.

**[0096]** The surface cleaning apparatus **100** includes an air treatment assembly **180**. The apparatus air flow path **150** extends through the air treatment assembly **180**. The air treatment assembly **180** is configured to treat the airflow (e.g., remove dirt and/or liquid from the airflow).

**[0097]** The air treatment assembly **180** may be removably mounted to the main body **130** and/or incorporated as part of the main body **130**. For example, the air treatment assembly **180** or a portion thereof may be removably mounted to the main body housing **132**. The air treatment assembly **180** may be openable when mounted to the main body housing and/or when removed therefrom. It will be appreciated that the air treatment assembly **180** or the portion thereof may be removeable from the main body housing **132** in any suitable way, such as translationally or rotationally. It will be appreciated that the air treatment assembly **180** or the portion

thereof may be removeable from the main body housing **132** in any direction, such as upwardly, downwardly, forwardly, rearwardly, or laterally. It will be appreciated that the air treatment assembly **180** or the portion thereof may be secured to the main body **130** in any suitable way, such as by releasable fasteners such as screws, clasps, or magnets, which may be releasable or overcome by a predetermined force to be applied by a user. In some embodiments, the air treatment assembly **180** or a part thereof is released by a user action, such as by a user interaction with a button or other toggle of a user interface **178**.

[0098] The air treatment assembly **180** includes one or more air treatment members **182** in one or more treatment stages. The exemplary assembly **180** of FIG. 4 includes two air treatment members, a first stage member **182a** and a second stage member **182b**. Any air treatment member or members known in the art may be used. For example, an air treatment stage of the surface cleaning apparatus may use one or more air treatment stage which uses or comprises an air treatment chamber (e.g., a cyclone, a non-cyclonic momentum separator or the like). In addition, a physical filter media (e.g., foam, felt, HEPA) or the like may be provided downstream from any or all air treatment chambers. The air treatment member **182** may be provided upstream or downstream from the suction motor.

[0099] The exemplary assembly **180** of FIG. 4 includes an air treatment member **182** having an air treatment chamber **184**. The air treatment chamber **184** includes walls **186**. It will be appreciated that any suitable form of chamber **184** may be used, such as a generally rectangular chamber or a generally cylindrical chamber. In some examples, the air treatment chamber **184** includes a longitudinal axis **188** extending between a first end **190** and a second end **192** opposite the first end **190**. The air treatment chamber may be an upwardly-downwardly extending chamber (i.e., the long dimension of the chamber). In some examples, as exemplified in FIG. 4, the first end **190** is a lower end and the second end **192** is an upper end when the apparatus upper end **116** is above the apparatus lower end **118**. Alternatively, in some examples the first end **190** is a front end and the second end **192** is a rear end (i.e., the axis **188** is generally horizontal when the top end **116** is above the bottom end **118**).

[0100] Referring still to FIG. 4, in some examples the chamber **184** includes a first end wall **194** at the first end **190** and a second end wall **196** at the second end **192**. The second end wall **196** may face the first end wall **194** across the chamber **184**. In some examples, the first and second end walls **194**, **196** are generally planar walls. In some examples, a sidewall **198** extends between the first and second ends **190**, **192**, and may extend between the first and second end walls **194**, **196**. The sidewall **198** may be a generally cylindrical wall and the longitudinal axis **188** may be a cyclone axis of rotation of, e.g., a cyclone chamber. However, it will be appreciated that the air treatment member **182** may not be a cyclone. In some examples, the air treatment member **182** is a non-cyclonic momentum separator.

[0101] The air treatment chamber **184** includes a chamber air inlet **202** and a chamber air outlet **204**. The exemplary chamber air inlet **202** of FIG. 4 includes an inlet port opening into the chamber **184**. Similarly, the exemplary chamber air outlet **204** of FIG. 4 includes an outlet port opening from the chamber **184**. It will be appreciated that the chamber air inlet **202** and the chamber air outlet **204** may each be any suitable inlet or outlet (e.g., a tangential inlet for

a cyclone), and may be at any particular location on the air treatment chamber **184**. In some examples, the chamber air inlet **202** and/or the chamber air outlet **204** are arranged at the first and/or second ends **190**, **192** of the chamber **184**. Referring to FIG. 4, the exemplary air inlet **202** is provided at the first end **190** and the exemplary air outlet **204** is provided at the second end **192**. Alternatively, the air outlet **204** and air inlet **202** may be provide at a common end, e.g., the upper end **192** (e.g., as exemplified in FIG. 32).

[0102] The air treatment member **182** may be openable to, e.g., to allow for collected material to be removed. For example, a chamber **184** may be opened by any means known in the vacuum cleaner art. For example, a first part of a wall defining the chamber may be moveable relative to a remainder of the wall which defines the chamber and it may be removeable therefrom. As exemplified herein, an openable air treatment chamber **184** includes an openable portion of a wall of the chamber, such as a door, which may be a lid. One or both of the member end walls (e.g., upper and/or lower walls), or a portion of one or each of the member end walls, may be openable. Opening an end of the chamber may facilitate access to the member air inlet and/or air outlet. Alternatively, or additionally, the member sidewall or a portion thereof may be openable. Referring to FIG. 2, the exemplary air treatment member **182** includes an openable door **181** moveable between an open position (FIG. 2) in which a dumping doorway (or opening) **183** is open and a closed position (FIG. 4) closing the dumping doorway. The door **181** may be removeable (e.g., the exemplary lid **181** of FIG. 30) and/or moveable (e.g., pivotable or rotatable about a hinge, translatable, etc.) between a closed and an open position. As exemplified in FIG. 2, the door **181** may be joined (mounted) to a wall of the air treatment assembly **180** (e.g., to the sidewall **198**) by a hinge about which it moves between the open and closed positions. As exemplified by the air treatment member **182** in FIG. 2, door **181** may be formed by the air treatment member end wall (e.g., the first end wall **194**). The member end wall may seat (abut) on the distal end of the member sidewall **198** when closed (see e.g., FIG. 4). It will be appreciated that the member end wall may be secured to the member sidewall in any suitable way, such as via a friction fit or a releasable fastener (e.g., a clasp or threaded fastener) and it may be removeable and/or rotationally (e.g., pivotally) openable.

[0103] Referring again to FIG. 4, the air treatment assembly **180** includes a dirt collection region **210**. It will be understood that the air treatment member **182** and dirt collection region **210** may be of any configuration suitable for separating dirt (i.e., dust and/or liquid) from an air stream and collecting the dirt and/or liquid. In accordance with this disclosure, a surface cleaning apparatus may use one or more of the different aspects of an air treatment member **182** and a dirt collection region **210** discussed herein.

[0104] As exemplified in FIG. 32, a dirt collection region **210** may be an internal region of an air treatment chamber **184**. The dirt collection region **210** may be at the bottom end of the air treatment chamber **184**. Alternatively, as exemplified by the air treatment member **210** of FIG. 4, a dirt collection region **210** may be in a dirt collection chamber **212** that is external to an air treatment chamber. If the dirt collection chamber **212** and the air treatment chamber **184** are discrete chambers, the dirt collection chamber **212** communicates with the air treatment chamber **184** via one or more dirt outlets **214** (e.g., an opening in a wall of the air

treatment chamber or a gap between walls of the air treatment chamber). The dirt outlet **214** may be at a lower end of the dirt collection chamber **212** as exemplified in FIG. 4. It will be appreciated that a dirt outlet **214** may be at various locations, depending, inter alia, on the location of the air inlet and/or air outlet of the chamber **184**.

[0105] The dirt collection chamber **212** may be of any suitable shape, such as generally cuboid or generally cylindrical. The exemplary dirt collection chamber **212** is generally cylindrical in shape with a long dimension extending along a longitudinal axis **201**. The longitudinal axis **201** extended between a first end **215** and a second end **217** opposite the first end **215**. The dirt collection chamber **212** may be an upwardly-downwardly extending chamber (i.e., the long dimension of the chamber). In some examples, as exemplified by the orientation of FIG. 4, the first end **215** is a lower end and the second end **217** is an upper end when the apparatus upper end **116** is above the apparatus lower end **118**.

[0106] The dirt collection chamber **212** may be of any suitable construction. The dirt collection chamber is formed by a set of dirt chamber walls **216**. The dirt chamber walls **216** may be flexible and/or rigid as discussed elsewhere herein in accordance with an aspect of this disclosure. The exemplary dirt collection chamber **212** of FIG. 4 includes a rigid container. The exemplary rigid container of FIG. 4 is lined by a flexible walled bag **213**. The collection chamber **212** may include a flexible walled bag **213** lining part or all of the axial length of a rigid walled container. Alternatively, a flexible walled bag may be provided without the rigid walled container, as exemplified in FIGS. 6 and 7. A bag can be thrown away with the dirt collected in the bag to reduce user interaction with the dirt.

[0107] The dirt collection chamber **212** is openable to enable the bag **213** to be removed. The dirt collection container **212** may be openable (by opening a door as discussed previously with respect to chamber **184**. Alternately or in addition, the dirt collection container **212** may be moveably mounted with respect to the a portion of the surface cleaning apparatus, such as air treatment member **182** or chamber **184** between an in use position and an emptying position. In the in use position, the bag **213** receives dirt from the air flow path **150** during a cleaning operation and in an emptying position, the dirt collection chamber **212** has been moved such that at least a portion of the chamber is open for emptying. The exemplary dirt collection chamber **212** of FIG. 1 is located at the first end (lower end in the orientation of FIG. 1) of the first air treatment chamber in the in-use position, and is moveable axially (downwardly in the orientation of FIG. 1) to an emptying position as illustrated in FIG. 2. Any suitable emptying position may be used, such as a removed position as exemplified in FIG. 2 or a position in which the chamber **212** remains attached to the air treatment member with a door or end opened by movement of the dirt collection chamber **212**. For example, in the orientation of FIG. 1, the upper end of the dirt collection chamber **212** may be pivotally mounted to the air treatment chamber **184** or the dirt collection chamber **212** may be mounted on telescopic members so as to be lowered to open the upper end thereof while the dirt collection chamber **212** is still attached to the chamber **184**.

[0108] It will be appreciated that the air treatment assembly **180** may include any suitable number of dirt collection

regions **210**, such as a dirt collection region **210** for each air treatment chamber **184** if a plurality of air treatment chambers **184** are included, or for each air treatment stage if a plurality of air treatment stages are included. It will be appreciated that, in some examples, multiple air treatment chambers may share a dirt collection region (e.g., two cyclones each with a dirt outlet opening into a common dirt collection chamber, such as the exemplary assembly **180** of FIG. 4), and/or an air treatment chamber may be in communication with two or more discrete dirt collection regions by a plurality of dirt outlets. Accordingly, the air treatment assembly **180** and/or an air treatment chamber thereof may include separate collection regions for different types of dirt (e.g., fine dust, coarse debris, and/or liquid), and/or separate dirt outlets to a common dirt collection chamber for different types of dirt. The exemplary apparatus **100** of FIGS. 10-13 includes a separate dirt collection region **210** for each air treatment stage of the air treatment assembly **180**.

[0109] As exemplified FIGS. 2 and 30, the dirt collection region **210** is an openable region for the collection of dirt or liquid separated by the air treatment chamber **184**. The air treatment chamber **184** and the dirt collection region **210** or a chamber thereof may be accessed through a common selectively openable doorway or they may be separately openable, such that one may be opened while the other remains closed. The exemplary dirt collection region **210** of FIG. 2 is openable while the air treatment chamber **184** remains closed, and the exemplary air treatment chamber **184** is selectively openable (via the door **181**) when the dirt collection region **210** is open or closed. Optionally if the dirt collection chamber is separate from the air treatment chamber **184**, the air treatment chamber **184** is opened prior to opening or removing the dirt collection chamber **212** so that dirt collected in the air treatment chamber **184** has an opportunity to move into the dirt collection chamber **212** prior to the dirt collection chamber **212** being opened. However, it will be appreciated that they may be opened concurrently. Alternately, as exemplified by FIG. 30 wherein a dirt collection region is provided internal of an air treatment chamber **184**, the dirt collection region **210** may be openable concurrently with the air treatment chamber **184** by removal, e.g., of the lid **181**.

[0110] Referring again to FIG. 4, the surface cleaning apparatus **100** also includes an air moving member **220**. The air moving member **220** is positioned in the apparatus air flow path **150**. The air moving member **220** is provided to generate air flow (e.g., vacuum suction) through the air flow path **150**. The air moving member **220** may include a suction motor and fan assembly **222**. The suction motor and fan assembly **222** includes a motor and at least one fan. At least one of the fan(s) is positioned in the apparatus air flow path **150**. Any suitable assembly **222** may be used. For example, the suction motor and fan assembly **222** may be a bypass assembly, in which the motor is sealed apart from the air flow path, e.g., to reduce the risk of exposure to liquid carried in the air flow path.

[0111] The air moving member **220** may be contained within a moving member housing **230**. The moving member housing **230** may form part of the outer surface of the main body housing **132**, or may be internal thereto. The moving member housing **230** may be of any suitable construction, including any of those exemplified herein.

[0112] The air moving member **220** in the illustrated example is positioned downstream from the air treatment

assembly **180**, although it will be appreciated that the air moving member **220** may be positioned upstream of the air treatment assembly **180** or an air treatment member thereof (e.g., a dirty air motor) in alternative embodiments. Accordingly, the air moving member may be a clean or dirty air motor and fan assembly.

[0113] As exemplified, in some embodiments the air moving member **220** rotates about a moving member axis of rotation **232** (e.g., a suction motor axis of rotation). In some examples, when the apparatus upper end **116** is positioned above the apparatus lower end **118**, the moving member axis of rotation **232** is oriented generally vertically and extends between the apparatus lower end **118** and the apparatus upper end **116**. In other examples, however, the moving member axis of rotation **232** may extend at any angle to the vertical, or it may extend horizontally. Accordingly, the air moving member **220** may be oriented in any direction within the surface cleaning apparatus **100**. The moving member axis of rotation **232** may be generally parallel to the member vertical axis **122**. In some examples, the moving member axis **232** is generally coaxial with the axis of rotation of at least one cyclone.

[0114] In some examples, the surface cleaning apparatus **100** also includes one or more filters **240** in an air flow path of the surface cleaning apparatus **100**. A filter **240** may be one or more of a foam filter, felt filter, HEPA filter, other physical filter media, electrostatic filter, and the like. Optionally, a filter **240** includes a series of screens, and, optionally, each downstream screen of the filter has finer pores than the preceding upstream screen. The filter **240** may be formed from any suitable physical, porous filter media and may have any suitable shape, including the examples disclosed herein.

[0115] The surface cleaning apparatus **100** may include a pre-moving member (pre-motor) filter upstream of the moving member **220** (e.g., the exemplary filter **240** of FIG. 4) and/or a post-moving member (post-motor) filter downstream of the moving member. A pre-moving member filter may remove dirt (e.g., fine dust) that could otherwise damage the moving member **220**, such as by accumulating on fan blades or interfering with motor movement. A post-moving member filter removes dirt (e.g., carbon dust from the motor) that would otherwise be released by the surface cleaning apparatus **100**. It will be appreciated that the surface cleaning apparatus **100** may have any suitable number of filters **240**.

[0116] A filter **240** may be provided in a filter housing **246**. The filter housing **246** may be of any suitable construction, including any of those exemplified herein. The filter housing **246** may be openable or accessible to allow the filter **240** to be cleaned and/or replaced. As exemplified, the moving member axis of rotation **232** may intersect the volume defined by one or more filter housings **246**. In some embodiments, the moving member axis of rotation **232** intersects the volume defined by at least one filter housing **246** of a pre-moving member filter **242** received in the main air flow path **150**. A filter housing **246** may form part of the outer surface of the main body housing **132**. A post-moving member filter may be located radially outwards of the moving member **220**, as exemplified by the post-moving member filter **240** illustrated in FIG. 4. A pre-moving member filter **240a** and/or a post-moving member filter **240b** may come in any suitable shape and be at any suitable location.

[0117] Referring now to FIG. 28, power may be supplied to the surface cleaning apparatus **100** (e.g., to components or elements such as the air moving member **220**) from an external source. As exemplified, surface cleaning apparatus **100** may include a power cord **250** that is connectable to household mains. Optionally, as exemplified, the power cord **250** may enter the main body housing **132** at a rear end **114** of the housing. However, it will be appreciated that the power cord may be provided at any location in the surface cleaning apparatus **100**. Referring now to FIG. 4, it will be appreciated that the power source may alternatively, or additionally, include one or more on-board energy storage member(s) **251** (e.g., a battery, a capacitor, optionally a plurality which may be provided in a pack such as a removeable pack).

[0118] An actuator is provided to turn the surface cleaning apparatus on and off. As exemplified in FIGS. 3 and 4, it will be appreciated that the surface cleaning apparatus **100** may include a user interface **178**. The user interface **178** may be part of a main control system **252** of the surface cleaning apparatus **100**. The main control system **252** controls operation of the air moving member. As exemplified, the user interface **178** may be single power on/off button. Alternately, it may be a touch screen interface and/or include a display screen. The main control system **252** may include a circuit joining the power on/off button to the air moving member **220** to control operation of the air moving member **220**. However, it will be appreciated that any suitable control system **252** may be used. For example, a control system may include a more complex user interface with multiple buttons, switches, and/or screens (e.g., one or more soft buttons provided on a touchscreen). As another example, the control system may include one or more onboard processors communicatively coupled to one or more on board data storage systems storing instructions, such as to respond to a user selection from between two or more operational modes (e.g., a carpet mode and a hard floor mode, which may determine a motor rotation speed and/or brush roll positioning) or to respond to sensor input from an onboard sensor (e.g., responding by changing the speed of rotation of the air moving member **220**).

[0119] As exemplified, the user interface **178** may be provided at the apparatus upper end **116**. Optionally, the user interface **178** is provided on a top surface of the main body housing **132**. The user interface may face upwardly (i.e., visible when looking down at the surface cleaning apparatus **100** when the apparatus is in use). A user interface at the apparatus upper end **116** is more readily accessible to a user than a user interface at the apparatus lower end **118**. As exemplified, the user interface may be provided adjacent the carry handle **134**, and optionally on the carry handle **134**. A user interface adjacent the carry handle may be readily accessible to a user that is already interacting with the carry handle. However, it will be appreciated that the user interface **178** may be provide at any position on the surface cleaning apparatus **100**.

[0120] Referring now to FIG. 24, the surface cleaning apparatus **100** may include one or more actuators **270**. Any moveable component of the surface cleaning apparatus **100** may be an operated device operated by an actuator. As exemplified in FIG. 24, an actuator **270** is drivingly connected to an operated device **272** (e.g., a bag positioning system, a door, a flap, etc. as discussed elsewhere herein) of the surface cleaning apparatus **100** to operate the operated

device **272**. The actuator **270** may be, e.g., mechanically drivingly connected (i.e., via a mechanical linkage), pneumatically drivingly connected (e.g., via an air flow path, such as an air flow path within a piston chamber pressurised by movement of a piston), magnetically drivingly connected or electromechanically connected, e.g., a solenoid.

[0121] An actuator may be an automatic actuator responsive to a sensed condition (e.g., a pressure of the air flow path **150** or a predetermined programmed condition such as a period of time elapsed since the air moving member was powered on), or the actuator may be a user-controlled actuator responsive to a user action (e.g., a button press, lever movement, accessory attachment or removal, or vacuum mode selection such a selection between a bag-in cleaning mode and a no-bag cleaning mode). The actuator **270** may be a powered actuator receiving power from a power source of the surface cleaning apparatus **100**. The actuator may be a condition-responsive actuator responsive directly to a change in a condition, such as a flexible diaphragm adjacent the air flow path **150** responding directly to pressure changes in the air flow path. The actuator **270** may be a manual actuator, such as moved by a lever or button that is operated by being slid or pushed by a user.

[0122] The actuator **270** may be communicatively coupled to the main control system **252** to send and/or receive information. The actuator may receive instructions and/or power from the main control system **252**. Alternatively, the actuator may be controlled by a discrete control system separate from the main control system **252**, such as a simple circuit. A simple circuit may not include a processor or a data storage device, such as a circuit with a toggle (e.g., a switch, slider, or button) that closes the circuit when activated and breaks the circuit when deactivated. For example, an actuator may itself close a circuit to actuate a solenoid (e.g., a conductive float raised by a water level).

[0123] The following is a discussion of a number of aspects, namely dirt outlets opening into dirt conduits, a dirt collection chamber below a uniflow cyclone, a screen between the air treatment chamber and a lower dirt collection chamber, a bag positioning system, a bag holding system, a bag holding system with an internal bag engaging member, a bag holding system with an external bag engaging member, a bag holding system with a roller, holding the bag using an adhesive, and a bag roll, which are disclosed herein. Each aspect may be used by itself or in combination with one or more of the other aspects disclosed herein.

#### Dirt Outlets Opening into Dirt Conduits

[0124] In accordance with this aspect of the disclosure, a dirt collection region receives separated material, e.g., dirt, from two or more air treatment stages. Each treatment stage has one or more air treatment chambers, each of which has one or more dirt outlets. The dirt outlets of each treatment stage deliver dirt to a dirt collection region, e.g., via one or more dirt conduits.

[0125] An advantage of this design is that only a single dirt collection region may need to be emptied and, in accordance with another aspect hereof, the dirt collection region may have a bag **213** that collects the dirt from some or all of the dirt outlets and is removeable for emptying.

[0126] Accordingly, the air treatment assembly may include two air treatment stages each comprising an air treatment chamber **184** wherein each air treatment chamber **184** has one or more dirt outlets. The dirt outlets may deliver

separated dirt from the air treatment chambers **184** to the dirt collection region **210** via one or more dirt conduits. For example, each dirt outlet may communicate with (e.g., open into) an inlet end of its own dirt conduit and the outlet end of each dirt outlet may communicate with (e.g., open into) the dirt collection region **210**. Accordingly, a dirt outlet of a second stage air treatment chamber may be connected to an inlet of a second stage dirt conduit and a dirt outlet of a first stage air treatment chamber may be connected to an inlet end of a first stage the dirt conduit. Alternately a dirt outlet of each air treatment stage may use a common dirt conduit. Accordingly, a dirt outlet of a second stage air treatment chamber may be connected to an inlet of a dirt conduit and a dirt outlet of a first stage air treatment chamber may be connected to the dirt conduit at a location downstream of the location at which the dirt outlet of the second stage air treatment chamber connects to the dirt conduit and the dirt collection region. Alternately, the first stage air treatment chamber may have a dirt outlet that opens into the dirt collection region and the second stage dirt outlet may open into an inlet end of a dirt conduit that extends to the dirt collection region (see, e.g., FIG. 4).

[0127] As exemplified in FIG. 4, and as discussed subsequently, the dirt collection chamber **212** is remote from the second stage air treatment chamber **184** with the first stage air treatment chamber **184** received between the second stage air treatment chamber **184** and the collection chamber **212**, and the dirt conduit extends along the first stage air treatment chamber **184**.

[0128] It will be appreciated that the dirt outlets and/or dirt conduits may be angularly spaced from one another around a longitudinal axis (e.g., a cyclone axis of rotation), leaving a space therebetween for an additional component, e.g., a spine of an upright vacuum cleaner, part of an air treatment chamber **184**, etc. For example, the surface cleaning apparatus may be an upright vacuum cleaner comprising a surface cleaning head, an upright section rotatable mounted to the surface cleaning head between in use and storage positions and a portable cleaning unit that is removably attached to the upright. Part of the upright section may be received between the dirt conduits, which may improve the stability of the connection between the apparatus **100** and the base.

[0129] An air treatment member including a dirt outlet upstream of a dirt conduit may be part of any cleaning stage of the apparatus **100**, such as the first and/or second air treatment stages. Referring to FIGS. 14-19, the second stage air treatment member **182b** includes a dirt outlet **214** upstream of a dirt conduit **221**. The dirt conduit **221** extends between the dirt outlet **214** and the dirt collection region **210**. The exemplary dirt conduit **221** of FIG. 17 extends from the dirt outlet **214** to the dirt collection region **210** to carry dirt ejected through the dirt outlet **214** to the dirt collection region **210**.

[0130] The air treatment member having a dirt outlet upstream of a dirt conduit may be near another nearby treatment member, optionally abutting the nearby treatment member. The nearby treatment member may be between the member having a dirt outlet upstream of a dirt conduit and the dirt collection region to which the dirt conduit leads. Accordingly, the dirt conduit **221** may extend along or adjacent the nearby treatment member. The exemplary second stage **182b** of FIGS. 14-19 is axially spaced above the first stage **182a**, and the dirt conduit **221** extends axially in

the same direction as the axis of rotation of the first stage cyclone. The exemplary conduit **221** extends axially along the treatment chamber **184** of the first member **182a**. As exemplified, the conduit **221** may be exterior to the nearby treatment member (first stage treatment member **182a**, as exemplified), e.g., so as not to interfere with air flow within the member.

[0131] It will be appreciated that if an air treatment stage has two dirt outlets, then the dirt outlets may be angularly spaced apart round the longitudinal axis of the air treatment stage (e.g., a cyclone axis of rotation).

[0132] The angularly spaced dirt conduits may be downstream from a common treatment chamber. Alternately or in addition, the angularly spaced dirt conduits may be downstream from different treatment chambers (e.g., two air treatment chambers of a single air treatment stage or a dirt conduit for a dirt outlet of air treatment chamber of one air treatment stage and a dirt conduit for a dirt outlet of air treatment chamber of another air treatment stage). Alternately, the angularly spaced dirt conduits may be downstream of a common dirt outlet (e.g., a dirt outlet opening into two or more diverging conduits or two or more diverging dirt conduit branches).

[0133] The exemplary second stage member **182b** of FIG. **16** includes two dirt outlets angularly spaced from one another. FIG. **16**, which is a perspective cross sectional view of the exemplary apparatus **100**, shows one of the two dirt outlets **214** of the second stage **182b**. The other half of the apparatus **100** is a mirror image of the illustrated half and has the second dirt outlet **214**. As exemplified, the dirt outlets **214** of the member **182b** of FIGS. **14-19** are angularly spaced apart about the cyclone axis **188** and each is upstream of a different dirt conduit **221**. FIGS. **18** and **19**, which is a horizontal sectional view through the first stage **182a**, show the two angularly spaced apart dirt conduits **214**.

[0134] In accordance with this aspect and as exemplified, the angular spacing of the dirt conduits **221** leaves space for another component between the dirt conduits **221**, namely a portion **223** of an upright section **108** that is optionally selectively positionable between the conduits **221**. As illustrated in FIG. **17**, the upwardly extending portion **223** of the upright section **108** is received between two dirt conduits **221** when the surface cleaning apparatus **100** is mounted to the upright portion **108**. The portion **223** of the upright section **108** may be a post (e.g., consisting of or comprising a rigid air flow conduit) extending generally parallel to the conduits **221** when received between the conduits **221**, as exemplified. Optionally, the portion **223** is received in a pocket **225** of the apparatus **100** (e.g., a pocket of the assembly **180**) for greater stability.

[0135] Optionally, one or more of the conduits may extend linearly. Accordingly, a dirt conduit **221** may extend along a conduit axis **227** (FIG. **18**). The conduit axis **227** may be generally parallel to the cyclone axis **188** of at least one treatment chamber.

[0136] The exemplary conduits **221** of FIGS. **18** and **19** are each downstream of a single dirt outlet **214**. A single dirt outlet per dirt conduit **221** may reduce re-entrainment of dirt. Although, it will be appreciated that more than one dirt outlet may open into a common dirt conduit **221** in some examples, e.g., a dirt outlet of a first stage member **182a** may open into the same dirt conduit **221** that is downstream of a second stage dirt outlet **214**.

[0137] As exemplified in FIGS. **18** and **19**, the first stage air treatment chamber **182a** has two dirt outlets **214**, each of which communicates with a dirt conduit **221a**. The dirt conduits **221a** are positioned beside and laterally outwardly of the dirt conduits **221** of the second stage air treatment chamber **182b**. It will be appreciated that the first stage may have a single dirt conduit **221a** or no dirt outlet and therefore no dirt conduit. The dirt conduits **221a** of the first stage air treatment chamber **182a** may be at any other locations, e.g., beside each other, both beside one of the second stage dirt conduits **221** or spaced from both second stage dirt conduits **221**.

[0138] The exemplary apparatus **100** of FIGS. **14-19** includes an openable door **181** to open the collection chamber **212** while the collection chamber **212** is coupled to the treatment assembly **180** in operation. The door **181** may be in a lower end of the collection chamber **212**. When the door **181** is opened, the dirt collection region **210** may extend out from the collection chamber **212**, e.g., into an extended collection area **210a** of the upright section **108** as exemplified in FIG. **17**. It will be appreciated that in this embodiment, a first stage dirt outlet may not be provided.

[0139] The exemplary apparatus **100** of FIG. **17** includes a flexible walled bag lining the collection chamber **212**, and the bag is extendable when the door **181** is moved from the closed position (FIG. **16**) to the open position (FIG. **17**) to enlarge the collection region **210**.

#### Dirt Collection Chamber Below a Uniflow Cyclone

[0140] In accordance with this aspect of the disclosure, a surface cleaning apparatus has a dirt collection chamber at the inlet end of a uniflow air treatment chamber (e.g., a uniflow cyclone), namely an air treatment chamber (e.g., a uniflow cyclone) wherein fluid enters at one end and exits at an axially opposed end.

[0141] When the apparatus **100** is in an in-use position to clean a floor, the first or inlet end is a lower end and the collection chamber is at an elevation lower than the cyclone (e.g., it underlies it as exemplified in FIG. **4**) such that dirt may travel into the collection chamber due to gravity. Arranging the dirt collection chamber at an end of a uniflow cyclone results in a compact width of the air treatment assembly.

[0142] Referring to FIG. **4**, the exemplary dirt collection chamber **212** is at the first end **190** of the air treatment chamber **184**. The exemplary air treatment chamber **184** of FIG. **4** is a uniflow cyclone chamber, with the air inlet **202** at the first end **190** and the air outlet **204** at the second end **192**.

[0143] The air treatment chamber **184** and the dirt collection chamber **212** meet at an interface **260**. In some examples, the interface **260** is an end-to-end interface where the air treatment chamber **184** and the dirt collection chamber **212** meet end-to-end. The end-to-end interface **260** extends across at least a portion of an end (e.g., the first end **190**) of the air treatment chamber **184** and at least a portion of an end of the dirt collection chamber **212**. As exemplified in FIG. **4**, the first stage air treatment member **182a** and the dirt collection chamber **212** have a common diameter and are axially aligned. Accordingly, the exemplary interface **260** of FIG. **4** extends across the full diameter of the first end **190** of the first stage air treatment member **182a** and a portion of the second end **217** of the dirt collection chamber **212**. A separation member **262** is arranged at the interface

**260** between the air treatment chamber **184** and the dirt collection chamber **212**. The separation member **262** separates the air treatment chamber **184** from the dirt collection chamber **212**.

[0144] In some examples, the separation member **262** is a moveable separation member. A moveable separation member may selectively open the cyclone chamber **184**. The moveable separation member **262** is moveable between a closed position (e.g., FIG. 4) and an open position (e.g., FIG. 5). In the closed position the separation member **262** is positioned at an interface of the dirt collection chamber and the air treatment chamber to separate the chambers. In the open position the separation member **262** is moved away from the interface. As exemplified, the separation member **262** may be moveable while the chambers **212**, **184** remain abutting. In the open position, the chambers **212**, **184** are opened to one another for transfer of dirt between them (e.g., dropping from the air treatment chamber into the dirt collection chamber).

[0145] The separation member **262** may be moveable in any suitable way. In some examples, the separation member **262** is removable (i.e., detachable from one or both of the chambers). In some examples, the separation member is movably connected to one of the dirt collection chamber and the air treatment member (e.g., via a hinge). The exemplary apparatus **100** of FIGS. 4 and 5 includes a hinge **264** coupling the separation member **262** to the air treatment member **182** such that the separation member **262** may pivot open.

[0146] The exemplary moveable separation member **262** of FIG. 4 is moveably connected to the air treatment chamber **184**. A separation member **262** connected to the air treatment chamber **184** stays with the assembly **180** when the dirt collection chamber **212** is removed from the apparatus **100**. This may leave the dirt collection chamber **212** unencumbered by the separation member **262** when the user removes the collection chamber **212** for dumping. As exemplified also in FIG. 2, the collection chamber **212** is an open-topped container when removed from the assembly **180**, leaving the user free to dump the collection chamber **212** without opening any further walls or doors.

[0147] The exemplary separation member **262** of FIGS. 4 and 5 is moved away from the interface **262** when in the open position (i.e., because the hinge **264** is at the rear side of the collection chamber **212** laterally of the sidewall of the treatment chamber **184**). It will be appreciated that in some examples the separation member **262** or a portion thereof (e.g., an attached end) may remain at the interface when in the open position. For example, the hinge **264** of the separation member **262** of FIG. 11 is at the interface **260** (it underlies part of the volume of the air treatment chamber **184**) and remains at the interface when the opposite end is moved out in the open position.

[0148] Referring to FIG. 4, when the separation member **262** is in a closed position, the exemplary dirt outlet **214** of the air treatment chamber **184** is formed between the separation member **262** and a wall of the air treatment chamber **184**. The exemplary dirt outlet **214** of FIG. 4 is formed between the exemplary cylindrical sidewall **198** and the separation member **262** when the separation member **262** is in the closed position. When the separation member **262** is opened, the dirt outlet **214** is opened such that bridging debris (e.g., large debris such as a hair ball) that may be trapped in the outlet **214** is released.

[0149] The separation member **262** may be any suitable member for separating the air treatment chamber and the dirt collection chamber. In some embodiments, the separation member **262** is a plate. The exemplary separation member **262** of FIGS. 4 and 5 is a generally planar plate. The plate **262** of FIGS. 4 and 5 extends generally perpendicular to the cyclone axis **188** and, optionally as exemplified, the separation member **262** may be air impermeable.

[0150] Alternatively, as also discussed subsequently, part or all of the separation member **262** may be porous. In some examples, a porous separation member **262** allows liquids (e.g., water) to pass through the separation member to exit the treatment chamber **184**. This may improve the separation of water. In some examples, a porous separation member **262** includes a screen. The screen prevents the passage of large debris while allowing liquid through. In some examples, the porous separation member **262** may be the dirt outlet **214**, without a dirt outlet formed by a spacing between the separation member **262** and the sidewall of the chamber **184**. Dirt that does not fit through the screen may collect in a collection region **210** that is within the chamber **184**, e.g., to be dumped from the chamber **184** when the separation member **262** is opened.

[0151] It will be appreciated that, optionally, the lower end of the chamber may be open (i.e., there may be no separation member **262**), which enables for the removal during operation of the apparatus **100** of large debris that would not fit through the screen.

[0152] As discussed previously and as exemplified in FIG. 4, the surface cleaning apparatus **100** may include a first air treatment stage (e.g., member **182a**) and a second air treatment stage (e.g., member **182b**) downstream from the first air treatment stage, wherein the first air treatment stage comprises the first stage air treatment chamber and the second air treatment stage overlies the first air treatment stage. Optionally, the central axes of the first and second stages are coaxial, as exemplified by axes **188** of the chamber **184** of the first and second members **182a**, **182b** of FIG. 4.

#### Screen Between the Air Treatment Chamber and a Lower Dirt Collection Chamber

[0153] In accordance with this aspect of the disclosure, a surface cleaning apparatus **100** has a porous member between the air treatment chamber and the dirt collection chamber. Accordingly, the dirt outlet from an air treatment chamber may comprise or consist of openings in the porous member.

[0154] An advantage of this aspect is that a liquid (e.g., water) may be collected in a dirt collection container and solid material may be collected on top of the porous member thereby enabling a single chamber **184** to separate liquid and solids and enable the solids and liquid to be separately emptied. Liquids such as water can pass through the porous member into the dirt collection chamber, while larger debris is blocked by the porous member. The porous member may allow for separate handling of liquids and larger debris. The liquids may pass through the porous member into a first collection region and the larger debris may be directed into a second, separate collection region (e.g., a separate region of a dirt collection chamber or kept in a large debris collection region within the air treatment chamber).

[0155] It will be appreciated this aspect allows differential collection of solids and liquids and enables differential



handling of solids and liquids. Accordingly, the solid and liquids may be collected in different regions so as to be emptyable concurrently or separately.

[0156] It will be appreciated that the porous member may be positioned at the interface between an air treatment chamber (which may be a uniflow air treatment chamber having a dirt collection chamber positioned there below as discussed previously) and the dirt collection chamber. The separation member 262 comprise or consist of the porous member. Alternatively, the porous member may be a separate member discrete from the separation member. The porous member may also be used to subdivide a single dirt collection container into a liquid collection region and a solid collection region.

[0157] The porous member may be a porous plate having one or more apertures formed therethrough and sized to allow the passage of liquids but block the passage of larger particulate debris, such as a pore size of less than 1 cm, less than 0.5 cm, or less than 1 mm. Optionally, the porous plate comprises or consists of a screen 265. The exemplary separation member 262 of FIG. 4A comprises a screen 265 mounted in an annular frame and extending across at least a portion of the extent of the separation member 262.

[0158] As exemplified in FIGS. 4 and 4A, the separation member 262, which is positioned at the interface between an air treatment chamber and the dirt collection chamber, may be a porous member. Alternatively, or in addition, the porous member may be discrete from the separation member 262 (see e.g., FIGS. 8 and 9). The separation member 262 may be non-porous, such as a generally continuous plate as discussed previously.

[0159] It will be appreciated that, as exemplified in FIGS. 4 and 4A, a chamber 184 may have a separation member 262 that is a porous member 263 as well as a dirt outlet 214. This embodiment enables liquid to pass through the porous member into the dirt collection chamber and also enables particulate matter, which may be separated by air flow in chamber 184, to travel to the same dirt collection chamber as the liquid or a different dirt collection container.

[0160] It will also be appreciated that, if a porous member 262 and a dirt outlet 214 are provided, they need not be at the same end of an air treatment chamber and the air treatment chamber need not be a uniflow chamber. For example, the lower end of the chamber 184 may have a porous member 263 (e.g., see FIG. 4) and the dirt outlet may be at an upper end of the chamber (see, e.g., the second stage chamber of FIG. 4).

[0161] As discussed previously with respect to separation member 262, a separation member which comprises or consists of a porous member 263 may overlie part or all of the dirt collection chamber. In some examples, the porous member 263 (e.g., the screen) overlies all of the dirt collection chamber 212.

[0162] Alternately or in addition to overlying part or all of the dirt collection chamber 212, the porous member 262 may subdivide a dirt collection chamber. For example, the forward portion of the exemplary dirt collection chamber of FIG. 4A may be separated from the rearward portion (e.g., by a dividing wall 267 that may extend generally vertically) to form a solids collection area 210b and a liquids collection area 210c. Alternately, as exemplified in FIGS. 8 and 9, a horizontal porous member 263 separates a solids collection area 210b from a liquids collection area 210c. The screen of the porous member 263 of FIGS. 8 and 9 is positioned

further from the second end 192 of the chamber 184 than the separation member 262. The collection area 210b is above the collection area 210c, e.g., to collect a heavier dirt (e.g., liquid) at a lower elevation than a lighter dirt (e.g., dust).

[0163] As discussed previously with respect to separation member 262, it will be appreciated that the porous member 263 maybe moveable between a closed position (FIG. 8) in which the porous member 263 is positioned at the interface 260 of the dirt collection chamber 212 and the first air treatment member 182a and an open position (FIG. 9) in which the porous member 263 is moved away from the interface 260. As exemplified, the porous member 263 may be moveably connected to the first air treatment member 182a. It will be appreciated that in some examples, the porous member 263 may be removeable from the member 182 (e.g., fully detached). In the open position, at least a portion of the porous member 263 is removed from the interface 260 (i.e., the end opposite the exemplified hinge 264 is swung downwardly). In some examples, as exemplified in FIG. 9, the porous member 263 is openable without opening the collection region 210 to dump materials past the porous member 263 without opening the collection region 210.

[0164] In some examples, the porous member 263 is moveable with the separation member 262. Referring to FIGS. 8 and 9, the exemplary member 263 is rigidly coupled to the separation member 262 to move as one about the hinge 264.

[0165] Accordingly, it will be appreciated that an air treatment assembly may comprise a single stage air treatment chamber (which may be a cyclone) wherein solid material is separated from a fluid flow through a dirt outlet 214 and liquid is separated through a porous member 263. In such a case, the screen may be provided, as exemplified, in a lower portion of the air treatment chamber and the dirt outlet may be at upper end or lower end of the air treatment chamber. The liquid and the dirt may be collected in the same dirt collection chamber or in separate dirt collection chambers that are individually removeable and/or emptyable.

[0166] Alternately, it will be appreciated that an air treatment assembly may comprise a single stage air treatment chamber (which may be a cyclone) wherein solid material is retained in the air treatment chamber and liquid is separated through a porous member 263. In such a case, the screen may be provided, as exemplified, in a lower portion of the air treatment chamber and dirt may accumulate at the lower end of the air treatment chamber. The liquid may be collected in a dirt collection chamber this is individually removeable and/or emptyable from the air treatment chamber.

[0167] Alternately, it will be appreciated that an air treatment assembly may comprise two air treatment stages wherein the first air treatment stage comprises a first stage air treatment chamber (which may be a cyclone) and the second stage comprises a second stage air treatment chamber (which may be a cyclone). In the first stage solid material may be separated from a fluid flow through a dirt outlet 214 or a dirt outlet may not be provided and dirt may be collected on the lower end of the first stage air treatment chamber. In any event, liquid is separated through a porous member 263. In the second stage, solid material may be separated from the fluid flow exiting the first treatment stage through a dirt outlet 214. As discussed previously, a second stage dirt

conduit may be used to convey dirt separated by the second air treatment stage to a dirt collection chamber. Similarly, a first stage dirt conduit may be used to convey dirt separated by the first air treatment stage to the same or a different dirt collection chamber or the dirt outlet of the first air stage air treatment chamber may open directly therein. In any such case, the screen may be provided, as exemplified, in a lower portion of the air treatment chamber and any dirt outlet may be at an upper end or lower end of its air treatment chamber. The liquid and the dirt may be collected in the same dirt collection chamber or in separate dirt collection chambers that are individually removeable and/or emptyable. For example, dirt separated by the first and second treatment stages may be collected in the same dirt collection chamber (which may be individually removeable and/or emptyable from a liquid collection region).

#### Bag Positioning System

[0168] In accordance with this aspect of the disclosure, a surface cleaning apparatus **100** having a bag positioning system to arrange the bag **213** within the dirt collection chamber **212** in an open position in which the bag is useable to collect material during operation of the surface cleaning apparatus. Accordingly, the system may position the bag in an arrangement in which it will be held during operation of the surface cleaning apparatus. During operation of the surface cleaning apparatus, the bag may be held in an open configuration by the same bag positioning system or by another system as described elsewhere herein.

[0169] The bag positioning system repositions the bag within the chamber, such as by moving a bottom of the bag down into a bottom of the chamber. The bag positioning system may open up the bag, e.g., opening up a bottom of the bag to line the bottom of the dirt collection chamber. The bag positioning system may be mechanical and/or pneumatic. A mechanical bag positioning system may comprise a mechanical member that is internal of the bag, e.g., arms selectively extendable down into the bag to open up the bag. A pneumatic bag positioning system may include a vacuum line opening from a bottom of the dirt collection chamber to draw the bag down into the chamber.

[0170] It will be appreciated that, a user may be required to arrange a rim of the bag to be secured in place during operation of the surface cleaning apparatus, such as on a rim of a dirt collection chamber **212**.

[0171] A mechanical bag engaging system may use one or more members that hold a portion of the bag in position in a dirt collection container **212** or the like by engagement with the inner surface of the bag. For example, if a plurality of members are positioned in the bag (e.g., see FIGS. 20-27), then the members may create an open internal frame that inhibit the bag closing during operation of the surface cleaning apparatus. Alternately or in addition, one or more of the members may position a portion of the bag against an inner surface of the dirt collection container whereby the portion of the bag is sandwiched between the member and the inner surface of the dirt collection container. It will be appreciated that even if a single internal member is provided, the member may sandwich the bag between the member and the inner surface of the dirt collection container.

[0172] It will be appreciated that any such member may be positionable between an extended (in use) position (see, e.g., FIG. 27) extending down into the container **212** and a retracted (bag removal) position (see, e.g., FIG. 25) removed

from the bottom of the container. For example, the members may be telescoping members or a drive mechanism may be provided to move the members between the in use and removal positions. Alternately, the members may be fixed in position, e.g., with respect to an air treatment chamber **184**, and positioned in the bag when a dirt collection container containing the bag is secured to the air treatment assembly. The mechanical members **292** may be secured to a member that is moveable away from the container, such as a removeable lid of the container.

[0173] The exemplary bag positioning system **290** of FIGS. 20-27 is a mechanical system and includes bag engaging members **292** (e.g., legs as exemplified) that engage an inner surface **215** of the bag **213**. The exemplary legs are generally straight cylindrical legs, although it will be appreciated that any suitable mechanical members may be used. The bag engaging members **292** are moveable (retractable) between an extended (in use) position (FIG. 27) extending down into the container **212** and a retracted (bag removal) position (FIG. 25) removed from the bottom of the container. When the bag engaging members are removed from the bottom of the container, a bag **213** can be arranged in the container **212** to be positioned by the bag engaging members.

[0174] The bag engaging members **292** may be mounted at any suitable location. Bag engaging members **292** which are internal to a bag in an extended position may be mounted adjacent an open end of the dirt collection chamber **212** and/or the bag **213**. The exemplary mechanical members **292** of FIGS. 20-27 are secured to a removeable frame **294**, and are moveable between an extended position (FIG. 27) and a retracted position (FIG. 25) by a drive system **296**.

[0175] The drive system **296** may be any suitable drive system comprising any suitable actuator **270**, such as a manually operated drive system with mechanical linkages between the mechanical members and a manual toggle (e.g., a lever), a pneumatic system with an air moving member, or an electro-mechanical drive system powered by an electric motor. The bag engaging members **292** may be biased to the retracted position or the extended position, or may be unbiased.

[0176] It will be appreciated that any number of mechanical members **292** may be used. In some examples, a single mechanical member **292** may be used, such as a leg with a wide foot that holds open the bottom of the bag **213**. However, more than one mechanical member **292** spaced apart may facilitate opening up the bottom of the bag **213** and/or allow for simpler mechanical members **292** (e.g., without a wide foot).

[0177] The exemplary bag positioning system **290** of FIGS. 28-33 is a pneumatic system. The pneumatic system includes a vacuum air flow path **300** on an exterior surface of the bag **213** to draw the bag **213** down into the container **212**. The vacuum line **300** includes an air inlet **302** opening into a bottom of the container **212** to draw the bag **213** down to the bottom of the container (see e.g., FIG. 32). The exemplary vacuum line **300** is selectively openable in fluid communication with the air moving member **220** (i.e., at the outlet end **304**) such that the air moving member is selectively operable to draw air through the vacuum line. Although, it will be appreciated that any suitable air moving member may be used, such as an additional air moving member discrete from the member used to move air through the air flow path **150**. When the vacuum line **300** is active,

the air flow in through the inlet end 302 draws the bag 213 down into the container 212. As exemplified in FIGS. 32 and 33, the vacuum line 300 draws the bag down to line the chamber 212.

[0178] In some examples, suction is applied until the bag 213 is in position. The air moving member may be turned off and/or the vacuum line 300 closed. In some examples, the inlet end 302 of the vacuum line 300 is closed once the bag is in position (e.g., if a portion of the line 300 is used for another purpose, such as to inflate the bag engaging member 292 as described elsewhere herein).

[0179] In some examples, once the bag 213 is positioned in the container 212 the bag 213 is held in that position. The bag may be held in position in any suitable way, such as any of the ways described herein.

#### Bag Holding System

[0180] In accordance with this aspect of the disclosure, a bag holding system of a surface cleaning apparatus. The bag holding system includes one or more bag securing members each comprising one or more bag engaging members which are moveable between a retracted bag insertion position in which a bag is positionable in the dirt collection chamber and an extended bag engaging position in which the bag is secured in the dirt collection chamber. The bag engaging members hold the bag in place to prevent the bag from being moved by, e.g., air flow through the surface cleaning apparatus drawings the sides of the bag inwardly.

[0181] The bag may be positionable in an air treatment chamber (e.g., lining a dirt collection region that is within an air treatment chamber) or a dirt collection chamber that is external to the air treatment chamber (e.g., lining a dirt collection chamber that is within an air treatment chamber).

[0182] As exemplified in FIGS. 28-34, a bag holding system 310 includes one or more bag securing members 298, each including one or more bag engaging members 292. The bag engaging member may be a mechanical member. The bag engaging member 292 may be an end of a rigid member (e.g., a leg as exemplified in FIGS. 27 and 35) or a flexible member (e.g., an inflatable member as exemplified in FIG. 34). A flexible member 292 is optionally elastic and deformed into the retracted and/or extended position.

[0183] The bag holding system may include a drive system 296 drivingly coupled to the bag securing member(s) 298 to move the bag engaging member(s) 292 between the retracted bag insertion position (e.g., FIG. 33) and the extended bag engaging (in use) position (e.g., FIG. 34). Any suitable drive system 296 or combination of drive systems may be used to move the bag engaging member(s) 292 between the retracted bag insertion position and the extended bag engaging position, such as a pneumatic system and/or a mechanical system. In some examples, the drive system 296 is a pneumatic system including an air flow path 312 and an air moving member (e.g., FIGS. 33 and 34). In some examples, the drive system is a mechanical system including a mechanical actuator 270 (e.g., a motor). A mechanical system may include a mechanical linkage 297 (e.g., FIGS. 35-38).

[0184] Referring to FIGS. 32-34, the exemplary drive system 296 is a pneumatic drive system coupled to a flexible bag engaging member 292. Any suitable pneumatic drive system may be used. The exemplary bag holding system 310 of FIGS. 32-34 includes a single, inflatable bag engaging member 292. The exemplary pneumatic drive system 296 of

FIGS. 32-34 moves the bag engaging member 292 by changing the air pressure in an interior 322 of the bag engaging member 292. The exemplary bag engaging member 292 of FIGS. 32 to 34 falls to a collapsed, retracted bag receiving position when the interior is not pressurized and extends radially outward when the interior 322 is pressurized (e.g., FIG. 34) to engage the bag 213. The air flow path 312 of the drive system is pressurized by an air moving member which drives the bag engaging member to the extended position.

[0185] The exemplary pneumatic drive system 296 of FIGS. 32 to 34 includes an air flow path 312 that is coupled to the pneumatic positioning system line 300, although it will be appreciated that the drive system 296 may be discrete from the positioning system line 300 in some examples. The actuator 270 of the exemplary pneumatic drive system 296 of FIGS. 32 to 34 is the main air moving member 220 of the air flow path 150. The exemplary pneumatic drive system 296 of FIGS. 32 to 34 inflates the bag engaging member 292 when the air flow path 150 is active. Optionally, the bag engaging member 292 is downstream of the motor in the air flow path 150 (i.e., the exhaust of the motor is used to inflate the bag engaging member). Although, it will be appreciated that the drive system 296 may be coupled to a discrete air moving member in some examples.

[0186] Alternatively, or additionally, in use, an air moving member may create a sub atmospheric pressure in the air flow path 312 of the drive system to drive the bag engaging member to the retracted position.

[0187] It will be appreciated that, if the pneumatic drive system 296 only moves the bag engaging member in one direction (e.g., from the retracted bag insertion position to the extended bag engaging position, or from the extended bag engaging position to the retracted bag insertion position), then the bag engaging member may be biased to the other position. For example, the bag engaging member 292 may be an elastic member that moves (e.g., is elastically deformed) to the bag engaging position when the interior is pressurized. When the pneumatic drive system 296 is deactivated (e.g., the motor 220 is deenergized), air may evacuate from the interior of the inflatable bag engaging member 292 and the inflatable bag engaging member 292 may return to its uninflated state, which may be the retracted bag insertion position.

[0188] In use, in some examples the air flow path 312 of the pneumatic drive system 310 of the bag engaging member 292 is active only when the air flow path 300 of the positioning system is active. Accordingly, energizing and deenergizing the motor 220 may consequently move the bag engaging member 292 between the bag engaging position and the bag insertion position.

[0189] It will be appreciated that a drive system 296 may drive a bag engaging member to engage engages an outer surface of the bag and drive a portion of the outer surface of the bag outwardly to hold the bag open. In such a case, in the bag engaging position, the bag is sandwiched between the bag engaging member and an inner surface of, e.g., a dirt collection container 212. The drive system 296 may be mechanical or pneumatic or electro-mechanical.

[0190] As exemplified in FIGS. 35-38, the exemplary bag engaging members 292 are moveable by a mechanical drive system between a first bag receiving position (FIG. 35) and a second bag engaging position (FIG. 36) by mechanical linkages 297 moved by an actuator 270 (e.g., a motor). Any

suitable mechanical, pneumatic or electro-mechanical drive system may be used. As exemplified, the bag securing member 298 includes a central cam 336 and linkages 297 in addition to the bag engaging members 292. The exemplary actuator 270 of FIGS. 35-38 rotationally drives the central cam 336, and the cam 336 has angled faces 338 for engagement with the linkages 297. Rotation of the cam 336 drives the linkages 297 outward to drive the bag engaging members 292 outward to engage the bag 213. The system 310 may be biased to the retracted bag insertion position. For example, bag engaging members 292 may each be in a fixed relationship with a respective linkage 297, and the linkages 297 may be biased to a retracted rest position, such as by an elastic member 332 coupling opposed linkages 297 and biasing the opposed linkages 297 together. In use, in the exemplary system 310 of FIGS. 35-38 as the cam is rotated the linkages 297 are driven radially outward and then allowed to return radially inward.

[0191] It will be appreciated that a drive system 296 may drive a bag engaging member to engage engages an outer surface of the bag and drive a portion of the outer surface of the bag inwardly to hold the bag open. In such a case, in the bag engaging position, the bag is sandwiched between the bag engaging member and an inner surface of, e.g., a dirt collection container 212. The drive system 296 may be mechanical or pneumatic or electro-mechanical.

[0192] As exemplified in FIGS. 50-51, the exemplary bag engaging members 292 are moveable inwardly by a pneumatic drive system between the first bag receiving position (FIG. 50) and the second bag engaging position (FIG. 51).

[0193] Referring now to FIGS. 50 and 51, the exemplary bag engaging member 292 is an elastic member. It will be appreciated that an elastic bag engaging member 292 may be deformed in any suitable way, such as by air pressure, an arm pushing the member 292, or a belt or cable. It will be appreciated that an elastic bag engaging member 292 may be in the retracted position at rest or in the extended position at rest or in a third position at rest, and may be deformed into the retracted position and/or the extended position by drive system 296. In some examples, the elastic member 292 is in the retracted position at rest, e.g., to allow the bag to be positioned without active actuation of the system 310. In some examples, the elastic member 292 is in the extended position at rest, e.g., to allow the bag to be held without active actuation of the system 310. The exemplary bag engaging member 292 of FIGS. 50 and 51 is a portion of a wall 216 (a sidewall 217 as exemplified) of the dirt collection chamber 212. The exemplary bag engaging member 292 of FIGS. 50 and 51 is an elastic member which is moved between the extended and retracted position by a mechanical drive system 296 comprising a belt 320 which is tightened to move the bag engaging member 292 into the extended position (e.g., deform the elastic member 292) and relaxed to allow the bag engaging member 292 to return to the retracted position (i.e., a rest position). It will be appreciated that, alternately, the elastic member may be an annular donut located inside the dirt collection container 212.

[0194] It will be appreciated that in other examples, other drive systems may be used. In some examples, a thermal drive system is used. For example, when activated, the drive system 296 may heat a member (e.g., a bimetallic strip) to change the shape of the member and secure the bag in place.

[0195] When actuated, the bag holding system 310 moves the bag engaging member(s) 292 between the retracted and

extended positions and/or vice versa. In some examples, when actuated the bag holding system 310 extends the bag engaging member(s) 292 to engage the bag 213. The bag holding system 310 may be actuated in any suitable way or combination of ways, such as manually and/or automatically. In some examples, the bag holding system is actuated by closing a door (e.g., a lid) of the dirt collection chamber 212. For example, the door may press a toggle (e.g., a pressure sensor) when closed to switch on the system 310 or the closing of the door may increase the pressure within the dirt collection chamber and a sensor may detect the change in pressure and activate the system 310 or a reed switch or the like may detect that the lid is closed and actuate the drive system. In some examples, the bag holding system is actuated by a bag lock actuation toggle, such as a button or switch at an exterior of the apparatus 100. The actuation toggle may be the same toggle used to turn on the apparatus 100 (e.g., a main power switch) or a dedicated bag lock toggle. In some examples, the bag holding system is actuated automatically during a cleaning operation. For example, the bag holding system 310 may be actuated when the surface cleaning apparatus is turned on, or the bag holding system 310 may be actuated sequentially after the bag positioning system positions the bag.

[0196] In some examples, in the retracted position and/or the extended position, the bag engaging member(s) 292 may hold the bag in place without the bag holding system 310 being operated, the bag holding system 310 is actuated only to move the bag engaging member(s) 292 into the position. For example, a mechanical drive system may move bag engaging member 292 into a position and then the drive system and/or bag engaging member 292 may lock in place to keep the bag engaging member in the position without requiring the drive system to remain actuated. Alternatively, or additionally, in some examples, the retracted position and/or the extended position of the bag engaging member(s) 292 is an unstable position and the bag holding system 310 is actuated periodically or continuously to hold the bag engaging member(s) 292 in the position. For example, an inflatable bag engaging member 292 may be periodically or continuously pressurized by the bag holding system 310 to remain sufficiently inflated for the bag engaging position.

[0197] A bag engaging member 292 may move in any suitable way between the retracted and extended positions. The exemplified movement of each bag engaging member 292 of FIGS. 20-27 is a linear movement path. However, it will be appreciated that any suitable path may be used, such as a path that is curved or includes changes in direction.

[0198] In some examples, the bag engaging member 292 does not move along a path, such as an inflatable member expanding or contracting rather than moving, although it will be appreciated that an inflatable member may both move and expand and contract.

[0199] If a bag holding system includes more than one member 292, the members 292 may move relative to one another in any suitable way. The exemplary members 292 of FIGS. 20-27 each move generally parallel to one another and are moved by a common drive system. However, it will be appreciated that in some examples, members 292 may move independently from one another (i.e., selective, such as allowing a user to move one at a time) and/or in different directions (e.g., a plurality of members each moving radially

outward or radially inwardly in different directions from a common axis, such as from the bottom of a leg) to engage the bag.

#### Bag Holding System with Internal Bag Engaging Member

[0200] In accordance with this aspect of the disclosure, a bag holding system of a surface cleaning apparatus in which the bag engaging member(s) are internal to the bag when in the bag engaging position.

[0201] In accordance with this aspect, bag engaging members extend into the bag from an open end thereof (e.g., they down into the bag) to prevent a portion of the bag, e.g., the bottom of the bag, from moving inwards and possibly upwardly towards the rim of the chamber it is in when the motor 220 is actuated. It will be appreciated that the bag engaging members may engage any portion or portions of the inner surface of the bag, optionally the sidewall when the bag is in an in use configuration.

[0202] As exemplified in FIGS. 20-27, the positioning system 290 also serves as a holding system 310. The bag engaging members 292 are each part of a bag securing member 298. The bag engaging members 292 are inside the bag 213 when the bag is lining the container 212. The members 292 hold the bag in position within the container 212. The members 292 remain in the extended position (FIG. 27) when the apparatus 100 is in operation to hold the bag 213 in position. It will be appreciated that alternatively an apparatus 100 may include a holding system 310 discrete from the positioning system 290.

[0203] The bag engaging members 292 may be fixed in position relative to a mounting location, e.g., a door or frame (e.g., a removeable lid or frame 294) that is moveable relative to the dirt collection chamber 212, and the members 292 are moveable between the retracted and extended positions by moving the door or other frame. For example, members 292 fixed to the frame 294 may be moveable to the retracted position by lifting the frame 294 away from the dirt collection chamber 212, and moveable to the engaging position by positioning the frame 294 on the dirt collection chamber 212.

[0204] Alternatively, or additionally, the bag engaging members 292 may be moveable relative to the mounting location. A bag engaging member 292 may be moved by any suitable drive system, such as any of the drive systems described elsewhere herein. The exemplary drive system 296 of FIG. 24 is a mechanical drive system with an actuator 270 (e.g., an electric rotary motor) mechanically coupled to the members 292 via mechanical linkages 297 to raise and lower the members 292 between a retracted position (FIG. 25) and an extended position (FIG. 27). The exemplary members 292 are moveable between the retracted position of FIG. 25 and the extended position of FIG. 27 by a combination of movement relative to the frame and movement of the frame relative to the dirt collection chamber 212. However, it will be appreciated that in some examples the bag engaging members 292 are moveable between the extended and retracted positions solely by movement relative to a mounting location (e.g., the frame 294) or solely by movement of the mounting location relative to the dirt collection chamber.

[0205] As exemplified, bag engaging member(s) 292, which are internal of the bag in the extended position, may move between the retracted and extended positions along a

path that includes at least a component (e.g., a vector component) that is generally parallel to an axis extending between an openable end of the dirt collection chamber and an opposite (e.g., bottom) end of the chamber 212. Optionally, the movement consists of movement that is generally parallel to an axis extending between an openable end of the dirt collection chamber and an opposite end of the chamber. The exemplified movement of the bag engaging members 292 of FIGS. 20-27 consists of movement that is generally parallel to an axis extending between an openable end of the dirt collection chamber and an opposite end of the chamber 212 (e.g., the longitudinal axis 201 exemplified). However, it will be appreciated that the members 292 may also or alternatively move in a different direction, such as perpendicular to the axis 201. For example, the members 292 may move into the chamber 212 generally parallel to the longitudinal axis 201 and then move radially outward, e.g., to engage the sidewall of the bag and thereby open the bag 213, or the members 292 may move into the chamber 212 downwardly and outwardly at an angle to the longitudinal axis 201.

[0206] Optionally, an internal mechanical member may sandwich the bag 213 against another member, such as an external member. The exemplary members 292 of FIGS. 20-27 sandwich the wall of the bag 213 against the sidewall of the chamber 212. However, it will be appreciated that in some examples the internal mechanical members 292 hold the bottom of the bag open and/or down within the chamber 212 without pushing the bag against another member. For example, the exemplary members 292 may create a frame that holds the bottom of the bag open regardless of whether there is clearance under the open bottom of the bag and the bottom and/or sides of the dirt collection chamber 212.

[0207] As discussed previously, any drive system may be used.

#### Wiper

[0208] In accordance with this aspect of the disclosure, a wiper is used to clean a bag positioning member and/or a bag securing member. Accordingly, any mechanical member that is internal of the bag and may have particulate matter thereon after a cleaning operation may be cleaned by a wiper.

[0209] As exemplified in FIGS. 20-27, the apparatus 100 includes a wiper to clean a bag securing member as the bag engaging member(s) are moved to the retracted position. The exemplary apparatus 100 of FIGS. 20-27 includes a wiper 280. Any suitable wiper 280 may be used. The exemplary wiper 280 includes a close-fitting resilient member 282, e.g., an annular member which may abut the bag engaging member as the bag engaging member is drawn thereby (e.g., retracted) to wipe off the bag engaging member as it is retracted. The wiper 280 may be made of a cloth material, bristles, or the like.

#### Bag Holding System With an External Bag Engaging Member

[0210] In accordance with this aspect of the disclosure, a bag holding system of a surface cleaning apparatus in which the bag engaging members are external of the bag when in the bag engaging position. As discussed previously, the bag

engaging members may move inwardly (e.g., generally radially inwardly) or outwardly (e.g., generally radially outwardly).

[0211] An external member holding the bag 213 avoids obstructing the interior of the bag 213 and separates the external member from the dirt collected in the bag 213. A moveable bag engaging member outside the bag simplifies protecting the moveable components from contamination by dirt that is within the bag. As discussed previously, the member or members may sandwich a portion of the bag to hold it in position.

[0212] Accordingly, one or more members may move outwardly to sandwich the bag, such as to sandwich the bag against a wall of the dirt collection chamber. Alternately, one or more members move inwardly to sandwich the bag, such as to sandwich the bag against a centrally located block. When the bag is installed, the centrally located block may be located in the dirt collection chamber and the bag may extend there over when installed (e.g., a block that is external of the bag). Alternately, the centrally located block may be a block that is internal to the bag such as a foot on the end of a leg secured to a lid or other frame. Optionally, the system includes both at least one first member that moves inwardly to sandwich a bag and at least one second member that moves outwardly to sandwich the bag, such as a first member and a second member that move relative to each other to sandwich the bag therebetween. As discussed previously, the bag engaging members may sandwich a bottom portion of the bag once the bag is arranged in the chamber 212 to hold the bottom portion of the bag in position in the chamber 212 (i.e., in a bottom end of the chamber 212).

[0213] In some examples, movement of bag engaging member(s) 292 into the external extended position includes at least a component (e.g., a vector component) that is radial, e.g., from an axis of rotation of a cyclone chamber. Optionally, the movement into the external extended position consists of radial movement.

[0214] Referring to FIGS. 28 to 34, and particularly FIGS. 32 to 34, the exemplary apparatus 100 includes a bag holding system 310 with an external bag engaging member 292. The bag engaging member 292 is moveable between a retracted bag receiving position (FIG. 33) and an extended bag engaging position (FIG. 34). The exemplary bag engaging member(s) 292 move outward, towards a sidewall of a chamber in which the bag is receivable. In some examples, the chamber in which the bag is receivable is a combined air treatment and dirt collection chamber, as exemplified in FIGS. 33 and 34. In the first position (e.g., a retracted bag insertion position) of FIG. 33 the exemplary bag engaging member 292 is spaced from a sidewall 216 of the chamber 212 by a first spacing 314. In the second position (e.g., an extended bag engaging position) of FIG. 34, the bag engaging member 292 is spaced from the sidewall 216 of the chamber 212 by a second spacing 316 less than the first spacing. When the bag engaging member 292 is in the first position, the bag 213 is positionable between the bag engaging member 292 and the wall 216, and when the bag engaging member 292 is in the second position a portion of a bag 213 is sandwiched between the bag engaging member 292 and the wall 216.

[0215] Similarly, the exemplary bag engaging members 292 of FIGS. 35 to 38 are moveable radially. The exemplary bag engaging members 292 of FIGS. 35 to 38 move out-

wardly into the extended position (FIGS. 36 and 38) and inwardly into the retracted position (FIGS. 35 and 37).

[0216] It will be appreciated that the bag engaging members may also or alternatively move inwardly into the extended bag engaging position. Referring to FIGS. 50 and 51, the exemplary bag engaging member 292 moves radially inwardly into the extended position (FIG. 51) and outwardly into the retracted position (FIG. 50).

[0217] As discussed previously, any drive system may be used.

#### Bag Holding System With a Roller

[0218] In accordance with this aspect of the disclosure, a bag holding system of a surface cleaning apparatus in which the bag securing member 298 includes a roller bag engaging member 292. The roller bag engaging member 292 is rotatable to engage an outer surface of the bag and thereby draw a portion (e.g., a bottom) of the bag past the roller and thereby, e.g., draw a bag into the container.

[0219] The portion of the bag 213 drawn past the roller is secured in position between the roller and an adjacent member. The adjacent member may be a second roller whereby the bag securing member 298 includes a pair of rollers to draw the portion of the bag between the rollers. However, it will be appreciated that a single roller 292 may be used. For example, the adjacent member may be a non-moveable wall that the roller engages in operation to secure a bag in position.

[0220] Referring now to FIGS. 39-43, the exemplary bag holding system 310 includes a bag securing member 298 which comprises a pair of roller bag engaging members 292. A mechanical drive system 296 is coupled to the rollers 292 to rotationally drive the rollers. The rollers 292 rotate towards one another to draw a portion 324 of the bag 213 past the rollers 292 between the two rollers 292. In use, a user positions the portion 324 of the bag 213 at the rollers 292 (FIGS. 40-42), such as by feeding a bottom portion of the bag into a slot 326 in the floor of the dirt collection chamber 212. When the drive system 296 is activated, the rollers 292 grip and pull the portion 324 of the bag 213 between them. A user may reverse the direction of the rollers to release the bag.

[0221] It will be appreciated that the outer surface of the rollers may be made of a non-slip material, such as a synthetic or natural rubber. Alternately, the outer surface of the roller may be textured.

[0222] As exemplified, the roller engages an external surface of the bag. The exemplary rollers of FIGS. 39-43 are arranged below the bag 213 to engage an external surface of the bag. The external surface may be a sidewall located above the bottom of the bag or, as exemplified, the portion engaged by the roller(s) may be a flange that extends downwardly from the bag.

#### Holding the Bag Using an Adhesive

[0223] In accordance with this aspect of the disclosure, a bag 213 is held in position during a cleaning operation of the surface cleaning apparatus 100 by an adhesive alone or in combination with any other bag holding system.

[0224] The adhesive joins a wall of the bag 213 to a surface of the dirt collection chamber 212 during the cleaning operation. The adhesive may be provided at a single location or a plurality of discrete locations that may be on a

common side of the bag or different sides. For example, the adhesive may be provided on the bottom of the bag and/or a sidewall of the bag or on an internal wall of the container to engage the bottom of the bag and/or a sidewall of the bag.

[0225] The adhesive may be provided on one or more of a disposable bag 213, on the wall of the chamber 212, on a mechanical bag engaging member 292.

[0226] Optionally, a bag engaging member may be operable to separate the bag wall and the wall of the dirt collection chamber 212 after the cleaning operation by pushing the bag wall away from the chamber wall to break the adhesive bond.

[0227] As exemplified in FIG. 27, the bag 213 is joined to a wall 216 of the dirt collection chamber 212 by an adhesive layer 370 that, in use, engages the bottom of the bag. The adhesive layer 370 forms an adhesive bond between the wall of the chamber 212 and a wall of the bag. The exemplary adhesive 370 is shown with a holding system 310 that includes bag engaging members internal to the bag 213 in the extended position, however it will be appreciated that the adhesive layer 370 may be used without another holding system 310.

[0228] The bag 213 and chamber wall may be brought together in any suitable way. For example, the user may manually move the bag wall against the chamber wall. In some examples, the bag positioning system 290 moves the bag wall and chamber wall together to allow the adhesive 370 to bond the bag wall and the chamber wall together. Any suitable bag positioning system 290 may be used, such as a mechanical or pneumatic system.

[0229] In some examples, the apparatus 100 includes an adhesion release system 372. The adhesion release system 372 includes one or more bag engaging members 292 moveable between a retracted position and an extended position. In the extended position, the bag engaging member(s) 292 of the adhesion release system 372 push the bag wall away from the chamber wall to which the bag is secured to break the connection provided by the adhesive layer 370. Any suitable bag engaging member may be used, including any of those described elsewhere herein. The bag engaging member(s) 292 of the adhesion release system 372 may be moved between the extended position and the retracted position by any suitable drive system, including any of the drive systems described herein.

### Bag Roll

[0230] In accordance with this aspect of the disclosure, a surface cleaning apparatus 100 is operable with a bag roll. A user may draw a bag from the roll and secure the bag in position, e.g., by positioning the end of the bag on the rim of the container so that the upper end of the bag is secured in position when the container is closed.

[0231] In accordance with this aspect, the surface cleaning apparatus 100 may include a roll holder to receive a bag roll. The bag roll may be any suitable roll of material to be used in lining the dirt collection chamber 212 with a bag, such as a roll of preformed bags, a continuous tube to be formed into bags, or a roll of sheeting to be formed into bags.

[0232] In some examples, the apparatus 100 includes a heat sealing member arranged to engage a portion of the roll, e.g., to heat seal a portion of plastic sheeting to another portion of plastic sheeting to form the edge of a bag 213. A heat sealing member may close a single edge of a bag, e.g., closing the bottom of a bag by sealing across a continuous

tube of plastic. A heat sealing member may close multiple edges of a bag, e.g., first sealing two sheets of plastic together to form a tube or sealing one edge of a doubled plastic sheet to the opposite edge to form a tube and concurrently or subsequently sealing across the tube to close a bottom of the bag. Alternately, or in addition, after use, the filled bag may be sealed by a heat sealing member.

[0233] Referring to FIGS. 44-51, the exemplary surface cleaning apparatus 100 includes a roll holder 340 to receive a bag roll 342. The exemplary bag roll 342 is received in a dispensing housing 344. The exemplary dispensing housing 344 and bag roll 342 are parts of a removeable cartridge 346, although it will be appreciated that the bag roll 342 may be removeable from the housing 344. Optionally, the housing 344 may be a part of the holder 340.

[0234] In use, the bag roll 342 is received in the roll holder 340. As exemplified in FIGS. 45 and 46, the cartridge 346 containing the bag roll 342 is positioned in the roll holder 340. The roll holder 340 may hold the bag roll 342 in any suitable way, such as by a releasable fastener (e.g., threaded fastener or electromagnet), adhesive, or friction fit. The holder 340 may hold the bag roll 342 directly or as exemplified in FIGS. 45 and 46, hold the cartridge 346. The exemplary holder 340 includes releasable clips 350 to releasably hold the cartridge 346.

[0235] The bag roll 342 may be any suitable bag roll. The bag material (e.g., plastic sheeting) may be preformed into bags which are rolled into the bag roll 342. For example, the bag roll 342 may be a roll of a plurality of open-topped bags secured end-to-end and separated by a line of weakness (e.g., a perforation line) at a bottom end of a preceding bag and a top end of a subsequent bag.

[0236] Alternatively, the bag roll 342 may include one or more sheets of material (e.g., plastic) to be formed into bags by the surface cleaning apparatus 100. The exemplary apparatus 100 of FIG. 50 includes a sealing member 352. The sealing member engages two sheets of bag material to join the two sheets together to form an edge of a bag. The sheets may be joined in any suitable way, such as by welding the sheets together or applying an adhesive between the sheets. In some examples, the bag material is a plastic material and the sealing member 352 is a heat sealing member. In use, the heat sealing member 352 welds the plastic sheets together to form the edge of a bag 213.

[0237] The edge formed by the sealing member 352 may be formed by joining two discrete sheets of bag material or by joining portions of a single sheet of bag material to one another. For example, referring to FIG. 50, the sealing member 352 may close the bottom 354 of an end 356 of plastic tube 358 to form a bag 213. Optionally, the heat sealing member 352 may also form the plastic tube 358, e.g., by joining lateral edges of a single sheet of bag material to one another to form a tube or by joining two or more sheets of bag material to one another to form a tube.

[0238] In use, a user may manually unroll an end of the bag roll 342 and raises a rim 218 of the bag material to the rim 219 of the dirt collection chamber 212, such as exemplified in FIGS. 47 and 48 (e.g., the user drapes the rim 218 over the rim 219). The bag material may be manually positioned in the chamber 212 or positioned by a positioning system 290. Once the bag material is positioned in the chamber, the sealing member 352 may form the bottom of the bag by sealing the end closed along edge 360, if a series of bags separated by a perforation line is not used. It will be

appreciated that the bottom of the bag 213 may be closed before a cleaning operation to prevent dirt from passing the edge, although in some examples the bottom of the bag 213 may be closed after a cleaning operation and before the user removes the bag 213.

[0239] During a cleaning operation of the apparatus 100, the bag material may be held in position by the roll holder 340 (e.g., in a cartridge held by the roll holder 340 or directly by the roll holder 340). Alternatively, or additionally, a separate bag holding system 310 is used, as exemplified in FIGS. 50-51. Any bag holding system discussed herein may be used.

[0240] Subsequent bags are installed by removing a used bag. Lifting the used bag up pulls a second bag upwardly and the user may then drape the rim of the second bag over the rim of the bin.

[0241] As used herein, the wording “and/or” is intended to represent an inclusive—or. That is, “X and/or Y” is intended to mean X or Y or both, for example. As a further example, “X, Y, and/or Z” is intended to mean X or Y or Z or any combination thereof.

[0242] While the above description describes features of example embodiments, it will be appreciated that some features and/or functions of the described embodiments are susceptible to modification without departing from the spirit and principles of operation of the described embodiments. For example, the various characteristics which are described by means of the represented embodiments or examples may be selectively combined with each other. Accordingly, what has been described above is intended to be illustrative of the claimed concept and non-limiting. It will be understood by persons skilled in the art that other variants and modifications may be made without departing from the scope of the invention as defined in the claims appended hereto. The scope of the claims should not be limited by the preferred embodiments and examples, but should be given the broadest interpretation consistent with the description as a whole.

#### Clause Set A

- [0243] 1. A surface cleaning apparatus comprising:
- [0244] (a) an air flow path extending from a dirty air inlet to a clean air outlet with a motor and fan assembly in the air flow path;
- [0245] (b) a first air treatment chamber having a first air treatment chamber air inlet, a first air treatment chamber air outlet and a dirt outlet, a first end and an axially opposed second end;
- [0246] (c) a dirt collection chamber that is moveably mounted from an in use position in which the dirt collection chamber is located at the first end of the first air treatment chamber to an emptying position; and,
- [0247] (d) a porous member positioned at an interface of the dirt collection chamber and the first air treatment chamber, the dirt outlet comprises openings in the porous member,

wherein a bag is positionable in the dirt collection chamber and is removeable from the dirt collection chamber when the dirt collection chamber is in the emptying position.

- [0248] 2. The surface cleaning apparatus of clause 1 wherein the porous member is moveably mounted between a closed position in which the porous member is positioned at the interface of the dirt collection

chamber and the first air treatment chamber and an open position in which the porous member is moved away from the interface.

- [0249] 3. The surface cleaning apparatus of clause 2 wherein the porous member is moveably connected to the first air treatment chamber
- [0250] 4. The surface cleaning apparatus of clause 1 wherein the first air treatment chamber air inlet is provided at the first end.
- [0251] 5. The surface cleaning apparatus of clause 4 wherein the first air treatment chamber air outlet is provided at the axially opposed second end.
- [0252] 6. The surface cleaning apparatus of clause 1 wherein the first air treatment chamber air outlet is provided at the axially opposed second end.
- [0253] 7. The surface cleaning apparatus of clause 1 wherein the porous member comprises a screen.
- [0254] 8. The surface cleaning apparatus of clause 1 wherein the first air treatment chamber air inlet is provided at the first end and the porous member is positioned further from the second end of the first air treatment chamber than the first air treatment chamber air inlet.
- [0255] 9. The surface cleaning apparatus of clause 1 wherein the screen porous member overlies all of the dirt collection chamber.
- [0256] 10. The surface cleaning apparatus of clause 7 wherein the screen overlies all of the dirt collection chamber.
- [0257] 11. The surface cleaning apparatus of clause 1 wherein the first air treatment chamber comprises a cyclone chamber.
- [0258] 12. The surface cleaning apparatus of clause 1 further comprising a second air treatment chamber downstream from the first air treatment chamber, the second air treatment chamber having a second dirt outlet wherein dirt separated by the second air treatment chamber is collectable in the dirt collection chamber.
- [0259] 13. The surface cleaning apparatus of clause 1 further comprising a second air treatment chamber downstream from the first air treatment chamber, the second air treatment chamber having at least two second air treatment chamber dirt outlets, each second air treatment chamber dirt outlet is upstream from a dirt conduit that extends axially along the first air treatment chamber.
- [0260] 14. The surface cleaning apparatus of clause 13 wherein the dirt conduits are exterior to the first air treatment chamber.
- [0261] 15. The surface cleaning apparatus of clause 14 wherein the dirt conduits are angularly spaced apart.
- [0262] 16. The surface cleaning apparatus of clause 1 wherein the surface cleaning apparatus comprises a first air treatment stage and a second air treatment stage downstream from the first air treatment stage, the first air treatment stage comprising the first stage air treatment chamber and the dirt collection chamber and the second air treatment stage overlies the first air treatment stage.
- [0263] 17. The surface cleaning apparatus of clause 16 wherein each of the first and second air treatment stages have a central longitudinal axis and the central longitudinal axes are generally coaxial.



[0264] 18. The surface cleaning apparatus of clause 1 wherein a mechanical member positioned internal of the bag when the bag is positioned in the dirt collection chamber secures the bag in position in the dirt collection chamber.

[0265] 19. The surface cleaning apparatus of clause 1 wherein a mechanical member positioned external of the bag when the bag is positioned in the dirt collection chamber secures the bag in position in the dirt collection chamber.

[0266] 20. The surface cleaning apparatus of clause 1 wherein releasable adhesive secures the bag in position in the dirt collection chamber.

#### Clause Set B

[0267] 1. A surface cleaning apparatus comprising:

[0268] (a) an air flow path extending from a dirty air inlet to a clean air outlet with a motor and fan assembly in the air flow path;

[0269] (b) a first air treatment stage positioned in the air flow path, the first air treatment stage comprising a dirt collection chamber that is moveably mounted from an in use position to an emptying position; and,

[0270] (c) a bag securing member comprising bag engaging members that are moveable from a retracted bag insertion position in which a bag is positionable in the dirt collection chamber and an extended bag engaging position in which the bag engaging members are positioned external of the bag and the bag is secured in the dirt collection chamber,

wherein the bag is removeable from the dirt collection chamber when the dirt collection chamber is in the emptying position.

[0271] 2. The surface cleaning apparatus of clause 1 wherein the bag engaging members are moveable outwardly from the retracted bag insertion position to the extended bag engaging position.

[0272] 3. The surface cleaning apparatus of clause 2 wherein, when the bag engaging members are in the extended bag engaging position, the bag is positioned between the bag engaging members and a wall positioned outwardly of the bag engaging members.

[0273] 4. The surface cleaning apparatus of clause 3 wherein the wall is a wall of the dirt collection chamber.

[0274] 5. The surface cleaning apparatus of clause 1 wherein the dirt collection chamber has an open end and a longitudinally spaced apart opposed closed end and the bag engaging members are moveable outwardly in a plane transverse to a longitudinal axis.

[0275] 6. The surface cleaning apparatus of clause 1 wherein the bag engaging members are moveable inwardly from the retracted bag insertion position to the extended bag engaging position.

[0276] 7. The surface cleaning apparatus of clause 6 wherein, when the bag engaging members are in the extended bag engaging position, the bag is positioned between the bag engaging members and a wall positioned inwardly of the bag engaging members.

[0277] 8. The surface cleaning apparatus of clause 1 wherein the dirt collection chamber has an open end and an opposed closed end and the bag engaging members are provided at the opposed closed end.

[0278] 9. The surface cleaning apparatus of clause 1 wherein the bag engaging members engage an outer surface of the bag.

[0279] 10. The surface cleaning apparatus of clause 1 wherein the dirt collection chamber has an open end and a longitudinally spaced apart opposed closed end, the bag engaging members form a recess exterior to the bag and at a lower end of the bag, and longitudinally extending surfaces of the bag engaging members engage longitudinally extending portions of the bag when the bag engaging members are in the extended bag engaging position.

[0280] 11. The surface cleaning apparatus of clause 1 wherein the bag securing member is inflatable.

[0281] 12. The surface cleaning apparatus of clause 1 further comprising a bag locking actuator that is drivingly connected to the bag engaging members.

[0282] 13. The surface cleaning apparatus of clause 12 wherein the bag locking actuator is actuated when the dirt collection chamber is moved to an in use position whereupon the bag engaging members are moved to the extended bag engaging position.

[0283] 14. The surface cleaning apparatus of clause 13 wherein the bag engaging members are biased to the retracted bag insertion position.

[0284] 15. The surface cleaning apparatus of clause 12 wherein the bag locking actuator is manually actuatable by a user.

[0285] 16. The surface cleaning apparatus of clause 12 wherein the bag locking actuator is pneumatically actuatable.

[0286] 17. The surface cleaning apparatus of clause 16 wherein, when the motor and fan assembly is actuated, the motor and fan assembly pressurizes a pneumatic line which drives the bag engaging members to the extended bag engaging position.

[0287] 18. The surface cleaning apparatus of clause 17 wherein the bag engaging members are biased to the retracted bag insertion position.

[0288] 19. The surface cleaning apparatus of clause 16 wherein, when the motor and fan assembly is actuated, the motor and fan assembly creates a sub atmospheric pressure in a pneumatic line which drives the bag engaging members to the extended bag engaging position.

[0289] 20. The surface cleaning apparatus of clause 1 wherein the bag engaging members are biased to the extended bag engaging position.

[0290] 21. The surface cleaning apparatus of clause 1 wherein the bag engaging members are biased to the retracted bag insertion position.

#### Clause Set C

[0291] 1. A surface cleaning apparatus comprising:

[0292] (a) an air flow path extending from a dirty air inlet to a clean air outlet with a motor and fan assembly in the air flow path;

[0293] (b) an air treatment stage positioned in the air flow path, the air treatment stage comprising a dirt collection chamber that is moveably mounted from an in use position to an emptying position; and,

[0294] (c) a bag roll holder,

wherein the dirt collection chamber has an open end and a longitudinally spaced apart opposed second end, the closed

end has an opening whereby, when a roll of bags is mounted to the bag roll holder, one of the bags is insertable through the opening into the interior of the dirt collection chamber.

[0295] 2. The surface cleaning apparatus of clause 1 wherein the bag roll holder is provided at the lower end of the dirt collection chamber.

[0296] 3. The surface cleaning apparatus of clause 1 wherein the open end of the dirt collection chamber has a rim and, when the dirt collection chamber is in the in use position with an upper end of the bag positioned overlying the rim, the upper end of the bag is secured in position between the dirt collection chamber and an air treatment chamber of the air treatment stage.

[0297] 4. The surface cleaning apparatus of clause 1 wherein the air treatment stage comprises an air treatment chamber and a plate, the air treatment chamber having a first end having a first dirt outlet and an axially opposed second end, the plate is moveably mounted between a closed position in which the plate is positioned at an interface of the dirt collection chamber and the air treatment chamber and an open position in which the plate is moved away from the interface.

[0298] 5. The surface cleaning apparatus of clause 4 wherein the plate is moveably connected to the air treatment chamber, wherein, when the plate is in the closed position, the dirt outlet comprises a space between the plate and a wall of the air treatment chamber.

[0299] 6. The surface cleaning apparatus of clause 4 wherein an air treatment chamber air inlet is provided at the first end and an air treatment chamber air outlet is provided at the opposed end.

[0300] 7. The surface cleaning apparatus of clause 1 further comprising a suction port provided in a wall of the dirt collection chamber.

[0301] 8. The surface cleaning apparatus of clause 7 further comprising the suction port is positioned upstream of the motor and fan assembly whereby, when the motor and fan assembly is actuated, the bag is positioned in the dirt collection chamber and the dirt collection chamber is in the in use position, a reduced air pressure is provided between an outer surface of the bag and the dirt collection chamber.

[0302] 9. A surface cleaning apparatus comprising:

[0303] (a) an air flow path extending from a dirty air inlet to a clean air outlet with a motor and fan assembly in the air flow path;

[0304] (b) an air treatment stage positioned in the air flow path, the air treatment stage comprising a dirt collection chamber that is moveably mounted from an in use position to an emptying position;

[0305] (c) a plastic roll holder; and,

[0306] (d) a heat sealing member,

wherein the dirt collection chamber has an open end and a longitudinally spaced apart opposed second end, the closed end has an opening whereby, when a plastic roll is mounted to the plastic roll holder, an end of a sheet of plastic of the plastic roll is insertable through the opening into the interior of the dirt collection chamber and the heat sealing member engages sides of the sheet of plastic sheet whereby an open ended bag is formed.

[0307] 10. The surface cleaning apparatus of clause 9 wherein the plastic roll holder is provided at the lower end of the dirt collection chamber.

[0308] 11. The surface cleaning apparatus of clause 9 wherein the open end of the dirt collection chamber has a rim and, when the dirt collection chamber is in the in use position with an upper end of the bag positioned overlying the rim, the upper end of the bag is secured in position between the dirt collection chamber and an air treatment chamber of the air treatment stage.

[0309] 12. The surface cleaning apparatus of clause 9 wherein the air treatment stage comprises an air treatment chamber and a plate, the air treatment chamber having a first end having a first dirt outlet and an axially opposed second end, the plate is moveably mounted between a closed position in which the plate is positioned at an interface of the dirt collection chamber and the air treatment chamber and an open position in which the plate is moved away from the interface.

[0310] 13. The surface cleaning apparatus of clause 12 wherein the plate is moveably connected to the air treatment chamber, wherein, when the plate is in the closed position, the dirt outlet comprises a space between the plate and a wall of the air treatment chamber.

[0311] 14. The surface cleaning apparatus of clause 12 wherein an air treatment chamber air inlet is provided at the first end and an air treatment chamber air outlet is provided at the opposed end.

[0312] 15. The surface cleaning apparatus of clause 9 further comprising a suction port provided in a wall of the dirt collection chamber.

[0313] 16. The surface cleaning apparatus of clause 15 further comprising the suction port is positioned upstream of the motor and fan assembly whereby, when the motor and fan assembly is actuated, the bag is positioned in the dirt collection chamber and the dirt collection chamber is in the in use position, a reduced air pressure is provided between an outer surface of the bag and the dirt collection chamber.

#### Clause Set D

[0314] 1. A surface cleaning apparatus comprising:

[0315] (a) an air flow path extending from a dirty air inlet to a clean air outlet with a motor and fan assembly in the air flow path;

[0316] (b) a first air treatment stage positioned in the air flow path, the first air treatment stage comprising a dirt collection chamber that is moveably mounted from an in use position to an emptying position; and,

[0317] (c) a bag securing member comprising bag engaging members that are moveable from a retracted bag insertion position in which a bag is positionable in the dirt collection chamber and an extended bag engaging position in which the bag engaging members are positioned internal of the bag and the bag is secured in the dirt collection chamber,

wherein the bag is removeable from the dirt collection chamber when the dirt collection chamber is in the emptying position.

[0318] 2. The surface cleaning apparatus of clause 1 wherein the dirt collection chamber has an open end and a longitudinally spaced apart opposed closed end and the bag engaging members are moveable in a direction of a longitudinal axis.

[0319] 3. The surface cleaning apparatus of clause 2 wherein the bag engaging members are moveable lon-

gitudinally towards the closed end as the bag engaging members move from the retracted bag insertion position to the extended bag engaging position.

[0320] 4. The surface cleaning apparatus of clause 3 wherein, when the bag engaging members are in the extended bag engaging position, the bag is positioned between the bag engaging members and a wall positioned outwardly of the bag engaging members.

[0321] 5. The surface cleaning apparatus of clause 4 wherein the wall is a wall of the dirt collection chamber.

[0322] 6. The surface cleaning apparatus of clause 4 wherein the wall is an outer wall of the dirt collection chamber.

[0323] 7. The surface cleaning apparatus of clause 2 wherein the bag engaging members are mounted adjacent the open end.

[0324] 8. The surface cleaning apparatus of clause 1 wherein the bag engaging members engage an inner surface of the bag.

[0325] 9. The surface cleaning apparatus of clause 1 wherein the bag securing member is inflatable.

[0326] 10. The surface cleaning apparatus of clause 1 further comprising a bag locking actuator that is drivably connected to the bag engaging members.

[0327] 11. The surface cleaning apparatus of clause 10 wherein the bag locking actuator is actuated when the dirt collection chamber is moved to an in use position whereupon the bag engaging members are moved to the extended bag engaging position.

[0328] 12. The surface cleaning apparatus of clause 11 wherein the bag engaging members are biased to the retracted bag insertion position.

[0329] 13. The surface cleaning apparatus of clause 10 wherein the bag locking actuator is manually actuatable by a user.

[0330] 14. The surface cleaning apparatus of clause 10 wherein the bag locking actuator is pneumatically actuatable.

[0331] 15. The surface cleaning apparatus of clause 14 wherein, when the motor and fan assembly is actuated, the motor and fan assembly pressurizes a pneumatic line which drives the bag engaging members to the extended bag engaging position.

[0332] 16. The surface cleaning apparatus of clause 15 wherein the bag engaging members are biased to the retracted bag insertion position.

[0333] 17. The surface cleaning apparatus of clause 14 wherein, when the motor and fan assembly is actuated, the motor and fan assembly creates a sub atmospheric pressure in a pneumatic line which drives the bag engaging members to the extended bag engaging position.

[0334] 18. The surface cleaning apparatus of clause 1 wherein the bag engaging members are biased to the extended bag engaging position.

[0335] 19. The surface cleaning apparatus of clause 1 wherein the bag engaging members are biased to the retracted bag insertion position.

[0336] 20. The surface cleaning apparatus of clause 1 wherein the bag engaging members comprise longitudinally extending members and the surface cleaning apparatus further comprises a bag engaging member

wiper, whereby the bag securing members are cleaned as the bag securing members move to the retracted bag insertion position.

1. A surface cleaning apparatus comprising:

(a) an air flow path extending from a dirty air inlet to a clean air outlet with a motor and fan assembly in the air flow path;

(b) a first air treatment chamber having a first end having a first air treatment chamber air inlet and a first dirt outlet and an axially opposed second end having a first air treatment chamber air outlet;

(c) a dirt collection chamber that is moveably mounted from an in use position in which the dirt collection chamber is located at the first end of the first air treatment chamber to an emptying position; and,

(d) a plate moveably mounted between a closed position in which the plate is positioned at an interface of the dirt collection chamber and the first air treatment chamber and an open position in which the plate is moved away from the interface, the plate is moveably connected to the first air treatment chamber, wherein, when the plate is in the closed position, the dirt outlet comprises a space between the plate and a wall of the first air treatment chamber,

wherein a bag is positionable in the dirt collection chamber and is removeable from the dirt collection chamber when the dirt collection chamber is in the emptying position.

2. The surface cleaning apparatus of claim 1 wherein the first air treatment chamber comprises a cyclone chamber.

3. The surface cleaning apparatus of claim 1 wherein, when the plate is in the closed position, the plate is positioned further from the second end of the first air treatment chamber than the first air treatment chamber air inlet.

4. The surface cleaning apparatus of claim 1 further comprising a second air treatment chamber downstream from the first air treatment chamber, the second air treatment chamber having a second dirt outlet wherein dirt separated by the second air treatment chamber is collectable in the dirt collection chamber.

5. The surface cleaning apparatus of claim 5 wherein the dirt collection chamber has a first portion that collects dirt separated by the first air treatment chamber and a separate second portion that collects dirt separated by the second air treatment chamber.

6. The surface cleaning apparatus of claim 1 further comprising a second air treatment chamber downstream from the first air treatment chamber, the second air treatment chamber having at least two second air treatment chamber dirt outlets, each second air treatment chamber dirt outlet is upstream from a dirt conduit that extends axially along the first air treatment chamber.

7. The surface cleaning apparatus of claim 6 wherein the dirt conduits are exterior to the first air treatment chamber.

8. The surface cleaning apparatus of claim 7 wherein the dirt conduits are angularly spaced apart.

9. The surface cleaning apparatus of claim 8 wherein the surface cleaning apparatus is an upright surface cleaning apparatus comprising a surface cleaning head and an upright section moveably mounted to the surface cleaning head between a storage position and a rearwardly inclined in use position and an air treatment assembly comprising the first and second air treatment chambers and the dirt collection chamber are removably mounted to the upright section and,

when the air treatment assembly is mounted to the upright section, at least a portion of the upright section is positioned between the angularly spaced apart dirt conduits.

**10.** The surface cleaning apparatus of claim **1** wherein the plate is porous.

**11.** The surface cleaning apparatus of claim **10** wherein the plate comprises a screen.

**12.** The surface cleaning apparatus of claim **1** further comprising a screen overlying the dirt collection chamber and the screen is positioned further from the second end of the first air treatment chamber than the plate.

**13.** The surface cleaning apparatus of claim **12** wherein the screen overlies all of the dirt collection chamber.

**14.** The surface cleaning apparatus of claim **12** wherein the screen is also moveably mounted between a closed position in which the screen is positioned at the interface of the dirt collection chamber and the first air treatment chamber and an open position in which the screen is moved away from the interface.

**15.** The surface cleaning apparatus of claim **14** wherein the screen is moveable with the plate.

**16.** The surface cleaning apparatus of claim **1** wherein the surface cleaning apparatus comprises a first air treatment

stage and a second air treatment stage downstream from the first air treatment stage, the first air treatment stage comprising the first stage air treatment chamber and the dirt collection chamber and the second air treatment stage overlies the first air treatment stage.

**17.** The surface cleaning apparatus of claim **16** wherein each of the first and second air treatment stages has a central longitudinal axis and the central longitudinal axes are generally coaxial.

**18.** The surface cleaning apparatus of claim **1** wherein a mechanical member positioned internal of the bag when the bag is positioned in the dirt collection chamber secures the bag in position in the dirt collection chamber.

**19.** The surface cleaning apparatus of claim **1** wherein a mechanical member positioned external of the bag when the bag is positioned in the dirt collection chamber secures the bag in position in the dirt collection chamber.

**20.** The surface cleaning apparatus of claim **1** wherein releasable adhesive secures the bag in position in the dirt collection chamber.

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